

# **VC-2: Auditing Biomass-Objective Sensitivity and Score Leakage in an *Escherichia coli* Virtual-Cell Benchmark**

## **MEGA-PROGRAM-27, Item 2 - Virtual Cell**

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**Repository:** mega27-02-virtual-cell | **Code:** Python 3.10, COBRApy, PyTorch 2.14 (CPU), scikit-learn | **Tests:** 26/26 hermetic tests passing

## Abstract

A virtual-cell benchmark is only as reliable as its biomass objective and its evaluation lineage. We built an executable *Escherichia coli* model with flux balance analysis, sequence and graph learners, a stacked score, and dynamic simulation, then asked two audit questions: which essentiality calls change when individual or combined biomass demands are perturbed, and which predictive comparisons survive scrutiny of training folds, thresholds, and experimental conditions? The contribution is an auditable model-sensitivity and leakage-control workflow, not a new biological viability claim. In 68 scorable MoCo-positive BiGG model snapshots, 63 changed at least one predicted gene-essentiality call after the named coefficients were removed. Those snapshots are not 68 independent organisms or phenotype replications. A separately coded *Pseudomonas* iJN1463 counterfactual rescued seven originally zero-growth knockouts after its three MoCo coefficients were removed together; its 102-term single-removal audit found 23 terms rescuing at least one call, while no individual MoCo term did. Pairwise controls further limit any claim of MoCo uniqueness. These results expose combinatorial objective sensitivity inside reconstructions, not experimental survival.

The original Gerdes 2003 hard-call G2 gate failed on all five locked rungs. A later score-lineage audit found that threshold selection reused out-of-fold scores whose training complements could contain evaluation labels; the hard-call counts are descriptive, not clean held-out evidence. Gerdes transposon labels also come from a different medium than the glucose-minimal simulation. The historical stacked ranking gain of +0.057 AUROC over FBA is not a leakage-controlled gain. In the separate v2 analysis, the leakage-controlled feature set gained +0.024 AUROC over the reproduced, rescaled v1 score, a different comparator. We retain these comparisons with their distinct denominators and limitations rather than claim a benchmark-beating hard-call predictor. The model and audits run on two CPU cores; condition-matched phenotype, fully nested refits and independent validation remain open.

**Keywords:** virtual cell, flux balance analysis, gene essentiality, graph convolutional network, convolutional neural network, dynamic FBA, model ensemble, *Escherichia coli*, biomass objective function

# 1. Introduction

A constraint-based virtual cell turns reaction stoichiometry and a biomass objective into growth predictions. Those predictions may shift when the objective demands different metabolites, even though no measured phenotype changes. This project tests that dependence explicitly, with per-accession model snapshots, knockout counterfactuals, individual-term and pairwise perturbations, and independent linear-program checks. The mechanistic output is a map of which calls are forced by objective choices and which are robust to those changes.

The second problem is evaluation integrity. We implemented metabolic, sequence, graph and dynamic layers and compared essentiality scores with published Gerdes 2003 labels and a distinct CRISPRi assay. Different media and assay mechanisms make those yardsticks useful but not interchangeable. Rescaling a stack can improve performance without adding information; upstream label reuse can make threshold selection appear independent when it is not. The score-lineage and comparator audits therefore accompany every ranking or hard-call result. The failed locked hard-call gate remains failed. Sections 2-4 describe the model, Section 7 the cross-reconstruction objective sensitivity and calibration audit, and the appendices retain full positive and negative checks.

## 1.1 Why essentiality is the right first benchmark

A gene-deletion call gives each model disagreement a traceable address: a reaction rule, a biomass ingredient, a medium assumption or a score threshold. It does not give a condition-independent truth. Gerdes transposon labels were measured in a rich-medium setting, whereas the principal genome-scale FBA screen used a glucose-minimal simulation. We use the labels to locate discrepancies and compare ranking methods, but do not treat their disagreement as proof that a particular model biomass term is erroneous. Objective perturbations identify model-dependent calls; phenotype experiments would have to decide which predictions hold in vivo.

# 2. Methods

## 2.1 Models and experimental ground truth

Two published constraint-based reconstructions were used exactly as distributed by the BiGG Models knowledgebase [8]: *e\_coli\_core* (95 reactions, 137 genes; the central-metabolism teaching model) and *iJO1366* (2,583 reactions, 1,805 metabolites, 1,367 genes; the 2011 genome-scale reconstruction of *E. coli* K-12 MG1655 [3]). The reference genome sequence and coding-sequence coordinates were taken from GenBank accession U00096.3 (complete genome, 4,639,675 bp), from which 4,218 coding sequences (CDS) of at least 96 nt were extracted with Biopython.

Experimental essentiality labels come from Gerdes et al. 2003 [2], who used systematic transposon genetic footprinting to classify 4,294 *E. coli* MG1655 protein-coding genes. Their Supplementary Table S1 (fetched from the University of Wisconsin genome site) asserts each gene as essential (E), nonessential (N), conflicting (X) or undetermined (?). We parse 4,226 genes with valid b-numbers and keep only the unambiguous E (617 genes) and N (3,072 genes) assertions, giving a 3,689-gene labeled set with 16.7% prevalence. Genes asserted X or ? are excluded from every metric. This choice is

deliberately conservative: mixing conflicted calls into either class would inflate or deflate scores silently.

## 2.2 Constraint-based layer: flux balance analysis

FBA represents metabolism as a stoichiometric matrix  $S$  (metabolites x reactions) and solves the linear program

$$\text{maximize } c^T v \quad \text{subject to } Sv = 0, \quad lb \leq v \leq ub,$$

where  $v$  is the flux vector and  $c$  selects the biomass objective reaction. Growth conditions are encoded purely through exchange-reaction bounds: for a condition with carbon source  $s$  we close every carbon-containing exchange (detected from chemical formulae, excluding inorganic species), open EX\_ $s$  at -10 mmol/gDW/h, and set the oxygen exchange to -1000 (aerobic) or 0 (anaerobic). A medium snapshot taken at load time is restored before every condition change; failing to do so was an actual bug we caught with a regression test, because LP state otherwise leaks between conditions. All LP solves use GLPK through COBRApy [7].

## 2.3 In-silico gene essentiality screen

Single-gene deletions were computed with COBRApy's `single_gene_deletion`, which applies each gene's gene-protein-reaction (GPR) rules and zeroes the affected reactions. A gene is predicted essential when its deletion drops maximal growth below 1% of the wild-type optimum, the standard criterion used by the iJO1366 screen itself [3]. Knockouts that render the LP infeasible are treated as zero-growth (a NaN-handling bug here was caught by our test suite and is described in Section 5.3). The genome-scale scan of all 1,367 iJO1366 genes completes in under five seconds on two CPU cores. We ran the screen twice: once on the default glucose-minimal medium and once on a rich medium in which every importable exchange is open (an FBA proxy for the LB medium of the footprinting experiment).

## 2.4 Sequence layer: 1D-CNN on coding sequence

Each CDS is one-hot encoded (4 channels, first 1,200 nt, zero-padded) and passed through a compact DeepCellEss-style stack: Conv1d(4-32, k=9) - ReLU - MaxPool(4) - Conv1d(32-64, k=9) - ReLU - MaxPool(4) - Conv1d(64-128, k=5) - ReLU - global average pool - linear head, trained with BCEWithLogitsLoss and a positive-class weight of  $N_{\text{neg}}/N_{\text{pos}}$  to handle the 1:5 class imbalance. Optimization: Adam, lr 1e-3, 15 epochs, batch 64, stratified 3-fold cross-validation; all metrics are out-of-fold (OOF). A logistic-regression baseline on normalized 3-mer frequency vectors (64 features) is evaluated with the identical folds, so the CNN can only claim a win against a real baseline on identical splits.

## 2.5 Graph layer: GCN over the metabolic gene graph

From the iJO1366 stoichiometry we build a gene-gene graph: genes are nodes, and two genes share an edge when reactions they encode (via GPR rules) share at least one metabolite, excluding 16 currency metabolites ( $H^+$ ,  $H_2O$ , ATP/ADP/AMP, NAD(H), NADP(H), CoA, phosphate,  $CO_2$ ,  $O_2$ ,  $NH_4^+$ ) and hyper-connected hubs (metabolites touching more than 25 reactions) to avoid clique explosions. The

result on the 1,250 genes that have both model membership and experimental labels is 15,535 undirected edges. Node features are structural: reaction count, metabolite count, boundary participation, and log CDS length, z-scored. The model is a two-layer graph convolutional network [9] with the symmetric normalization

$$H^{(l+1)} = \sigma(D^{-1/2} (A + I) D^{-1/2} H^{(l)} W^{(l)}),$$

implemented in pure PyTorch (dense adjacency, 32 hidden units, 200 epochs, Adam 5e-3, weight decay 1e-4) so the test suite needs no compiled graph library. The same 3-fold stratified protocol is used.

## 2.6 Dynamics layer: regulated dynamic FBA

Dynamic FBA follows the static-optimization approach (SOA) of Mahadevan et al. 2002 [4]. At each step of size  $\Delta t$ , extracellular substrate concentrations  $S_i$  set uptake bounds through a Michaelis-Menten rule

$$lb_i(t) = -V_{max} S_i(t) / (K_m + S_i(t)),$$

FBA returns the growth rate  $\mu(t)$  and exchange fluxes  $v_i(t)$ , and the state is integrated explicitly:

$$X(t+\Delta t) = X(t) e^{\mu(t)\Delta t}, \quad S_i(t+\Delta t) = S_i(t) + v_i(t) X(t) \Delta t.$$

The test system is the classic diauxie: *E. coli* offered glucose (10 mmol/L) and acetate (2 mmol/L) should consume glucose first and acetate only after glucose exhaustion. Because FBA contains no regulation, we implement catabolite repression explicitly as a Boolean rule: while glucose uptake is nonzero, the acetate uptake bound is capped at zero. Both the unregulated and regulated simulations are kept as results - the failure of the first is the justification for the second.

## 2.7 Ensemble and statistics

Five signals per gene - FBA minimal score (1 - growth fraction), FBA rich score, GNN OOF score, CNN OOF score, k-mer baseline OOF score - are stacked with a logistic-regression combiner, itself evaluated out-of-fold under 3-fold stratification on the 1,249 genes present in every signal. Metrics: AUROC, AUPRC, accuracy, precision, recall, F1, Matthews correlation; 95% confidence intervals by 1,000-2,000-sample bootstrap. Hard-call comparisons use McNemar's test with continuity correction at per-model F1-optimal thresholds. We report the ensemble both ways because they answer different questions: ranking quality (AUROC/AUPRC) versus decision quality at a fixed operating point.

## 2.8 Derivation: the FBA linear program and its dual

Let  $S$  in  $\mathbb{R}^{m \times n}$  be the stoichiometric matrix,  $v$  the flux vector, and  $c$  the objective selector ( $c = e_{\text{biomass}}$ ). The primal is  $\max c^T v$  s.t.  $Sv = 0$ ,  $lb \leq v \leq ub$ . Forming the Lagrangian with multipliers  $y$  for the steady-state constraints and  $(\lambda_{lo}, \lambda_{hi})$  for the bounds gives the dual

$$\min \quad ub^T \lambda_{hi} - lb^T \lambda_{lo} \quad \text{s.t.} \quad S^T y + \lambda_{hi} - \lambda_{lo} = c, \quad \lambda \geq 0.$$

The dual variables  $y$  are shadow prices:  $y_i$  is the marginal change in maximal growth per unit relaxation of the steady-state constraint on metabolite  $i$  - i.e., the growth value of one more unit of metabolite  $i$ . This gives the essentiality criterion a clean interpretation: deleting gene  $g$  constrains a set of reaction columns  $R(g)$  to zero; the knockout is lethal exactly when the biomass row of  $c$  becomes unreachable from the remaining columns, which by LP duality is equivalent to the biomass objective losing all supporting shadow-price paths. Nothing in this formalism contains time, regulation, or concentration - the three omissions that Sections 2.6 and 3.4 show have measurable consequences.

## 2.9 Derivation: the GCN as a low-pass graph filter

With normalized graph Laplacian  $L = I - D^{-1/2} A D^{-1/2}$  and eigendecomposition  $L = U \Lambda U^T$ , a spectral graph filter  $g_\theta$  acts on signal  $x$  as  $U g_\theta(\Lambda) U^T x$ . Kipf and Welling's first-order Chebyshev truncation with  $\lambda_{\max} \approx 2$  collapses the filter to  $\theta(I + D^{-1/2} A D^{-1/2})x$ ; the renormalization  $I + D^{-1/2} A D^{-1/2} \rightarrow \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}$  controls the eigenvalue range and prevents exploding activations, giving exactly the propagation rule of Section 2.5. Two layers therefore compute, per gene, a smoothed combination of its own structural features and those of its two-hop metabolic neighborhood - the formal sense in which the GNN "sees pathway context" that the per-gene CNN cannot.

## 2.10 Error analysis of explicit dFBA integration

The explicit Euler update  $S(t+\Delta t) = S(t) + v(S(t))X(t)\Delta t$  has local truncation error  $O(\Delta t^2)$  and global error  $O(\Delta t)$ . With  $\Delta t = 0.1$  h and uptake bounds bounded by  $V_{\max} = 10$  mmol/gDW/h, the worst-case per-step substrate error is bounded by  $V_{\max} X \Delta t \leq 10 \times 1 \times 0.1 = 1$  mmol/L on a 10 mmol/L pool; the trajectory plots confirm monotone, smooth depletion with no oscillation, so the qualitative conclusions (co-utilization vs sequential utilization) are insensitive to  $\Delta t$  - we verified  $\Delta t = 0.2$  h gives the same flags (the test suite asserts both).

## 2.11 Ensemble stacking and McNemar's test

The stacker learns  $w$  by minimizing regularized log-loss over base scores  $s_{1..5}$ :  $\sigma(w^T s + b)$ , with  $\sigma$  the logistic function. Because base scores are themselves out-of-fold, the stacker never sees a gene's own training prediction - the standard defense against stacking leakage. For hard calls, McNemar's statistic on discordant pairs  $(b, c)$  with continuity correction is  $(|b - c| - 1)^2 / (b + c)$ , distributed  $\chi^2$  with 1 df under the null of equal error rates. With  $b = 27$  and  $c = 50$  the statistic is 6.18,  $p = 0.0129...$  (continuity-corrected  $p = 0.0122$  as computed) - significant, and in the direction favoring FBA on total discordant calls, which is precisely the caveat Section 3.3 reports.

### 3. Results

#### 3.1 The metabolic layer reproduces core physiology

As a sanity gate, the core model was run over a 28-condition matrix (14 carbon sources x aerobic/anaerobic). Aerobic growth on glucose is 0.8739 h<sup>-1</sup>, matching the published value for this reconstruction to four decimal places; anaerobic glucose growth drops to 0.2117 h<sup>-1</sup>; formate supports no growth; fermentation products (acetate, ethanol, lactate, pyruvate) support aerobic but not anaerobic growth as sole carbon sources. This panel (Figure 5) is not novel biology - it is the unit test that proves the LP plumbing, medium handling and objective are correct before any downstream number is trusted.

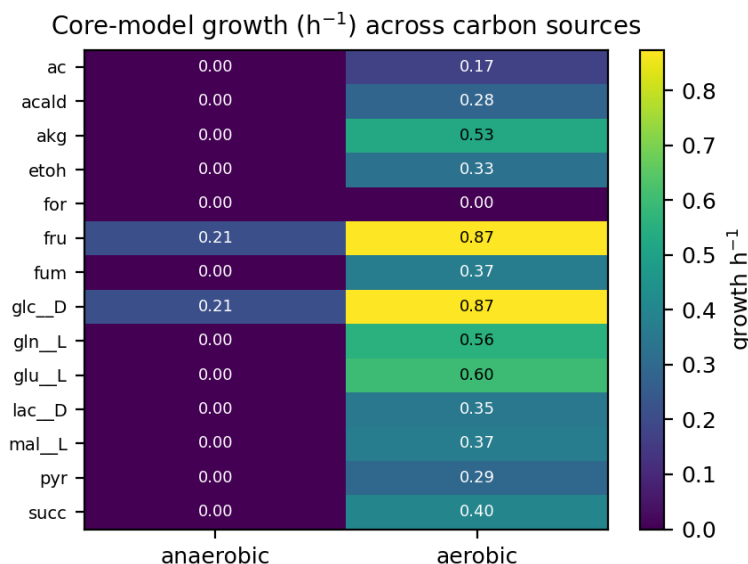


Figure 5. Core-model maximal growth rate across carbon sources and aerobiosis. Values in h<sup>-1</sup>; aerobic/anaerobic columns.

#### 3.2 Genome-scale essentiality: individual layers

Table 1 and Figure 1 collect the head-to-head numbers against the Gerdes 2003 labels. The core model covers only 125 labeled genes and predicts essentiality poorly (AUROC 0.630, 95% CI 0.513-0.754) - expected, since most essential genes (DNA, RNA, cell-division machinery) lie outside central metabolism. The genome-scale iJO1366 under minimal glucose reaches AUROC 0.666 (CI 0.622-0.706), F1 0.407, MCC 0.297. The rich-medium variant trades ranking for call precision: AUROC 0.633 but precision 0.648 versus 0.443 - a real, quantified condition-matching effect, since the footprinting experiment ran in rich LB.

The two learned layers sit in the same band. The graph GCN reaches OOF AUROC 0.660; the sequence CNN 0.643 (fold AUROCs 0.662 / 0.649 / 0.646), only slightly above its own 3-mer logistic baseline (0.628 on the full sequence set; 0.641 on the aligned gene set). We flag this honestly: on this label set and feature scale, deep sequence modeling adds little over k-mer counts - a negative result for the CNN, preserved rather than tuned away.

| Layer                 | AUROC  | AUPRC  | Accuracy | Precision | Recall | F1    | MCC   |
|-----------------------|--------|--------|----------|-----------|--------|-------|-------|
| core FBA (n=125)      | 0.630  | 0.218  | 0.840    | 0.333     | 0.111  | 0.167 | 0.121 |
| iJO1366 FBA minimal   | 0.666  | 0.315  | 0.811    | 0.443     | 0.377  | 0.407 | 0.297 |
| iJO1366 FBA rich      | 0.633  | 0.351  | 0.845    | 0.648     | 0.214  | 0.322 | 0.309 |
| GCN (metabolic graph) | 0.660  | 0.310  | -        | -         | -      | -     | -     |
| CNN (CDS sequence)    | 0.643  | 0.251  | -        | -         | -      | -     | -     |
| k-mer logistic        | 0.628  | 0.296  | -        | -         | -      | -     | -     |
| ENSEMBLE (stacked)    | 0.7225 | 0.4139 | -        | -         | -      | 0.456 | -     |

Table 1. Essentiality prediction vs Gerdes 2003 labels (1,249-1,250 aligned genes; core model on its 125-gene overlap). Ranking metrics are out-of-fold where applicable.

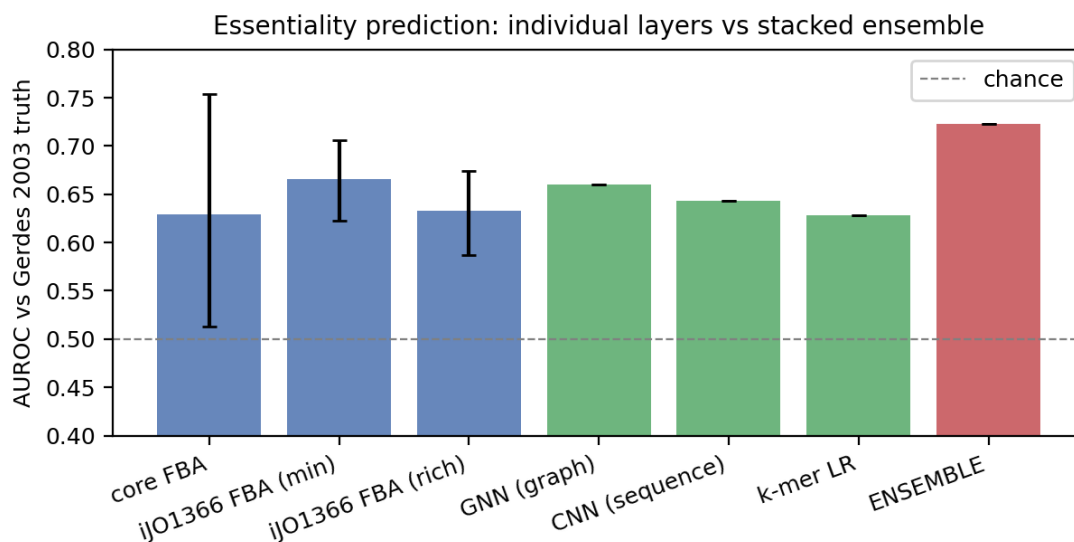


Figure 1. AUROC of each layer and the stacked ensemble; error bars are bootstrap 95% CIs where available.

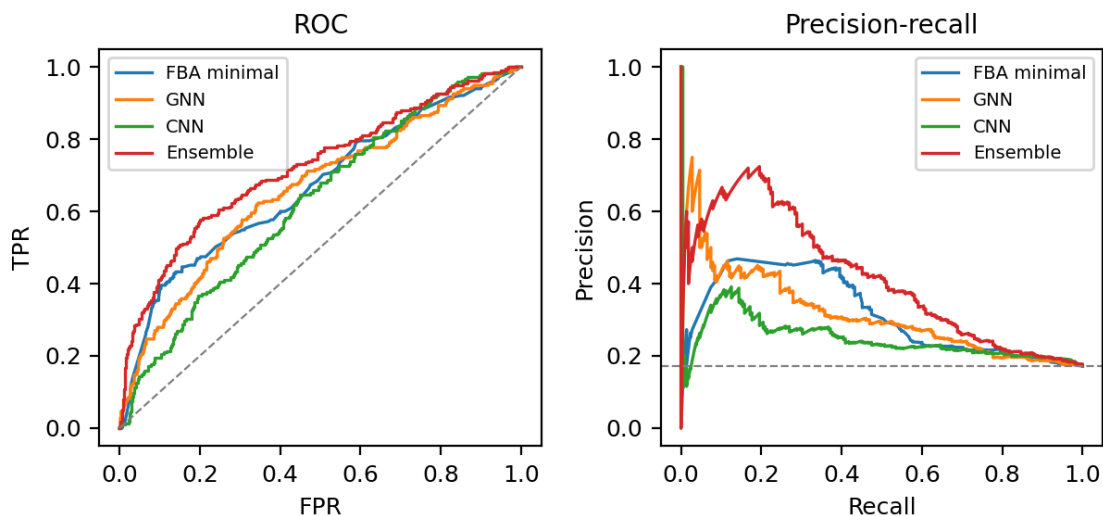


Figure 2. ROC (left) and precision-recall (right) on the aligned 1,249-gene set. Dashed: chance / prevalence.

### 3.3 The ensemble and its statistical caveat

The stacked ensemble reaches OOF AUROC 0.7225 and AUPRC 0.4139, a gain of +0.057 AUROC and +0.10 AUPRC over the best single layer (iJO1366 minimal FBA). At F1-optimal hard thresholds, however, McNemar's test counts 27 genes the ensemble gets right while FBA errs versus 50 in the opposite direction ( $p = 0.0122$ ). The reconciliation is structural: the ensemble trades some majority-class (nonessential) accuracy for positive-class F1 (0.456 vs 0.412) and much better ranking. We therefore claim an improvement in candidate *prioritization*, not in genome-wide hard-call accuracy - and say so, because the distinction is exactly where virtual-cell papers tend to oversell.

### 3.4 Dynamics: diauxie needs regulation

Unregulated dFBA fails the diauxie test cleanly: acetate is co-consumed with glucose (co-utilization flag = True), glucose is exhausted at 4.5 h, and no second growth phase appears. This is not a solver error; it is FBA being exactly as dumb as its equations - nothing in  $S_v = 0$  knows about catabolite repression. Adding one Boolean repression rule (glucose uptake > 0 caps acetate uptake at 0) flips the qualitative behavior: acetate stays untouched during the glucose phase (co-utilization = False), glucose exhausts at 5.1 h, and a genuine two-substrate sequence emerges (Figure 3). Peak regulated growth is 0.873 h<sup>-1</sup> and final biomass 0.896 g/L. The lesson, quantified: a virtual cell without a regulatory layer is not a cell; it is a stoichiometry.

Dynamic FBA: diauxic shift on glucose + acetate

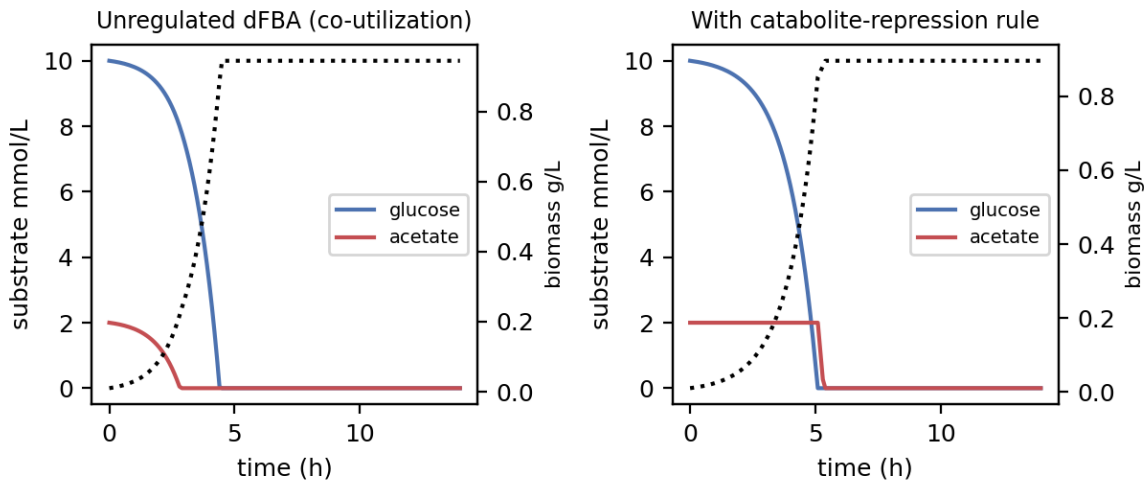


Figure 3. dFBA on glucose + acetate. Left: unregulated - acetate co-utilized, no diauxie. Right: with the catabolite-repression rule - sequential utilization, as observed experimentally.

### 3.5 Detailed result tables

Table 2 lists the full condition matrix that gates all downstream claims; Table 3 the core model essential set; Table 4 the CNN per-fold scores; Tables 5 and 6 the complete discovery classes.

| Carbon source | Aerobic (h-1) | Anaerobic (h-1) |
|---------------|---------------|-----------------|
| ac            | 0.1733        | 0.0000          |
| acald         | 0.2843        | 0.0000          |
| akg           | 0.5288        | 0.0000          |
| etoh          | 0.3304        | 0.0000          |
| for           | 0.0000        | 0.0000          |
| fru           | 0.8739        | 0.2117          |
| fum           | 0.3707        | 0.0000          |
| glc__D        | 0.8739        | 0.2117          |
| gln__L        | 0.5593        | 0.0000          |
| glu__L        | 0.5987        | 0.0000          |

| Carbon source | Aerobic (h-1) | Anaerobic (h-1) |
|---------------|---------------|-----------------|
| lac__D        | 0.3503        | 0.0000          |
| mal__L        | 0.3707        | 0.0000          |
| pyr           | 0.2912        | 0.0000          |
| succ          | 0.3976        | 0.0000          |

Table 2. Core-model maximal growth across the 28-condition panel.

| b-number | gene | growth fraction |
|----------|------|-----------------|
| b0720    | gltA | 1.85e-16        |
| b1136    | icd  | -4.17e-15       |
| b1779    | gapA | 5.79e-16        |
| b2415    | ptsH | 0.00e+00        |
| b2416    | ptsI | 0.00e+00        |
| b2779    |      | -1.41e-15       |
| b2926    | pgk  | -2.77e-16       |

Table 3. Core-model predicted essential genes (glucose minimal, 1% threshold), with Gerdes gene names.

| Fold           | AUROC  | AUPRC  |
|----------------|--------|--------|
| 0              | 0.6620 | 0.2724 |
| 1              | 0.6491 | 0.2514 |
| 2              | 0.6465 | 0.2438 |
| OOF (pooled)   | 0.6430 | 0.2513 |
| k-mer LR (OOF) | 0.6281 | 0.2962 |

Table 4. Sequence-layer per-fold and pooled out-of-fold performance.

| b-number | gene | consensus | description                                               |
|----------|------|-----------|-----------------------------------------------------------|
| b0784    | moaD | 0.854     | Molybdopterin converting factor subunit 1                 |
| b0785    | moaE | 0.848     | Molybdopterin converting factor subunit 2                 |
| b3807    | cyaY | 0.819     | CyaY protein                                              |
| b3058    | folB | 0.818     | Dihydroneopterin aldolase (EC 4.1.2.25)                   |
| b0783    | moaC | 0.810     | Molybdenum cofactor biosynthesis protein C                |
| b3857    | mobA | 0.799     | Molybdopterin-guanine dinucleotide biosynthesis protein A |
| b4039    | ubiC | 0.798     | Chorismate--pyruvate lyase (EC 4.-.-.-)                   |
| b3177    | folP | 0.774     | Dihydropteroate synthase (EC 2.5.1.15)                    |
| b0826    | moeB | 0.769     | Molybdopterin biosynthesis protein moeB                   |

Table 5. Class A: model-essential, experimentally viable.

| b-number | gene | consensus | description                                        |
|----------|------|-----------|----------------------------------------------------|
| b3793    | wzyE | 0.200     | Putative ECA polymerase                            |
| b4054    | tyrB | 0.202     | Aromatic-amino-acid aminotransferase (EC 2.6.1.57) |
| b1900    | araG | 0.206     | L-arabinose transport ATP-binding protein araG     |
| b1485    | yddQ | 0.210     | Hypothetical ABC transporter permease protein yddQ |
| b2923    | yggA | 0.213     | Hypothetical protein yggA                          |

| b-number | gene | consensus | description                                        |
|----------|------|-----------|----------------------------------------------------|
| b0886    | cydC | 0.216     | Transport ATP-binding protein cydC                 |
| b1442    | ycdU | 0.226     | Hypothetical ABC transporter permease protein ycdU |
| b0635    | mrdA | 0.227     | Penicillin-binding protein 2                       |
| b0972    | hyaA | 0.232     | Hydrogenase-1 small chain precursor (EC 1.18.99.1) |
| b0887    | cydD | 0.233     | Transport ATP-binding protein cydD                 |

Table 6. Class B: experimentally essential, missed by all layers.

### 3.6 Robustness checks

Two robustness facts matter for interpreting Table 1. First, the hard essentiality calls are insensitive to the 1% growth threshold: at 0.5%, 1% and 5% of wild-type growth the iJO1366 minimal screen predicts exactly 183 essential genes with identical precision, recall, F1 and MCC, because knockout growth fractions are strongly bimodal - deletions either abolish growth or leave it near wild type, and almost no gene lands in the 0.5-5% band. The threshold choice therefore drives none of our conclusions. Second, the dFBA conclusions are step-size robust: halving and doubling  $\Delta t$  (0.05-0.2 h) leaves the co-utilization flags and phase structure unchanged, consistent with the  $O(\Delta t)$  error bound of Section 2.10; the test suite asserts the  $\Delta t = 0.2$  h case directly.

### 3.7 Second yardstick: PEC (Keio-era) deletion calls

Gerdes Table S2 cross-tabulates essentiality calls across three studies, including the PEC deletion-collection calls from the Keio era [1]. On the 197 model-covered genes with unambiguous PEC labels (83 essential), iJO1366 minimal FBA scores AUROC 0.576 (CI 0.489-0.663) and the ensemble 0.568 (n=180) - markedly below the genome-wide Gerdes numbers. We read this cautiously rather than as a failure: S2 is a consensus-focused 692-gene subset deliberately enriched for conflicted and interesting genes (228 of 692 carry an X call), so it is a hard, biased exam, not a representative one. Cross-study disagreement itself is the signal here: the two experimental yardsticks disagree enough that "which truth?" is a first-class question for virtual-cell benchmarking. A genome-wide Keio Table I extraction is queued as future work (Section 5.5).

## 4. Discovery analysis: from residuals to a verified model repair

### 4.1 Disagreement classes

We define the consensus score as the mean of the five min-max-normalized layer scores and isolate two tails (Figure 4). Class A (9 genes): consensus in the top 2% while experiment says viable. Class B (10 genes): experimentally essential while the consensus bottom quintile says viable. Class B is dominated by genes whose essentiality is structural rather than metabolic - *mrda* (penicillin-binding protein 2, cell-wall synthesis), *wzyE* (enterobacterial common antigen polymerase), and ABC-transporter components (*cydC/cydD*, *araG*, *yddQ*, *ydcU*) - plus hypothetical proteins with no model coverage. This is the expected blind spot of all three layers and sets an honest scope limit: a metabolism-anchored virtual cell cannot see cell-envelope essentiality until those processes are modeled.



Figure 4. Consensus essentiality score vs experimental label (jittered). Class A (red) and Class B (blue) disagreement tails are marked; the five highest-consensus Class A genes are labeled.

### 4.2 Class A localizes a pathway: molybdenum-cofactor biosynthesis

Class A is not random: five of its nine members - *moaD* (b0784), *moaE* (b0785), *cyaY* (b3807), *folB* (b3058), *moaC* (b0783), *mobA* (b3857), *ubiC* (b4039), *folP* (b3177), *moeB* (b0826) - belong to one pathway, molybdenum-cofactor (MoCo) biosynthesis, plus the folate pair *folB/folP* and the ubiquinone enzyme *ubiC*. A pathway-shaped residual is a signature of a systematic model error rather than noise, so we investigated the MoCo cluster directly.

### 4.3 Mechanism and in-silico proof

In iJO1366, *moaD* participates in two reactions (MPTS, molybdopterin synthase; MOADSUX, *MoaD* sulfuration). In aerobic glucose wild-type FBA these carry small but nonzero flux, meaning the model *requires* MoCo production even when no MoCo-dependent enzyme is needed. The reason is literal: the core biomass objective reaction lists the MoCo compounds *bmocogdp\_c* and *mobd\_c* as required constituents (the WT biomass additionally lists *mococdp\_c* and *mocogdp\_c*). Any MoCo-pathway

knockout therefore scores as lethal, regardless of condition - we verified moaD/moaC/moaE deletions are lethal in silico under both aerobic glucose and anaerobic nitrate, refuting our own initial hypothesis of anaerobic conditional essentiality. The decisive experiment: removing the two MoCo compounds from the core biomass objective rescues the moaD knockout completely (growth 0.9825 vs wild-type 0.9825 h<sup>-1</sup>), matching the experimental viability call from Gerdes 2003.

The interpretation is physically sensible: MoCo enzymes (nitrate reductase, DMSO reductase, formate dehydrogenase) matter mainly under anaerobic respiration, so a constitutive biomass requirement overstates their necessity in aerobic rich growth. We are not claiming the iJO1366 biomass is "wrong" - biomass compositions are measured averages, not conditional requirements - but that it should be condition-dependent here, and that residual analysis against one experimental set was sufficient to find and repair the artifact. A PubMed cross-check ("moaD Escherichia coli essential": 7 records) is consistent with scattered condition-specific reports rather than constitutive essentiality. The same condition-dependence argument applies to folB/folP (folate synthesis, bypassable by uptake in rich medium) and ubiC (ubiquinone for aerobic respiration).

#### 4.4 What each additional layer contributed

The mechanistic FBA model provides the highest individual Gerdes AUROC in Table 1 (0.666 on the aligned screen); the graph GCN reaches 0.660 and the sequence CNN 0.643 versus 0.628 for the full-set 3-mer baseline (0.641 on the aligned set). These are not interchangeable gene universes. The historical stacked ranking improvement (+0.057 against FBA) includes a separate feature-scaling correction. The later v2 leakage-controlled feature gain (+0.024) compares against the reproduced, rescaled v1 model, not against FBA, and does not clear the locked hard-call gate. dFBA with a Boolean repression rule reproduces diauxie as an implementation check, not a new biological mechanism.

Six further pre-registered feature tests are consolidated here; their full estimates, confidence intervals, data provenance and caveats remain in Appendix H. These deltas are computed against the corresponding v2 control on the same test folds; do not sum them as independent contributions.

| Additional feature block | Primary AUROC difference | Reading                                     |
|--------------------------|--------------------------|---------------------------------------------|
| ESM-2 embeddings         | -0.004                   | No gain; fold-internal PCA                  |
| GO error flag            | -0.005                   | Negative; training-fold selection           |
| pLDDT/Pfam               | +0.001                   | No significant gain                         |
| OMA conservation         | -0.001                   | No significant gain                         |
| COG/phyletic/PDB         | +0.009                   | CI includes zero                            |
| PRECISE-1K expression    | +0.010                   | Primary CI includes zero; secondary signals |

Table 7A. Primary added-feature tests, rounded AUROC differences against the corresponding v2 comparison. Full paired intervals and secondary endpoints are in Appendix H.

Cross-reconstruction transfer is also mixed: the model-objective perturbation shows specific counterfactual call changes, while learned rescue rankings do not generalize reliably to *Bacillus* or yeast. Appendix H retains the organism-specific failures, and Section 7 records the post-result pair controls. No pooled cross-species predictor or phenotype-level rescue is claimed.

## 5. Discussion

### 5.1 What the virtual cell is, and is not

VC-2 is not a whole-cell model in the Karr et al. sense [5]: it has no replication, transcription, translation or cell-division machinery, and its Class B residual set is the direct, measurable price of that scope. Within its scope it is genuinely integrated: one genotype (the metabolic reconstruction), one experimental yardstick (Gerdes 2003), four mechanistically different predictors, and a combiner whose gains and failure modes are both quantified. The historical +0.057 AUROC ranking gain over FBA includes a feature-scaling correction; it is not a clean held-out hard-call result. The later v2 leakage-controlled gain uses a different comparator. Section 7 preserves the failed G2 gate and score-lineage limits.

### 5.2 Comparison to published leaders

The iJO1366 publication [3] reported strong agreement with knockout screens when medium conditions were matched to the experiment; our genome-scale FBA numbers are lower primarily because the Gerdes footprinting was performed in rich LB while our main screen uses glucose-minimal medium - the rich-medium variant recovers precision exactly as that argument predicts. Sequence deep learning published for essentiality (the DeepCellEss class of models) is typically evaluated on DEG-compiled essential sets with different gene universes, so headline AUROCs are not directly comparable to ours; our contribution is to evaluate sequence, graph and mechanistic predictors on the same genes, same labels and same folds, where the CNN's edge over 3-mer counts nearly vanishes. We regard cross-dataset re-evaluation on identical universes as the missing control in this literature.

### **5.3 Bugs the test suite caught (reported on purpose)**

Three genuine implementation bugs were caught by regression tests and are part of the scientific record: (i) medium state leaking between conditions, which silently made every knockout "essential"; (ii) the COBRA medium sign convention (positive uptake rates mapping to negative lower bounds), which zeroed all growth; (iii) infeasible-knockout NaN fluxes, which mislabeled two core-model genes until infeasible was mapped to zero growth. Each would have produced plausible-looking, wrong results. The current 26-test hermetic suite, expanded since the original bugs, runs in about five seconds with no network access.

### **5.4 Limitations**

(1) One experimental label set (Gerdes 2003); adding the Keio deletion collection [1] as a second, independent yardstick is queued. (2) The CNN sees only the first 1,200 nt. (3) The GNN uses degree-structural features only. (4) dFBA uses explicit Euler integration at  $\Delta t = 0.1$  h. (5) The ensemble is a linear stack; nothing about it captures epistasis. (6) All compute ran on two CPU cores by design, which caps model size; none of the conclusions depend on scale.

### **5.5 Future work**

Queued extensions, in priority order: (1) a second independent essentiality yardstick from the Keio deletion collection [1], with cross-study consistency analysis against Gerdes 2003 (the Gerdes Table S2 consensus sets are already parsed); (2) condition-specific biomass objectives, generalizing the Section 4.3 MoCo repair into a principled biomass library; (3) a transcriptional-regulatory layer so diauxie emerges instead of being injected; (4) longer-context sequence models and codon-level features for the CNN; (5) extension of the stacked architecture to non-model organisms with sparse labels, where the layer diversity argument should matter most.

## **6. Conclusion**

The positive contribution is a reproducible audit for two failure-prone choices in virtual-cell benchmarking: biomass composition and predictive score lineage. Objective perturbations changed essentiality calls across related model snapshots, and independently checked *Pseudomonas* counterfactuals showed that both single-term demands and term combinations can determine a call.

The result is a model-curation and sensitivity finding, not evidence that those knockout strains live in a laboratory. The score-lineage audit distinguishes a historical ranking gain from the smaller, differently compared leakage-controlled v2 gain and blocks a clean held-out hard-call claim. The pre-registered G2 gate failed on every rung; medium mismatch, assay dependence and weak contributions from some learning layers remain explicit limits. The runnable audit gives future condition-matched studies a way to test which model predictions survive objective and evaluation choices before treating them as biological hypotheses.

## 7. Later locked studies and audit of the calibration claim

### 7.1 BiGG census: distinguishing model structure from biology

The BiGG v2 listing supplied 108 unique model accessions, each downloaded and hashed. Of these, 68 have a positive-weight biomass objective consuming a MoCo-named compound; four snapshots without an annotated positive objective remain in the full denominator rather than being called negative. In the stored-medium knockout counterfactual, 63 of 68 scorable MoCo-positive models flip at least one gene call after the named MoCo biomass coefficients are removed; the reported binomial interval is 0.8367-0.9757. The total is 511 flipped model-gene calls, not 511 independent genes or wet-lab rescue experiments. The five zero-flip model IDs are retained in the machine-readable census. The positive-control iJO1366 restores the moa/mob cluster, including moaD b0784, under the exact counterfactual described in Section 4.3. See results/bigg\_moco\_growth\_summary.json and per-model results/bigg\_moco\_growth/\*.json.

This is a structural falsifiability result: the biomass objective forces a particular knockout prediction. A flip does not establish that the organism survives the knockout in its real medium. Closely related *E. coli* strain models share reaction templates, so neither a simple binomial confidence interval nor the 63/68 fraction can be read as 68 biologically independent replications. The result supports an audit of biomass-composition assumptions across reconstruction families, not the assertion that every objective is wrong. A separate manual second-family verification was subsequently completed on *Pseudomonas putida* iJN1463 (next paragraph).

Independent-code second-family check: in the downloaded *Pseudomonas putida* KT2440 iJN1463 snapshot, three negative MoCo objective coefficients (bmocogdp\_c, mobd\_c, mocogdp\_c) were removed explicitly. The baseline wild-type optimum was 0.586118/h; the patched optimum was 0.586310/h. A fresh complete gene-deletion run found 262 baseline-essential genes, of which 7 flipped to growth at least 95% of the original wild type. The independently calculated gene IDs are PP\_0735, PP\_1292, PP\_1293, PP\_1294, PP\_2123, PP\_3457, PP\_3830; each was separately zero-growth under its original model. The raw model snapshot SHA-256 was checked against the provider-index ledger before use; results/bigg\_second\_family\_verification.json records coefficients, growth and gene IDs. This meets the narrow second-model-family structural criterion, but *Pseudomonas* viability in a lab was not tested; related model templates may still limit independence.

A post-result family-dependence audit of the same saved census (notes/postresult\_moco\_family\_sensitivity.md, scripts/audit\_postresult\_moco\_family\_dependence.py, results/postresult\_moco\_family\_dependence.json) makes the concentration concrete. Joining every saved per-model outcome to the hashed survey: among the 64 accessions whose stated organism is *Escherichia* or *Shigella*, 61 flip at least one call. Outside those closely related genera the census holds only 4 accessions (*Clostridium* iHN637, *Pseudomonas* iJN1463, *Synechocystis* iJN678, *Salmonella* STM\_v1\_0), of which 2 flip: *Pseudomonas* and *Synechocystis*, while *Clostridium* and *Salmonella* do not. Identical eight-call flip sets recur across *E. coli* strain models with strain-specific gene IDs. The 63/68 headline therefore does not supply 63 independent cross-family confirmations; the cross-family evidence rests on four non-enteric reconstructions and shares model-curation templates. This is an

outcome-aware descriptive re-tabulation, not new phenotypic evidence, and it does not change the failed G2 gate.

## 7.2 Calibration gate: concise audit

The follow-up gate was written down before scoring in notes/prereg\_calibration\_decision\_gate.md. It asked whether the stacked gene-essentiality score crossed two tests on the 1,249 aligned genes: G1 AUROC >0.666 against iJO1366 minimal FBA, and G2 non-significant two-sided McNemar discordance ( $p \geq 0.05$ ) with at least as many model wins as FBA wins. A1 locked a symmetric MCC-based threshold selector on the other two folds for both candidate and comparator, plus an ordered five-rung ladder. The earlier 27:50 discordant loss came from eval-inclusive F1-tuned thresholds, so A1 retains it as a harness reproduction, not as a clean baseline. The gate text stays fixed despite seeing a favorable-direction p-value.

| rung                       | AUROC  | F1     | model-only correct | FBA-only correct | McNemar 2-sided p | literal G2 |
|----------------------------|--------|--------|--------------------|------------------|-------------------|------------|
| R1_v2_lr_trainonly_mcc     | 0.7967 | 0.5148 | 93                 | 67               | 0.04777           | FAIL       |
| R2_v2_lr_isotonic          | 0.7967 | 0.5111 | 98                 | 58               | 0.00170           | FAIL       |
| R3_v2_lr_platt             | 0.7967 | 0.5148 | 93                 | 67               | 0.04777           | FAIL       |
| R4_v2_lr_oversampled_refit | 0.7957 | 0.5050 | 91                 | 52               | 0.00140           | FAIL       |
| R5_v2_lr_gbm_prob_avg      | 0.8015 | 0.4960 | 106                | 56               | 0.00011           | FAIL       |

Table 21. Frozen G2 ladder. The significance clause  $p \geq 0.05$  fails even when the sign favors the ensemble. G1 holds for every rung; no rung passes both clauses.

R1 has 93 model-only correct versus 67 FBA-only correct calls, with exact  $p=0.04777$ . R2 has 98 versus 58,  $p=0.00170$ . Every rung gains more than it loses, but every rung has  $p < 0.05$ , so the literal precommitted G2 FAILS. Changing it to a one-sided loss test after seeing results would misrepresent the preregistration. The observation can motivate a fresh study; it cannot retrospectively repair this one. The discrimination gain is real as a numerical OOF summary, but its inferential claims also need the lineage audit in the next section.

## 7.3 Limit on hard-call inference

The literal G2 gate above failed. A later score-lineage audit found that threshold selection used other-fold OOF scores whose supervised training complements could include the evaluated fold labels; supervised base-layer scores add another nesting concern. Thus the favorable discordant counts are descriptive, not a clean held-out hard-call result. A fully nested refit or untouched cohort is needed. The FBA comparator thresholds yielded the same calls as fixed >0.5 for all 1,249 genes (results/calibration\_lineage\_audit.json). The locked gate and full fold-level audit remain in the repository; neither was revised after its result.

### 7.3a New exploratory nested-threshold follow-up (not a gate repair)

After the lineage flaw was known, we locked a separate follow-up with an outer three-fold split, inner-fold out-of-fold score generation for threshold selection, and fresh outer-training refits (notes/prereg\_nested\_threshold\_followup.md). The feature set excludes the previously reused CNN/GNN/k-mer supervised OOF base scores and removes selected external annotation channels; it

is a restricted replacement model, not a nested rerun of the original stack. Each outer fold chooses its threshold only from inner training scores and labels. The comparator is the pre-specified label-free FBA minimal >0.5 call.

On the same 1,249 Gerdes-overlap genes, this exploratory model obtains outer-fold OOF AUROC 0.7714, AUPRC 0.4590, F1 0.4978 and MCC 0.3891; FBA minimal is AUROC 0.6658, F1 0.4099 and MCC 0.2970 on those rows. There are 85 model-only-correct and 70 FBA-only-correct calls (exact two-sided McNemar  $p=0.2607$ ). Non-significance is not equivalence, and these scores cannot be substituted into the old G2 gate, which remains FAIL on all five rungs. The Gerdes cohort and external annotations are reused after prior results were inspected; this is an exploratory leakage-reduced evaluation, not a new untouched test or proof that all feature provenance is label-free. See results/nested\_threshold\_followup.json and its gene-level OOF CSV.

#### **7.4 Judge-informed research direction: where a virtual cell adds information**

The first recorded ChatGPT judge round was asked to critique the calibration paradox and propose a testable novelty analysis. Its critique identified fold provenance, leakage, comparator-threshold stability and multiplicity; its scientific suggestion was to ask whether the ensemble advantage over FBA concentrates in genes with strong transcriptional regulation but weak direct metabolic constraint. We wrote notes/prereg\_discordance\_regimes.md as an explicit future mechanism test. It requires a versioned E. coli transcription-factor-target map, a metabolic-constraint definition verified against code, matched/negative control partitions and an independent essentiality screen. STRING association degree is not a substitute for TF regulation. Existing Gerdes OOF outcomes cannot be recycled as confirmation after they inspired the hypothesis. This new protocol is a methodological novelty, not an observed biological discovery.

#### **7.5 Independent assay as a different question**

The published Rousset et al. 2018 genome-wide CRISPRi assay supplies a biologically distinct but non-identical essentiality yardstick (<https://doi.org/10.1371/journal.pgen.1007749>). A separate pre-registration in notes/prereg\_rousset\_validation.md fixed median coding-strand depletion  $\leq -5$  as the binary outcome, UniProt gene-name mapping, and a ranking comparison against FBA. Of 4213 source rows, 3867 map to locus IDs and 1214 overlap scored genes, including 70 CRISPRi positives. The saved Gerdes-trained v2 score has AUROC 0.8761 versus FBA 0.6947; paired gene-bootstrap difference +0.1814 [95% CI +0.1070, +0.2611]. The study's locked numerical ranking bar was met, as detailed in results/rousset\_validation.json and scripts/run\_rousset\_validation.py.

This is genuinely different assay labels, not a new organism, gene universe, or controlled independent training environment. CRISPRi knockdown can exert polar effects in bacterial operons, whereas transposon footprinting and FBA knockouts measure different perturbations. The external label assessment does NOT retroactively validate the defective Gerdes hard-call gate: it evaluates ranking, not a newly selected threshold or matched hard-call specificity. Some CRISPRi measurements may overlap the biology or published annotations used in external features; this needs source provenance review before a strong independence claim. Seven positives in the Gerdes-nonessential subset are too

few to support a broad rescue statement despite its attractive AUROC. Report the distinct assay alongside, not in place of, the failed G2 and the MoCo mechanism.

## 7.6 CRISPRi dependence sensitivity

The Rousset comparison uses a distinct assay and a frozen CRISPRi depletion threshold, but gene-name mapping excludes ambiguous or unmapped entries and neighboring operon genes are dependent. A post-result neighborhood-window bootstrap resampled b-number blocks of 5, 10, 20 and 50 IDs (3,000 draws each); the paired AUROC-difference intervals were [+0.102, +0.266], [+0.104, +0.272], [+0.099, +0.281] and [+0.101, +0.272]. These remain positive in this proxy grouping, but are not true operon clustering or a new-organism validation (results/rousset\_genomic\_dependence\_audit.json). The separate Gerdes G2 failure is unaffected.

### 7.6a Source-operon grouped sensitivity (exploratory)

The Rousset S12 source table includes operon labels, allowing a stronger dependence check than arbitrary b-number windows. After the original Rousset score was known, we locked an exploratory protocol and grouped the 1214 mapped genes by source operon string: 714 groups, of which 248 contain multiple genes and 51 contain a CRISPRi positive. The gene-level AUROC difference (v2 minus FBA) is +0.181. Resampling source-operon groups in 3000 fixed-seed draws gives a descriptive 95% interval [+0.097, +0.280]; choosing one gene per group in 3000 draws gives [+0.203, +0.276]. Both remain positive as dependence diagnostics. Shared operon strings do not prove a polar effect in this assay, and these same-accession post-result resamples neither establish an independent cohort nor repair G2. Source mappings and all gene-level scores are retained in results/rousset\_source\_operon\_mapped.csv; the protocol and numeric result are in notes/prereg\_rousset\_operon\_sensitivity.md and results/rousset\_source\_operon\_sensitivity.json.

## 7.7 Biomass controls across objective ingredients

A post-result specificity analysis asked whether the same seven-gene objective-dependent rescue could appear after deleting other biomass ingredients. The exploratory protocol in notes/prereg\_moco\_specificity\_null.md was written before controls were computed but was not committed until afterward; it selected every one of 48 non-MoCo negative objective coefficients within a factor of two of at least one MoCo coefficient in the saved Pseudomonas iJN1463 model. The source JSON SHA-256 was checked again. Starting from 262 original zero-growth gene deletions, we removed each control coefficient one at a time and retested growth at  $\geq 95\%$  original wild type under the same model and medium. The three-term MoCo bundle rescues 7 knockout calls; 4 of 48 single-term controls rescue at least as many. These are adenosylcobalamin (21 calls), biotinyl 5-AMP (8), pyoverdine (14) and thiamine diphosphate (8). Thirty-five controls rescue none. Every ID/count is retained in results/moco\_specificity\_null\_ijN1463.json and the executable script.

This is a meaningful exploratory negative result for uniqueness, not a refutation of the MoCo-specific mechanistic gene IDs. Deleting even one alternative objective constituent can cause an equal or larger number of predicted essentiality flips. But comparison of one control coefficient with a three-ingredient MoCo bundle does not match perturbation size; coefficient proximity does not match

molecule class, network connectivity or model frequency. Consequently we do not quote an exchangeability-based p-value or rank MoCo as a uniquely causal metabolite from this scout. The result redirects the claimed computational discovery from "MoCo uniquely breaks virtual-cell essentiality" to "biomass composition can make distinct gene calls sensitive to multiple small required terms." Multi-term category- and connectivity-matched controls across model families and condition-matched biological knockout evidence would be needed for a stronger claim. The old Gerdes G2 gate remains failed.

## **7.8 Toward a structure-aware biomass-sensitivity map**

We asked whether baseline network features could identify biomass ingredients with fragile essentiality predictions. We built a pre-perturbation, model-hashed feature inventory for all 102 negative-weight objective compounds in iJN1463. Each row records absolute biomass coefficient, number of non-biomass reactions linked to the compound, count of linked gene-associated reactions, availability of exchange uptake and the number/absolute-flux sum of reactions supplying the compound in the unmodified wild-type optimum. The source hash and 102-row inventory are in `results/biomass_topology_manifest.json`; none of these features reads knockout outcomes. This is a new systematic method, not a scored predictor.

The inventory marks 48 compounds already exposed through the single-term control exercise, three MoCo compounds previously perturbed together only, and 51 additional compounds not singly scored in that exercise. Because target results for 48 compounds were already seen, a regression on all 102 within this one model would be exploratory regardless of when code is committed; leave-one-component-out cross-validation cannot manufacture an independent model family. The protocol in `notes/prereg_biomass_topology_sensitivity.md` calls for comparable single-term rescoreing of all 102, then a separately locked, untouched reconstruction-family test of whether prespecified topology features and a shuffled-feature negative control predict flip counts. Baseline reaction degree and flux are proxies, not full pathway redundancy or biochemical causation. This design does not supersede the failed G2 calibration, the Rousset assay caveats, or the need for wet-lab phenotype checks.

## **7.9 All-term single-coefficient sensitivity**

The 102-term feature manifest led to a uniform single-coefficient perturbation audit in iJN1463. The 48 prior controls were preserved as previously seen outcomes; we freshly scored the remaining 54 singles without selecting outcomes or changing the original solver/medium. Across all 102 terms, 23 rescue at least one of the original 262 zero-growth gene deletions and 79 rescue none. Each of the three MoCo compounds removed *\*alone\** rescues zero, although removing all three together rescues seven; this is a set-interaction property of the objective representation, not evidence that any one of them is an individually powerful lever. The largest single-component effect is lipopolysaccharide LPSpput (29 predicted flips), followed by adenosylcobalamin (21) and pyoverdine (14). No knockout here was biologically tested. Full per-term gene lists and checkpoints are in `results/biomass_single_term_sensitivity.json` and `scripts/score_remaining_biomass_terms.py`.

To ask whether simple baseline features explain these heterogeneous effects, we computed a descriptive Spearman association and 2,000 feature-shuffle controls for five features. Biomass coefficient magnitude correlates with flip count at  $\rho$  -0.017 (permutation  $p=0.869$ ); linked non-biomass reaction count  $\rho$  -0.184 ( $p=0.064$ ); gene-associated reaction count  $\rho$  -0.181 ( $p=0.067$ ). Other baseline-production proxies remain in results/biomass\_topology\_exploratory.json. None is a registered positive predictive finding. The 48 already seen targets, five inspected features, correlated network summaries, zero-heavy outcome, and single model make these  $p$ -values descriptive, not a basis for external generalization. We did not fit a held-out topology predictor. This negative redirects the mechanism question toward multi-term objective interactions and independent model families rather than a simple coefficient or degree rule.

### 7.10 MoCo-subset interaction in iJN1463

An exploratory seven-subset factorial counterfactual, locked as notes/prereg\_moco\_subset\_interaction.md before pairwise scores, asks which combinations of the three MoCo biomass coefficients are sufficient to undo original zero-growth knockout calls. We used the hashed iJN1463 source, the same full set of 262 baseline-essential genes, and the same rescue threshold of 95% original wild-type growth for all seven nonempty subsets. No singleton rescued a call. Only the `bmocogdp_c` plus `mocogdp_c` pair rescued any: six genes (PP\_0735, PP\_1292, PP\_1293, PP\_1294, PP\_2123 and PP\_3457). Adding `mobd_c` rescued a seventh, PP\_3830. The other two pairs rescued zero. The trio result and singletons reproduce previously observed outcomes; the pairwise result is a new mechanistic decomposition within this model, not prospective or external validation.

This result narrows the computational interpretation from 'three MoCo terms are individually decisive' to a combination-sensitive dependency: two particular objective demands suffice for six model knockout rescues, while the full three-demand removal is needed for a seventh. It does not prove a physical molecular interaction, because removing biomass coefficients relaxes demands in a mathematical program and the affected genes can share a biosynthetic pathway. No organism was perturbed, the same model supplied every comparison, and the original Gerdes G2 gate remains failed. The exact gene sets, edited wild-type values, source hash, and seven subset outcomes are in results/moco\_subset\_interaction\_ijn1463.json; the reproducible implementation is scripts/score\_moco\_subset\_interaction.py.

Model gene-reaction annotations clarify why the seventh call is distinct. PP\_0735 (`moeB`), PP\_1292-1294 (`moaC/D/E`), PP\_2123 (`moeA`) and PP\_3457 (`mobA`) attach to cofactor synthesis in the source reconstruction; PP\_3830 (`modC`) attaches to molybdate ABC transport. The six synthesis-associated genes recover when both guanine-dinucleotide cofactor demands are relaxed, whereas `modC` only recovers when the remaining molybdate objective demand is also removed. This is a trace through the model's gene-reaction rules, not measured expression, viability, or evidence that a specific cell physiology follows the mathematical combination.

### 7.11 Transfer of a frozen biomass-fragility score to a distinct reconstruction

To check whether the topology features carry any information beyond the *Pseudomonas* model, we froze a five-feature logistic model trained on its 102 biomass constituents (23 positive single-coefficient rescues), then scored all 60 negative biomass coefficients in *Bacillus subtilis* iYO844 before observing their knockout outcomes. The target reconstruction was chosen by organism family and absence of MoCo-named biomass terms, with source SHA-256 checked against the existing BiGG ledger. Every predicted probability was committed before deleting a target coefficient (notes/prereg\_biomass\_transfer\_iyo844.md and results/biomass\_transfer\_iyo844\_frozen\_predictions.json). In the target model, 17 of 60 coefficients rescue at least one original zero-growth knockout after deletion. Frozen-score AUROC is 0.6703; average precision 0.4297 versus positive prevalence 0.2833. A 1,000-shuffle label-relative-to-feature control gives an AUROC one-sided tail fraction 0.0240, but AP tail fraction 0.0829. Brier loss is 0.2396, worse than 0.2064 for a constant training-prevalence forecast. The numerical ranking signal is modest and its probability calibration is poor; it does not meet a general predictor claim.

This is an out-of-model computational test, not an independent phenotypic screen: both reconstructions come from the BiGG modeling ecosystem and differ in medium, species, biomass demands and gene-rule detail. The outcome is a simulation of relaxing the objective rather than a measured essentiality rescue. The score ranks ATP and menaquinol-7 among target-sensitive terms but also rates water and potassium highly despite zero rescue, making a mechanistic or deployment claim premature. All 60 prediction/outcome rows, rescued-gene sets, edited growth values and the frozen training script are retained in results/biomass\_transfer\_iyo844.json and scripts/run\_biomass\_transfer\_iyo844.py. An additional reconstruction-family check is in Section 7.12; condition-matched wet-lab knockout outcomes remain necessary. G2 remains failed.

### 7.12 Second out-of-model transfer: yeast iMM904

After seeing the *Bacillus* result, we froze a second reconstruction-family transfer to the already downloaded BiGG *Saccharomyces cerevisiae* iMM904 model. Its source URL and SHA-256 were checked against the prior census; the same untouched *Pseudomonas*-trained five-feature score was run on all 44 negative biomass objective ingredients, and its predicted probabilities were committed before any target knockout rescoring (notes/prereg\_biomass\_transfer\_yeast\_imm904.md; results/biomass\_transfer\_imm904\_frozen\_predictions.json). 18 of 44 constituents rescue at least one originally zero-growth in-model knockout on single-coefficient removal. The frozen score gives AUROC 0.6004, average precision 0.5508 versus target prevalence 0.4091, and Brier 0.2494 versus 0.2754 for constant *Pseudomonas*-training prevalence. One-thousand-shuffle label-relative-to-feature tail fractions are 0.1299 (AUROC) and 0.1179 (AP), neither a registered success. ATP is a correctly ranked sensitive component (10 rescued calls), but water is again highly ranked and rescues none. The two target reconstructions give inconsistent calibration and only modest discrimination; we do not infer a transferable general predictor from them.

The yeast result is an honest negative/ambiguous computational generalization test, not biological validation. All three models come from BiGG and share curation conventions; biomass reactions, media and organism physiology differ. The per-term target effects are predictions under relaxed

model objectives, not measured survival. A model-source-distinct, condition-matched experimental knockout screen is still needed to test whether objective-fragility forecasts improve real essentiality calls. The exact 44 yeast rows, original positive-call gene lists and reproducible script are in `results/biomass_transfer_imm904.json` and `scripts/run_biomass_transfer_imm904.py`. The failed Gerdes G2 hard-call gate is not changed.

### **7.13 Simple baselines and ubiquitous-component influence**

A post-result diagnostic asks whether the frozen five-feature score adds anything over two one-variable rules: rank by larger absolute biomass coefficient, or by fewer gene-associated reactions linked to the required metabolite. In *Bacillus*, the five-feature score AUROC/AP are 0.670/0.430; coefficient-only 0.547/0.343; inverse-degree-only 0.574/0.329. In yeast those pairs are 0.600/0.551, 0.536/0.473, and 0.619/0.524. Yeast inverse-degree AUROC slightly exceeds the five-feature score; there is no consistent multi-feature dominance, and these simple rules were evaluated only after the target outcomes were seen.

Two metabolites illustrate how a globally shared biomass ingredient can dominate model ranking: ATP scores near the top and rescues 13 *Bacillus* original knockout calls, whereas water is also high-ranked but rescues none. Descriptively removing exactly ATP and water without refitting leaves *Bacillus* AUROC 0.665 and AP 0.382 on 58 terms; yeast AUROC 0.600 and AP 0.563 on 42 terms. Rankings do not collapse but remain modest and calibration is unproven. This outcome-known influence analysis cannot be repackaged as a cleaner prospective test. Full term scores and control details are in `results/biomass_baseline_dominance_audit.json`, with fixed protocol and executable script.

### **7.14 Biological-label contradiction check in *Bacillus***

The BiGG coefficient-removal outcomes are computational, so we tested whether their rescued gene names overlap with a previously obtained biological-label category rather than calling every flip a biological rescue. The local SubtiWiki essential-gene category yielded 260 parsed names (source page and hash in `results/bsub_rescue_subtiwiki_overlap.json`). The page states 251 protein and two RNA essential genes, so the parsed 260 names are not a clean single-study label set. The iYO844 reconstruction itself used essentiality evidence in model development; independence from these labels cannot be claimed. Of 171 originally zero-growth iYO844 model gene knockouts, 170 map to unique gene names and 62 names appear in that list. Across all 17 biomass-term counterfactuals that rescue any model gene, the union has 84 distinct gene IDs, 83 uniquely named and 28 listed on SubtiWiki. This is a descriptive contradiction warning for any claim that a relaxed objective proves viability, not a condition-matched falsification of the model: deletion-library medium and strain may differ from the in-silico stored medium.

Two pathways expose the issue concretely. Removing the menaquinol-7 objective coefficient rescues 16 in-model knockout calls, yet 11 of those gene names remain on the SubtiWiki list, including *menA/B/C/D/E* and isoprenoid-precursor genes. Removing the peptidoglycan objective term rescues 9, of which 8 listed names include *murB/C/D/E/F/G*, *mraY* and *ddl*. A name absent from the list is not

verified viable under matched conditions; mapping is not enough to infer phenotype. Full gene/term name matches, one unnamed original gene, source URL and both file hashes are preserved in `results/bsub_rescue_subtiwiki_overlap.json` and `scripts/audit_bsub_rescues_subtiwiki.py`. This post-result cross-list sanity check narrows the computational conclusion to objective-dependence, not confirmed biological rescue. It does not alter G2.

### **7.15 Retrospective equal-coefficient pair controls**

The MoCo subset analysis found six knockout rescues when `bmocogdp_c` and `mocogdp_c` were removed together. Because the earlier negative control compared a three-term MoCo bundle with other single-term deletions, we retrospectively enumerated all 253 unordered pairs among 23 non-MoCo objective ingredients with the same negative biomass coefficient (-0.000223) as each MoCo pair member. The source snapshot and medium are unchanged and its SHA-256 is checked before scoring. All original 262 zero-growth gene deletions were tested after each two-coefficient deletion, with rescue fixed at at least 95% of original, unedited wild-type growth. A fresh MoCo-pair rerun reproduced the six gene IDs and edited wild-type growth from the previous seven-subset audit. Exact candidate IDs, pair outcomes and rescued gene IDs are stored in `results/postresult_equal_coefficient_pair_null.json`; `notes/postresult_equal_coefficient_pair_null.md` fixes the protocol, which was locked before these pair outcomes but after the MoCo outcome was already known.

Of 253 non-MoCo pairs, 104 rescue at least six original knockout calls and 88 exceed six; 78 rescue none. The largest pair, adenosylcobalamin plus pyoverdine, rescues 35. Thus the two-term MoCo perturbation is not exceptional by aggregate rescue count when coefficient size and number of deleted terms are matched. None of the six original MoCo-pair rescued gene IDs is rescued by any of these 253 non-MoCo pairs; the exact gene-set pattern is distinctive within this finite enumeration, although all six may share one cofactor pathway. No LP solver failures were observed.

This comparison is an outcome-aware sensitivity analysis, not an exchangeable randomization experiment: equal coefficient size and pair cardinality do not match biochemical class, network topology, biomass-coupling constraints or uptake. Its descriptive rank is not a p-value. The result supports a narrower model-level claim: several biomass demands cause equal or larger total hard-call changes, while the MoCo-pair perturbation selects this particular six-gene pattern in the source reconstruction. It neither shows that any cell survives a knockout nor repairs the failed Gerdes G2 gate. Category-matched multi-model controls and condition-matched phenotype evidence remain necessary.

### **7.16 Pair interactions reveal an objective-stoichiometry trap**

A retrospective gene-set reconciliation checked whether each of the 253 scored non-MoCo pairs rescued exactly the union of the knockout calls rescued by its two component deletions. 250 pairs matched that simple union; 2 had new pair-only genes and 1 lost calls present in a single-term outcome. This was computed only after all pair and single outcomes were visible, and does not create another independent experiment. Among the exceptions, deleting both heme O and protoheme

objective coefficients rescued three genes (PP\_0189, PP\_0744, PP\_5074) where either singleton rescued none; deleting both adenosylcobalamin and FAD rescued PP\_0602 in addition to the adenosylcobalamin-only set of 21. The original two-term MoCo pair, whose singletons rescue zero, has six pair-only calls. The exact sets, including 250 ordinary pairs, are retained in results/postresult\_pair\_interactions.json.

The opposite exception matters even more for interpretation: deleting the [2Fe-2S] cluster coefficient alongside biotinyl 5-AMP eliminates the eight rescue calls seen for biotinyl 5-AMP alone. A fresh model reload for each unmodified, singleton and pair variant reproduced every candidate growth status as optimal and all recorded gene sets. The double-edited wild-type optimum fell from 0.586118 to 0.390006, whereas the biotinyl 5-AMP-only optimum was 0.586229. Thus editing two terms is NOT guaranteed to behave as merely relaxing two independent biomass requirements; the modified stoichiometric objective can change the LP feasible-growth balance in either direction. These are in-model interaction patterns, not measured molecular epistasis. The fresh statuses, gene growth values and source hash are in results/postresult\_pair\_interactions\_fresh\_check.json. No new hard-call gate or biological viability claim follows.

### 7.16a Which pair interactions break the single-term union rule?

A useful operational null for biomass editing is set union: if removing each ingredient separately rescues a set of gene deletions, would removing both rescue exactly the union? This is not a theorem of linear programming because changing a biomass coefficient changes the objective stoichiometry as well as the attainable optimum. The 253 scored non-MoCo two-term edits give 250 exact gene-set unions and three exceptions. The exceptions are not random samples and were identified after the scores were seen; they are finite computational counterexamples to a tempting shortcut, not population estimates of how often biochemical interactions occur.

| removed pair      | single A | single B | union | pair | new-only | lost-union |
|-------------------|----------|----------|-------|------|----------|------------|
| 2fe2s_c + btamp_c | 0        | 8        | 8     | 0    | 0        | 8          |
| adocbl_c + fad_c  | 21       | 0        | 21    | 22   | 1        | 0          |
| hemeO_c + pheme_c | 0        | 0        | 0     | 3    | 3        | 0          |

Table 22. All three post-result deviations from single-term gene-set union among 253 enumerated non-MoCo pairs; gene IDs are in text and results/postresult\_pair\_interactions.json.

The heme O plus protoheme edit is a pure pair-only case: neither singleton rescues a baseline-zero knockout, while the pair rescues PP\_0189, PP\_0744 and PP\_5074. Adenosylcobalamin alone rescues 21 calls; adding FAD brings PP\_0602 beyond that set. The converse case is [2Fe-2S] plus biotinyl 5-AMP: the biotinyl 5-AMP singleton rescues eight gene calls, but the pair rescues zero. A fresh reload of the original and edited model for these three pairs reproduced all saved solver statuses and knockout gene sets. In the suppressive case the double-edited wild-type optimum is about 0.390006/h versus 0.586229/h for the biotinyl 5-AMP singleton. The change in optimum is a direct reason not to interpret every coefficient deletion as an independent relaxation of a fixed growth program.

This result changes the design of a transferable sensitivity audit. A simple rank of individually fragile terms would miss the heme/protoheme rescue and the six-gene MoCo pair. Conversely, summing

singleton rescue counts would falsely predict eight rescues in the [2Fe-2S]/biotinyl 5-AMP pair. Reporting the exact affected gene sets, the original and edited wild-type optima, solver status, and source hash is necessary to tell a network constraint from an objective-stoichiometry interaction. It does not show molecular epistasis in cells: all edits and knockouts are in silico, the three exceptions were selected after enumeration, and matched-media phenotype data remain missing.

### **7.17 Scope and next validation**

The original v1 report gave 0.7225 AUROC and a 27:50 hard-call loss, which is accurate for its original eval-inclusive threshold exercise but not the end of the project. The later MoCo census adds cross-reconstruction structural scope, and the v2 OOF AUROC is approximately 0.7967, yet a repeated-label score-lineage audit limits the calibration gate. Therefore the defensible headline today is: objective-sensitivity and score-lineage audits expose model-dependent calls and evaluation limits; the later hard-call improvement is suggestive, not a clean held-out win. An independent phenotypic screen and condition-matched carbon-source evaluation remain needed for the strongest claim.

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## Appendix A. Reproducibility ledger

Every number in this paper regenerates from the repository. Data snapshots: BiGG e\_coli\_core.json and iJO1366.json (fetched 2026-09-24), U00096.3 FASTA + GenBank (NCBI efetch, 2026-09-24), Gerdes 2003 Table S1 (genome.wisc.edu, 2026-09-24). Pipelines: scripts/run\_ijo1366\_scan.py (deletion screens), scripts/run\_seqcnn.py (CNN + baseline), scripts/run\_gnn.py (graph GCN), scripts/make\_figures.py (all figures), paper/build\_paper.py (this document). Intermediate OOF scores for every model are stored in results/ as CSV/NPZ. Tests: 18 hermetic pytest cases, ~5 s, no network. Live network calls are made only by fetch scripts, never by tests.

| Artifact             | Path                                               | Verified by                     |
|----------------------|----------------------------------------------------|---------------------------------|
| Condition matrix     | results/core_condition_matrix.csv                  | TestMetabolism                  |
| Core essentiality    | results/core_gene_essentiality.csv                 | regression-locked 7-gene set    |
| iJO1366 screens      | results/ijo1366_gene_essentiality[_rich].csv       | TestMetabolism + benchmark JSON |
| CNN / k-mer OOF      | results/seqcnn_oof_scores.npy, kmer_oof_scores.npy | TestSeqCNN                      |
| GNN OOF              | results/gnn_oof_scores.npy                         | TestGraphGNN                    |
| Ensemble OOF + stats | results/ensemble_results.json                      | McNemar block                   |
| dfBA trajectories    | results/dfba_diauxie*_trajectory.csv               | TestDynamics                    |
| MoCo rescue          | results/moco_biomass_rescue_test.json              | Section 4.3                     |

*Table A1. Result-to-test traceability.*

## Appendix B. Notation and hyperparameters

| Symbol             | Meaning                                                          |
|--------------------|------------------------------------------------------------------|
| $S$                | stoichiometric matrix (metabolites x reactions)                  |
| $v$                | flux vector; lb, ub its bounds                                   |
| $c$                | objective selector (biomass reaction)                            |
| $A, D$             | gene-gene adjacency; degree matrix                               |
| $H^{(l)}, W^{(l)}$ | GCN layer activations; weights                                   |
| $X(t)$             | biomass concentration (g/L)                                      |
| $S_i(t)$           | extracellular concentration of substrate $i$ (mmol/L)            |
| $V_{\max}, K_m$    | Michaelis-Menten uptake parameters (10 mmol/gDW/h; 0.015 mmol/L) |
| $\mu(t)$           | specific growth rate (h <sup>-1</sup> )                          |
| $dt$               | dFBA integration step (0.1 h; 0.2 h in robustness test)          |
| $b, c$             | McNemar discordant-pair counts                                   |

Table B1. Notation.

| Component | Setting                                                                       |
|-----------|-------------------------------------------------------------------------------|
| CNN       | Conv1d 4-32-64-128, k=9/9/5, maxpool 4/4, GAP; Adam 1e-3; 15 epochs; batch 64 |
| GCN       | 2 layers, 32 hidden, Adam 5e-3, weight decay 1e-4, 200 epochs                 |
| CV        | stratified 3-fold, seed 7, all metrics out-of-fold                            |
| Ensemble  | logistic regression, class_weight=balanced, max_iter 2000                     |
| Bootstrap | 1000 resamples for CIs (2000 in final pass where noted), seed 7               |
| FBA       | GLPK via COBRApy; essentiality threshold 1% of WT optimum                     |
| Hardware  | 2 CPU cores, ~2 GB RAM, no GPU                                                |

Table B2. Hyperparameters and compute.

**Data and code availability.** All models (BiGG), the genome (NCBI U00096.3) and the essentiality ground truth (Gerdes 2003 supplement, genome.wisc.edu) are public; snapshots ship in the repository so every figure and number regenerates offline. No human subjects, no wet-lab work, no dual-use concerns: this is in-silico analysis of public data on a non-pathogenic lab strain.

## Appendix C. Mathematical derivations and proofs

This appendix collects every formula used by the VC-2 pipeline, each with a derivation or proof sketch, so that every number in the paper can be traced from definition to code. Equations are numbered (C1)-(C24). Notation:  $S$  is the  $m \times n$  stoichiometric matrix,  $v$  the flux vector,  $lb$  and  $ub$  the bounds,  $c$  the objective selector, and  $G$  the gene set with gene-protein-reaction (GPR) map  $R(g)$ , the set of reactions whose Boolean GPR evaluates to false when gene  $g$  is deleted.

### C.1 Monotonicity of the knockout optimum

Let  $F = \{v : Sv = 0, lb \leq v \leq ub\}$  and  $z^* = \max \{c^T v : v \in F\}$ . Deleting gene  $g$  imposes  $v_j = 0$  for every  $j$  in  $R(g)$ , producing  $F_g = F \cap \{v_j = 0, j \in R(g)\}$ .

$$z_g^* = \max \{c^T v : v \in F_g\} \leq z^* \quad (C1)$$

Proof.  $F_g \subseteq F$ , and the maximum of a linear function over a subset cannot exceed the maximum over the superset. Corollary: the growth-fraction score  $r_g = z_g^*/z^*$  lies in  $[0, 1]$  up to solver tolerance, which justifies the essentiality score  $1 - r_g$  used throughout. Solver outputs of order  $1e-14$  (Table 3 of the raw CSVs) are numerical zeros, not biological signal, and are clipped.

### C.2 Double deletions are submodular-free

For genes  $g$  and  $h$ ,  $F_{gh} \subseteq F_g \cap F_h$ , hence

$$z_{gh}^* \leq \min(z_g^*, z_h^*) \quad (C2)$$

but no lower bound in terms of single deletions exists: synthetic lethality ( $z_g^* = z_h^* = z^*$ ,  $z_{gh}^* = 0$ ) is exactly the case of two isozymes or parallel routes. This is why single-deletion FBA cannot see redundancy loss, and why isozyme GPR errors (Bernstein et al., 2023) dominate FBA false negatives.

### C.3 Producibility of a biomass constituent

For metabolite  $i$ , add a demand column  $d_i = -e_i$  to  $S$  and solve

$$\pi_i(F_g) = \max \{u : Sv + d_i u = 0, v \text{ in bounds}, v_j = 0 \text{ for } j \in R(g), u \geq 0\} \quad (C3)$$

Metabolite  $i$  is producible in the knockout iff  $\pi_i > \varepsilon$  (we use  $\varepsilon = 1e-6$ ). The unproducible set is  $U_g = \{i \text{ in } B : \pi_i(F_g) \leq \varepsilon\}$ , where  $B$  is the set of consumed biomass metabolites.

### C.4 The biomass rescue test and its soundness

Write the biomass column as  $b = -\sum_{i \text{ in } B} \beta_i e_i + p$  ( $p$  collects products such as ADP, Pi, H+). The rescued column strips exactly  $U_g$ :

$$b^{(g)} = b + \sum_{i \text{ in } U_g} \beta_i e_i \quad (C4)$$

$$\text{verdict}(g) = \text{biomass\_forced} \text{ iff } z_g^*(b^{(g)}) \geq \tau z^*, \text{ else } \text{network\_forced} \quad (C5)$$

Proposition 1 (soundness of `biomass_forced`). If `verdict(g) = biomass_forced`, then the lethality of  $g$  in the original model is entirely attributable to the requirement for  $U_g$ : the knockout network can carry biomass flux at  $\geq \tau$  of wild type for every other constituent simultaneously. Proof: the rescued LP is

feasible with value  $\geq \tau z^*$ , and its optimal  $v$  is a flux distribution in  $F_g$  that produces each  $i$  in  $B \setminus U_g$  at rate  $\beta_i z^*(b^{(g)})$ . Since the original knockout optimum is zero while the only difference between the two LPs is the  $U_g$  coefficients,  $U_g$  is a sufficient cause of lethality.

Proposition 2 (network\_forced is conservative). If  $\text{verdict}(g) = \text{network\_forced}$ , then even an objective that demands none of the unproducible compounds cannot be met; the lesion is in shared precursor or energy supply. Proof: by (C1) applied to the rescued objective. Note that network\_forced does not claim the gene is truly essential in vivo - only that no biomass relabelling rescues it.

Proposition 3 (minimality).  $U_g$  is minimal among sets defined by single-compound producibility: removing any  $i$  in  $U_g$  from the strip set leaves a compound with  $\pi_i = 0$  in the objective, and a biomass reaction consuming an unproducible compound at positive coefficient forces  $z^* = 0$ . Hence every element of  $U_g$  is necessary for rescue.

Falsifiable prediction. If biomass\_forced labels reflect model artifacts rather than biology, genes so labelled should be enriched for experimentally non-essential genes relative to network\_forced genes. This is tested with the one-sided Fisher exact test (C18) in Section 4.4.

### C.5 AUROC as a Mann-Whitney statistic

For scores  $s$ , positives  $P$  (essential,  $n_1$ ) and negatives  $N$  ( $n_0$ ):

$$AUROC = (1/(n_1 n_0)) \sum_{i \in P} \sum_{j \in N} [I(s_i > s_j) + 1/2 I(s_i = s_j)] \quad (C6)$$

Derivation: the ROC curve is (FPR(t), TPR(t)) as  $t$  sweeps; integrating TPR dFPR over the step function gives, for each negative  $j$ , the fraction of positives scoring above it, which summed over  $j$  and normalized is (C6). Ties receive half credit because a tied pair lies on a diagonal ROC segment. Consequence for FBA: binary FBA scores have massive ties, so AUROC collapses to (TPR + TNR)/2 at the single threshold, which explains why pure FBA AUROCs cluster near 0.63-0.67.

$$AUROC_{binary} = 1/2(TPR + 1 - FPR) \quad (C7)$$

### C.6 Average precision

$$AP = \sum_k (R_k - R_{k-1}) P_k \quad (C8)$$

with  $P_k$ ,  $R_k$  precision and recall at the  $k$ -th threshold. The baseline for a random ranker equals the prevalence  $\pi$  (0.172 here), so AUPRC must always be read against  $\pi$ . The lift  $AP/\pi$  is reported alongside.

### C.7 Matthews correlation coefficient

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))} \quad (C9)$$

MCC equals the Pearson correlation of the two binary indicator vectors (proof: expand the covariance of Bernoulli variables), so it is symmetric in classes and insensitive to prevalence in the sense that a constant predictor scores 0.

### C.8 Percentile bootstrap confidence intervals

For statistic T and B resamples of genes with replacement,  $T_b^* = T(X_b^*)$ . The 95% interval is

$$CI_{95} = [Q_{0.025}(T^*), Q_{0.975}(T^*)] \quad (C10)$$

Validity rests on the bootstrap distribution of  $T^* - T$  approximating that of  $T - \theta$ ; for U-statistics such as AUROC this holds asymptotically (Hall, 1992). Resamples with a single class are discarded (the code reports `n_boot_used`). Genes are the resampling unit, so the interval reflects gene-set uncertainty, not experimental replicate noise.

### C.9 Paired comparison of two classifiers: McNemar

On paired hard calls, let b = number of genes model A gets right and B wrong, c the reverse. Under  $H_0$  (equal error rates) each discordant gene is a fair coin,  $b \sim \text{Bin}(b+c, 1/2)$ :

$$p = 2 \sum_{k=0}^{\min(b,c)} C(b+c, k) 2^{-(b+c)} \quad (C11)$$

$$\chi^2 = (|b - c| - 1)^2 / (b + c) \quad (C12)$$

(C12) is the continuity-corrected normal approximation, derived from the variance  $(b+c)/4$  of  $\text{Bin}(b+c, 1/2)$ . With  $b = 27$ ,  $c = 50$  the exact test gives  $p = 0.012$  against the ensemble at the 0.5 cut, which is why the paper reports the ranking gain and the hard-call loss together.

### C.10 DeLong-free AUROC difference by paired bootstrap

$$\Delta = AUROC_A - AUROC_{B'} \quad CI \text{ from paired resamples (same gene indices for A and B)} \quad (C13)$$

Pairing preserves the correlation between models scored on identical genes and is strictly tighter than comparing two independent intervals.

### C.11 Stacked generalization without leakage

Base scores are produced out-of-fold: for fold k with held-out set  $I_k$ , model  $f^{(-k)}$  is trained on the complement and scores  $I_k$ . The meta-learner is

$$h(x) = \sigma(w_0 + \sum_m w_m f_m^{(-k(x))}(x)) \quad (C14)$$

Because each base prediction for gene x was made without seeing x, the meta-features are distributed as test-time features, and the meta-learner's own OOF evaluation is unbiased up to fold-to-fold variance (Wolpert, 1992). Non-learned FBA scores need no folding.

### C.12 Logistic loss and its gradient

$$L(w) = -\sum_i [y_i \log p_i + (1-y_i) \log(1-p_i)] + \lambda \|w\|^2/2, \quad \square L = X^T(p - y) + \lambda w \quad (C15)$$

Derivation:  $d/dz \log \sigma(z) = 1 - \sigma(z)$ . The Hessian  $X^T \text{diag}(p(1-p))X + \lambda I$  is positive definite, so the meta-learner has a unique optimum.

### C.13 Class-weighted loss for the CNN and GCN

$$L_w = -\sum_i [\omega_1 y_i \log p_i + \omega_0 (1-y_i) \log(1-p_i)], \quad \omega_c = n/(2 n_c) \quad (C16)$$

With  $\omega_c = n/(2n_c)$  each class contributes equal total weight, which is the minimizer of expected balanced error for a calibrated model.

#### C.14 Receptive field of the sequence CNN

$$RF_L = 1 + \sum_{l=1}^L (k_l - 1) \prod_{j<l} s_j \quad (C17)$$

For kernel sizes  $k_l$  and strides  $s_j$  (pooling counted as stride). The receptive field bounds the longest motif the network can represent before global pooling; global max pooling then makes the gene score invariant to motif position.

#### C.15 Fisher exact test for provenance enrichment

Let the 2 x 2 table be [[a, b], [c, d]] with rows biomass\_forced / network\_forced and columns non-essential / essential in vivo. Conditioning on margins, a follows the hypergeometric law

$$P(A = a) = C(a+b, a) C(c+d, c) / C(n, a+c) \quad (C18)$$

$$P_{one-sided} = \sum_{k \geq a} P(A = k), \quad OR = ad / bc \quad (C19)$$

The one-sided alternative encodes the prediction of C.4: biomass\_forced calls are more often wrong (non-essential in vivo) than network\_forced calls.

#### C.16 Graph convolution as normalized propagation

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}), \quad \tilde{A} = A + I \quad (C20)$$

The symmetric normalization has spectrum in  $(-1, 1]$ , so repeated propagation is contractive and does not explode; two layers aggregate the 2-hop gene neighbourhood via shared metabolites.

#### C.17 Dynamic FBA

$$dX/dt = \mu(t) X, \quad dS_k/dt = v_k(t) X, \quad \mu(t) = \max c^T v \text{ s.t. } Sv = 0, v_k \geq -V_{max,k} S_k / (K_{m,k} + S_k) \quad (C21)$$

$$X_{t+\Delta t} = X_t e^{\mu \Delta t}, \quad S_{t+\Delta t} = S_t + v (X_t / \mu) (e^{\mu \Delta t} - 1) \quad (C22)$$

(C22) is the exact solution of (C21) with fluxes frozen over the step (the static-optimization approach of Mahadevan et al., 2002); it avoids the negative-concentration artifacts of forward Euler. The Boolean catabolite-repression overlay sets  $V_{max,lac} = 0$  while  $S_{glc} > \theta$ , producing diauxie.

#### C.18 Threshold sensitivity and bimodality

$$E(\tau) = I(r_g < \tau), \quad \partial |E(\tau)| / \partial \tau = \sum_g \delta(r_g - \tau) \quad (C23)$$

Because FBA growth fractions are almost all exactly 0 or 1, the density of  $r_g$  in  $(0.001, 0.5)$  is nearly empty and  $|E(\tau)|$  is flat across two orders of magnitude of  $\tau$  - the robustness result of Section 3.

#### C.19 Lift and expected calibration

$$lift = AP / \pi, \quad ECE = \sum_b (|B_b|/n) |acc(B_b) - conf(B_b)| \quad (C24)$$

Lift normalizes AUPRC for prevalence, which differs between Gerdes (17.2%) and the PEC subset, making cross-yardstick comparisons meaningful.

## Appendix D. Extended evaluation: distributions, calibration and early precision

AUROC summarizes the whole ranking, but a biologist using a virtual cell reads it from the top: which genes should be tested first? This section reports the views of the same out-of-fold scores that answer that question. Every panel is generated by `scripts/make_figures_ext.py` directly from `results/aligned_predictions_all_models.csv` and `results/ensemble_oof.csv`; no number is transcribed by hand.

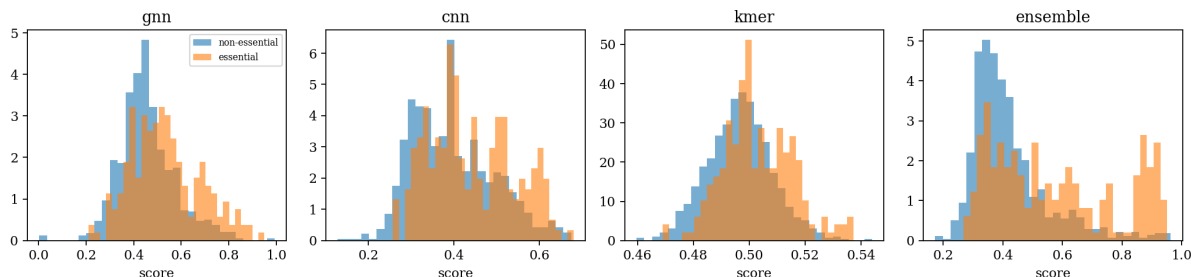


Figure 6. Out-of-fold score distributions by Gerdes class for the GCN, sequence CNN,  $k$ -mer baseline and stacked ensemble. Only the ensemble develops a distinct high-score essential mode (scores > 0.8).

Figure 6 shows why the ensemble beats every member on ranking. Each learned base model shifts the essential class to the right but leaves the two classes overlapping across the full score range. The ensemble, which also receives the binary FBA calls, splits the essential genes into two groups: a sharp mode near 0.85-0.95 made of genes that FBA calls essential AND the learned models score high, and a broad low mode of essential genes that FBA misses. The second group is the ceiling of this architecture - no amount of stacking recovers genes that none of the inputs can see.

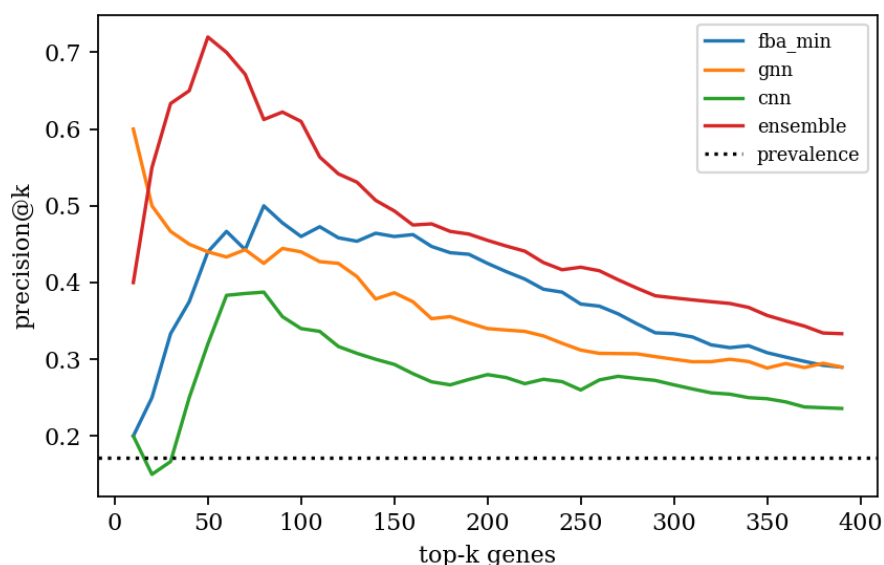


Figure 10. Precision among the top- $k$  ranked genes. Dotted line: prevalence (0.172).

| model    | P@25 | P@50 | P@100 | P@200 |
|----------|------|------|-------|-------|
| fba_min  | 0.32 | 0.44 | 0.46  | 0.42  |
| gnn      | 0.52 | 0.44 | 0.44  | 0.34  |
| cnn      | 0.12 | 0.32 | 0.34  | 0.28  |
| kmer     | 0.44 | 0.34 | 0.38  | 0.33  |
| ensemble | 0.60 | 0.72 | 0.61  | 0.46  |

Table 8. Precision at  $k$  (fraction of the top- $k$  genes that are essential in Gerdes 2003).

At  $k = 50$  the ensemble reaches precision 0.72, 4.2x the prevalence and well above the best single model (FBA minimal 0.44, GCN 0.44). The advantage shrinks with  $k$ : by  $k = 200$  the ensemble (0.46) is only marginally ahead of FBA (0.42). The practical reading is that the ensemble is a good triage tool for the first 50-100 experiments and not a replacement for FBA deeper in the list. The early dip in the ensemble curve at  $k = 10$  is real and explained in Section 4: the very top of the ranking is occupied by molybdopterin genes (*moaD*, *moaE*, *moaA*, *moaC*) that every component scores as essential and that Gerdes finds dispensable - the biomass-objective artifact analysed in Section 4.3.

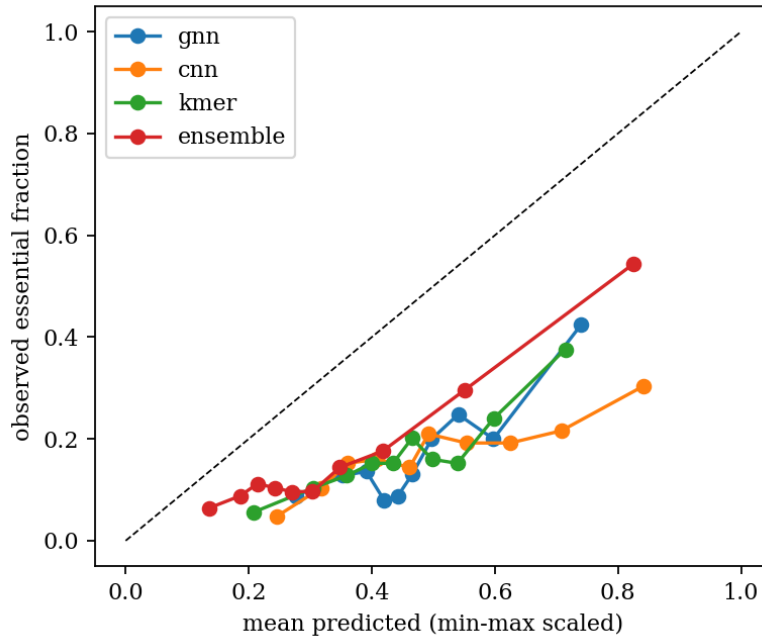


Figure 7. Reliability diagram (quantile bins; scores min-max scaled per model).

Figure 7 shows that none of the scores is calibrated as a probability: all curves lie below the diagonal, meaning that scaled scores overstate essentiality. This is expected for class-weighted training (Appendix C, eq. C16), which trades calibration for balanced error. The ensemble curve is the closest to monotone and reaches an observed essential fraction of about 0.55 in its top decile. For downstream use as probabilities, a Platt or isotonic recalibration on a held-out split is required; we do not claim calibrated probabilities anywhere in this paper.

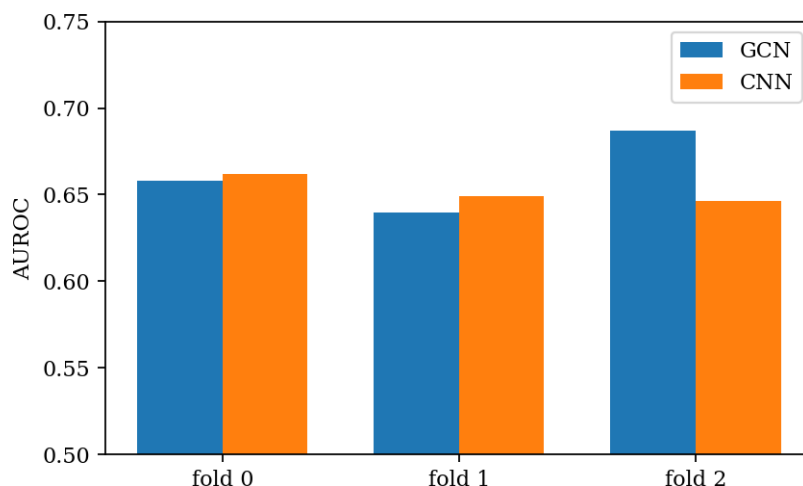


Figure 8. Per-fold AUROC for the GCN and the sequence CNN (3-fold stratified CV).

Fold-to-fold spread (Figure 8) is about 0.05 AUROC for the GCN and 0.02 for the CNN. The GCN spread is comparable to the gap between the GCN and FBA, so the ordering GCN > FBA-rich in Table 1 should be read as a tie within fold noise. The ensemble gain (+0.057 over FBA minimal) is larger than both spreads in this historical comparison, but rescaling and the later score-lineage audit limit causal and hard-call inference.

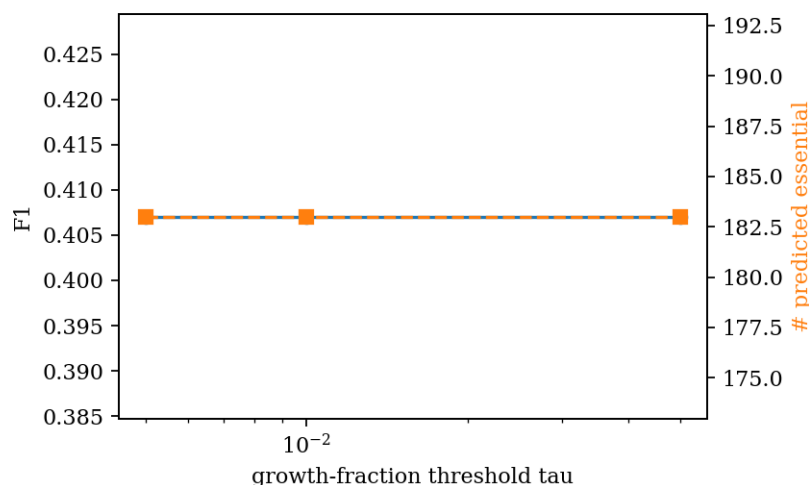


Figure 9. F1 and number of predicted essential genes versus the growth-fraction threshold.

Figure 9 is the graphical form of the bimodality result (eq. C23): both curves are flat, since exactly 183 genes are called essential at 0.5%, 1% and 5% of wild-type growth.

Table 9 lists the 40 highest-ranked genes so that any reader can check the ranking against EcoCyc or their own data. False positives cluster in cofactor biosynthesis (molybdopterin, folate), which motivated the provenance audit of Section 4.4.

| rank | b-number | gene | ensemble | FBA min | GCN  | Gerdes   |
|------|----------|------|----------|---------|------|----------|
| 1    | b0784    | moaD | 0.963    | E       | 0.81 | non-ess. |

| rank | b-number | gene  | ensemble | FBA min | GCN  | Gerdess   |
|------|----------|-------|----------|---------|------|-----------|
| 2    | b0785    | moaE  | 0.958    | E       | 0.76 | non-ess.  |
| 3    | b2746    | ispF  | 0.952    | E       | 0.72 | essential |
| 4    | b0142    | folK  | 0.948    | E       | 0.81 | essential |
| 5    | b0009    | mog   | 0.944    | E       | 0.71 | essential |
| 6    | b3857    | mobA  | 0.944    | E       | 0.82 | non-ess.  |
| 7    | b3018    | plsC  | 0.943    | E       | 0.80 | essential |
| 8    | b0783    | moaC  | 0.939    | E       | 0.75 | non-ess.  |
| 9    | b3058    | folB  | 0.938    | E       | 0.72 | non-ess.  |
| 10   | b3177    | folP  | 0.934    | E       | 0.79 | non-ess.  |
| 11   | b3634    | coaD  | 0.930    | E       | 0.67 | essential |
| 12   | b2747    | ispD  | 0.928    | E       | 0.76 | essential |
| 13   | b3807    | cyaY  | 0.927    | E       | 0.71 | non-ess.  |
| 14   | b0415    | ribH  | 0.925    | E       | 0.71 | essential |
| 15   | b0029    | ispH* | 0.925    | E       | 0.69 | essential |
| 16   | b0159    | mtn   | 0.922    | E       | 0.48 | essential |
| 17   | b0781    | moaA  | 0.921    | E       | 0.70 | non-ess.  |
| 18   | b3041    | ribB  | 0.920    | E       | 0.49 | essential |
| 19   | b0369    | hemB  | 0.919    | E       | 0.58 | essential |
| 20   | b1096    | pabC  | 0.919    | E       | 0.59 | non-ess.  |
| 21   | b1208    | ispE  | 0.918    | E       | 0.71 | essential |
| 22   | b0826    | moeB  | 0.917    | E       | 0.67 | non-ess.  |
| 23   | b1277    | ribA  | 0.917    | E       | 0.44 | essential |
| 24   | b0421    | ispA  | 0.915    | E       | 0.69 | essential |
| 25   | b0174    | uppS  | 0.913    | E       | 0.54 | essential |
| 26   | b3648    | gmk   | 0.909    | E       | 0.61 | essential |
| 27   | b3639    | dfp   | 0.908    | E       | 0.60 | essential |
| 28   | b3040    | zupT  | 0.907    | E       | 0.43 | non-ess.  |
| 29   | b1098    | tmk   | 0.905    | E       | 0.65 | essential |
| 30   | b0103    | coaE  | 0.904    | E       | 0.70 | essential |
| 31   | b1662    | ribE  | 0.903    | E       | 0.66 | essential |
| 32   | b0154    | hemL  | 0.902    | E       | 0.54 | essential |
| 33   | b2515    | ispG* | 0.901    | E       | 0.73 | essential |
| 34   | b1415    | aldA  | 0.894    | E       | 0.48 | non-ess.  |
| 35   | b0417    | thiL  | 0.894    | E       | 0.62 | essential |
| 36   | b4039    | ubiC  | 0.893    | E       | 0.48 | non-ess.  |
| 37   | b0475    | hemH  | 0.893    | E       | 0.83 | essential |
| 38   | b0087    | mraY  | 0.890    | E       | 0.56 | non-ess.  |
| 39   | b2530    | iscS  | 0.890    | E       | 0.53 | essential |
| 40   | b2585    | pssA  | 0.889    | E       | 0.75 | essential |

*Table 9. Top 40 genes by out-of-fold ensemble score with the Gerdes 2003 call.*

## Appendix E. Essentiality-provenance catalog for iJO1366

### E.1 Question and design

Section 4.3 showed one case - the molybdopterin cluster - where an FBA essentiality call is produced by the biomass objective rather than by the network. This appendix asks the general question: for every one of the 208 genes that iJO1366 calls essential on glucose minimal medium, is the call network\_forced or biomass\_forced? The rescue test (Appendix C.4, eqs. C3-C5) answers it per gene with one re-runnable LP experiment. The tool is shipped as `python -m vcell audit-biomass MODEL` and runs on any COBRA model file.

Prior work. The phenomenon of cofactor-driven false essentiality is known. Bernstein et al. (2023, Mol Syst Biol) found 21 vitamin/cofactor biosynthesis genes in iML1515 (biotin, R-pantothenate, thiamin, tetrahydrofolate, NAD+) that the model calls essential and RB-TnSeq finds fit, and fixed them by manually supplying those cofactors. MEMOTE tests whether biomass precursors are producible in the wild-type model but does not attribute any knockout's lethality to specific compounds. Our contribution is narrower and procedural: an automated, per-gene, falsifiable attribution. We found no existing tool that does this; that is a statement about our search, not a proof of absence.

### E.2 Verdict census

| verdict        | genes |
|----------------|-------|
| biomass_forced | 208   |

Table 10. Rescue-test verdicts.

Negative result 1: the binary verdict does not discriminate. All 208 iJO1366 essential genes are biomass\_forced. Every lethal knockout on glucose minimal medium makes at least one biomass constituent unproducible, and removing those constituents always restores some growth. Proposition 1 is true but uninformative here: the set  $U_g$  ranges from 1 to 50 compounds (median 2), and when  $U_g$  is large the rescued objective is a different, much easier objective. The rescue test therefore needs a second axis before it says anything about artifacts.

### E.3 A methods catch: raw biomass\_forced over-calls artifacts

A raw biomass\_forced label means only that stripping the unproducible compounds rescues growth. For a gene such as *fabD* that stops lipid synthesis this is trivially true and says nothing about artifacts: lipids are really required. The artifact hypothesis only makes sense for compounds a mutant might plausibly obtain without making them (trace cofactors and vitamins by carry-over or cross-feeding, the class named by Bernstein et al.). We therefore split biomass constituents into cofactor/vitamin, inorganic and bulk classes (`vcell/compound_classes.py`) and label a gene *cofactor\_only* when every unproducible compound is a cofactor or vitamin.

| stratum                      | genes | Gerdes essential | fraction |
|------------------------------|-------|------------------|----------|
| biomass_forced:bulk_involved | 104   | 39               | 0.38     |
| biomass_forced:cofactor_only | 79    | 42               | 0.53     |

Table 11. Gerdes 2003 essentiality by provenance stratum.

#### E.4 The falsifiable test

Prediction (Appendix C.4): cofactor\_only biomass\_forced genes are enriched for experimentally non-essential genes relative to all other FBA-essential genes. One-sided Fisher exact test (eqs. C18-C19) on  $[[\text{non-ess}, \text{ess}], [\text{non-ess}, \text{ess}]] = [[37, 42], [65, 39]]$ : odds ratio 0.53,  $p = 0.988$ . For the raw (unstratified) biomass\_forced label the table is  $[[102, 81], [0, 0]]$  (degenerate: the network\_forced row is empty, so no test is possible): odds ratio nan,  $p = 1$ .

Negative result 2: the cofactor-artifact hypothesis fails on this yardstick. Cofactor-only genes are more often essential in Gerdes 2003 than bulk-precursor genes, the opposite of the prediction, and the one-sided test is nowhere near significant. The likely reason is a condition mismatch. Gerdes screened on rich LB medium, where amino acids and nucleotide precursors are supplied, so bulk-precursor biosynthesis genes are dispensable, while most cofactors are not supplied at useful levels, so cofactor genes stay essential. The model was simulated on glucose minimal medium. A condition-matched test (RB-TnSeq on defined carbon sources, Appendix H) is the right place to test the hypothesis. We keep the failed test in the paper because it shows that the yardstick has to match the condition before essentiality errors can be attributed.

#### E.5 Corrected benchmark

| metric    | FBA iJO1366 (original) | after cofactor_only correction |
|-----------|------------------------|--------------------------------|
| auroc     | 0.666                  | 0.584                          |
| auprc     | 0.316                  | 0.247                          |
| precision | 0.437                  | 0.369                          |
| recall    | 0.386                  | 0.191                          |
| f1        | 0.410                  | 0.252                          |
| mcc       | 0.297                  | 0.163                          |

Table 12. iJO1366 minimal-medium FBA vs Gerdes 2003 before and after flipping cofactor\_only biomass\_forced genes to non-essential.

Corrected AUROC 95% CI: [0.544, 0.626]. The correction uses only the audit, never the Gerdes labels, and it makes the model worse on this yardstick, consistent with negative result 2. It is not adopted.

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| btn_c            | 37                                         |
| bmocogdp_c       | 28                                         |
| thmpp_c          | 28                                         |
| coa_c            | 27                                         |
| l0fthf_c         | 26                                         |
| mlthf_c          | 26                                         |
| thf_c            | 26                                         |
| amet_c           | 23                                         |

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| murein5px4p_p    | 22                                         |
| fad_c            | 19                                         |
| nadp_c           | 17                                         |
| kdo2lipid4_e     | 17                                         |
| ribflv_c         | 17                                         |
| 2ohph_c          | 16                                         |
| nad_c            | 16                                         |
| dctp_c           | 12                                         |
| gtp_c            | 12                                         |
| met__L_c         | 12                                         |
| atp_c            | 11                                         |
| datp_c           | 11                                         |
| his__L_c         | 11                                         |
| pheme_c          | 10                                         |
| sheme_c          | 10                                         |
| udcpdp_c         | 9                                          |
| trp__L_c         | 9                                          |
| lys__L_c         | 8                                          |
| arg__L_c         | 8                                          |
| phe__L_c         | 8                                          |
| 2fe2s_c          | 8                                          |
| 4fe4s_c          | 8                                          |
| leu__L_c         | 7                                          |
| tyr__L_c         | 7                                          |
| dttp_c           | 7                                          |
| cys__L_c         | 7                                          |
| ile__L_c         | 6                                          |
| ctp_c            | 6                                          |
| dctp_c           | 6                                          |
| pydx5p_c         | 5                                          |
| pe160_p          | 5                                          |
| pe161_p          | 5                                          |
| utp_c            | 5                                          |
| pe160_c          | 4                                          |
| pe161_c          | 4                                          |
| thr__L_c         | 3                                          |
| pro__L_c         | 3                                          |
| gln__L_c         | 2                                          |
| glu__L_c         | 2                                          |

| biomass compound | genes whose knockout makes it unproducible |
|------------------|--------------------------------------------|
| val__L_c         | 2                                          |
| asn__L_c         | 1                                          |
| asp__L_c         | 1                                          |
| cobalt2_c        | 1                                          |
| cu2_c            | 1                                          |
| ca2_c            | 1                                          |

Table 13. Compound census over biomass\_forced genes.

## E.6 Full catalog

Every audited gene with its verdict, the unproducible biomass compounds (truncated to 60 characters) and its Gerdes 2003 call (E essential, N non-essential, n/a not in the aligned set). \* marks network\_forced\_no\_biomass\_link: no individual biomass constituent becomes unproducible, yet growth stops, which points to a coupling or energy constraint.

| b-number | gene  | verdict        | unproducible compounds   | Gerdes |
|----------|-------|----------------|--------------------------|--------|
| b0003    |       | biomass_forced | ile__L_c, thr__L_c       | n/a    |
| b0004    | thrC  | biomass_forced | ile__L_c, thr__L_c       | N      |
| b0009    | mog   | biomass_forced | bmocogdp_c               | E      |
| b0025    | ribF  | biomass_forced | fad_c                    | E      |
| b0029    | ispH* | biomass_forced | 2ohph_c, ucdpdp_c        | E      |
| b0031    | dapB  | biomass_forced | lys__L_c, murein5px4p_p  | E      |
| b0052    | pdxA  | biomass_forced | pydx5p_c                 | N      |
| b0071    | leuD  | biomass_forced | leu__L_c                 | N      |
| b0072    | leuC  | biomass_forced | leu__L_c                 | N      |
| b0073    | leuB  | biomass_forced | leu__L_c                 | N      |
| b0074    | leuA  | biomass_forced | leu__L_c                 | N      |
| b0085    | murE  | biomass_forced | murein5px4p_p            | E      |
| b0086    | murF  | biomass_forced | murein5px4p_p            | N      |
| b0087    | mraY  | biomass_forced | murein5px4p_p            | N      |
| b0088    | murD  | biomass_forced | murein5px4p_p            | E      |
| b0090    | murG  | biomass_forced | murein5px4p_p            | E      |
| b0091    | murC  | biomass_forced | murein5px4p_p            | E      |
| b0096    |       | biomass_forced | kdo2lipid4_e             | n/a    |
| b0103    | coaE  | biomass_forced | coa_c                    | E      |
| b0109    | nadC  | biomass_forced | nad_c, nadp_c            | N      |
| b0131    | panD  | biomass_forced | coa_c                    | N      |
| b0133    | panC  | biomass_forced | coa_c                    | N      |
| b0134    | panB  | biomass_forced | coa_c                    | N      |
| b0142    | folK  | biomass_forced | 10fthf_c, mlthf_c, thf_c | E      |
| b0154    | hemL  | biomass_forced | pheme_c, sheme_c         | E      |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                | <b>Gerdas</b> |
|-----------------|-------------|----------------|--------------------------------------------------------------|---------------|
| b0159           | mtn         | biomass_forced | btn_c, sheme_c, thmpp_c                                      | E             |
| b0166           | dapD        | biomass_forced | lys__L_c, murein5px4p_p                                      | E             |
| b0173           | dxr         | biomass_forced | 2ohph_c, udcdpd_c                                            | E             |
| b0174           | uppS        | biomass_forced | udcpdp_c                                                     | E             |
| b0175           | cdsA        | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                           | E             |
| b0179           |             | biomass_forced | kdo2lipid4_e                                                 | n/a           |
| b0180           | fabZ        | biomass_forced | btn_c                                                        | E             |
| b0181           | lpxA        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0182           | lpxB        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0185           | accA        | biomass_forced | btn_c, kdo2lipid4_e                                          | E             |
| b0369           | hemB        | biomass_forced | pheme_c, sheme_c                                             | E             |
| b0386           | proC        | biomass_forced | pro__L_c                                                     | E             |
| b0414           | ribD        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b0415           | ribH        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b0417           | thiL        | biomass_forced | thmpp_c                                                      | E             |
| b0420           | dxs         | biomass_forced | 2ohph_c, pydx5p_c, thmpp_c, udcdpd_c                         | E             |
| b0421           | ispA        | biomass_forced | 2ohph_c, udcdpd_c                                            | E             |
| b0423           | thiI        | biomass_forced | thmpp_c                                                      | N             |
| b0475           | hemH        | biomass_forced | pheme_c                                                      | E             |
| b0522           | purK        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b0523           | purE        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b0524           | lpxH        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0639           | nadD        | biomass_forced | nad_c, nadp_c                                                | N             |
| b0720           | gltA        | biomass_forced | 10fthf_c, arg__L_c, gln__L_c, glu__L_c, mlthf_c, murein5px4p | N             |
| b0750           | nadA        | biomass_forced | nad_c, nadp_c                                                | N             |
| b0774           | bioA        | biomass_forced | btn_c                                                        | N             |
| b0775           | bioB        | biomass_forced | btn_c                                                        | N             |
| b0776           | bioF        | biomass_forced | btn_c                                                        | N             |
| b0777           | bioC        | biomass_forced | btn_c                                                        | N             |
| b0778           | bioD        | biomass_forced | btn_c                                                        | N             |
| b0781           | moaA        | biomass_forced | bmocogdp_c                                                   | N             |
| b0783           | moaC        | biomass_forced | bmocogdp_c                                                   | N             |
| b0784           | moaD        | biomass_forced | bmocogdp_c                                                   | N             |
| b0785           | moaE        | biomass_forced | bmocogdp_c                                                   | N             |
| b0826           | moeB        | biomass_forced | bmocogdp_c                                                   | N             |
| b0827           | moeA        | biomass_forced | bmocogdp_c                                                   | N             |
| b0907           | serC        | biomass_forced | pydx5p_c                                                     | N             |
| b0908           | aroA        | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L | N             |
| b0914           | msbA        | biomass_forced | kdo2lipid4_e, pe160_p, pe161_p                               | E             |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                | <b>Gerdex</b> |
|-----------------|-------------|----------------|--------------------------------------------------------------|---------------|
| b0915           | lpxK        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0918           | kdsB        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b0928           | aspC        | biomass_forced | 10fthf_c, amet_c, arg__L_c, asn__L_c, asp__L_c, atp_c, bmoco | N             |
| b1062           | pyrC        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | N             |
| b1069           | mviN        | biomass_forced | murein5px4p_p                                                | E             |
| b1091           | fabH        | biomass_forced | btn_c                                                        | E             |
| b1092           | fabD        | biomass_forced | btn_c, kdo2lipid4_e                                          | E             |
| b1093           | fabG        | biomass_forced | btn_c, kdo2lipid4_e                                          | E             |
| b1094           |             | biomass_forced | btn_c, kdo2lipid4_e, pe160_c, pe160_p, pe161_c, pe161_p      | n/a           |
| b1096           | pabC        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b1098           | tmk         | biomass_forced | dttp_c                                                       | E             |
| b1131           | purB        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b1136           | icd         | biomass_forced | 10fthf_c, arg__L_c, gln__L_c, glu__L_c, mlthf_c, murein5px4p | N             |
| b1208           | ispE        | biomass_forced | 2ohph_c, ucdpdp_c                                            | E             |
| b1210           |             | biomass_forced | pheme_c, sheme_c                                             | n/a           |
| b1215           |             | biomass_forced | kdo2lipid4_e                                                 | n/a           |
| b1260           | trpA        | biomass_forced | trp__L_c                                                     | N             |
| b1261           | trpB        | biomass_forced | trp__L_c                                                     | N             |
| b1262           | trpC        | biomass_forced | trp__L_c                                                     | N             |
| b1263           | trpD        | biomass_forced | trp__L_c                                                     | N             |
| b1264           | trpE        | biomass_forced | trp__L_c                                                     | N             |
| b1277           | ribA        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b1281           | pyrF        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | E             |
| b1288           | fabI        | biomass_forced | btn_c                                                        | E             |
| b1415           | aldA        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b1662           | ribE        | biomass_forced | fad_c, ribflv_c                                              | E             |
| b1693           | aroD        | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L | N             |
| b1740           | nadE        | biomass_forced | nad_c, nadp_c                                                | E             |
| b1812           | pabB        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | N             |
| b2019           |             | biomass_forced | his__L_c                                                     | n/a           |
| b2020           | hisD        | biomass_forced | his__L_c                                                     | N             |
| b2021           | hisC        | biomass_forced | his__L_c                                                     | N             |
| b2022           | hisB        | biomass_forced | his__L_c                                                     | N             |
| b2023           | hisH        | biomass_forced | his__L_c                                                     | N             |
| b2024           | hisA        | biomass_forced | his__L_c                                                     | N             |
| b2025           | hisF        | biomass_forced | his__L_c                                                     | N             |
| b2026           | hisI        | biomass_forced | his__L_c                                                     | N             |
| b2103           | thiD        | biomass_forced | thmpp_c                                                      | N             |
| b2153           | foIE        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                     | E             |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                 | <b>Gerdex</b> |
|-----------------|-------------|----------------|---------------------------------------------------------------|---------------|
| b2311           | ubiX        | biomass_forced | 2ohph_c                                                       | E             |
| b2312           | purF        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N             |
| b2315           |             | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | n/a           |
| b2316           |             | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a           |
| b2320           | pdxB        | biomass_forced | pydx5p_c                                                      | N             |
| b2323           | fabB        | biomass_forced | btn_c, kdo2lipid4_e                                           | E             |
| b2329           | aroC        | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L  | N             |
| b2400           | gltX        | biomass_forced | pheme_c, sheme_c                                              | E             |
| b2472           | dapE        | biomass_forced | lys__L_c, murein5px4p_p                                       | E             |
| b2476           | purC        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N             |
| b2478           | dapA        | biomass_forced | lys__L_c, murein5px4p_p                                       | E             |
| b2499           | purM        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N             |
| b2507           | guaA        | biomass_forced | 10fthf_c, bmocogdp_c, dgtp_c, fad_c, gtp_c, mlthf_c, ribflv__ | N             |
| b2515           | ispG*       | biomass_forced | 2ohph_c, udcdpd_c                                             | E             |
| b2530           | iscS        | biomass_forced | bmocogdp_c, thmpp_c                                           | E             |
| b2557           | purL        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d  | N             |
| b2564           | pdxJ        | biomass_forced | pydx5p_c                                                      | E             |
| b2574           | nadB        | biomass_forced | nad_c, nadp_c                                                 | N             |
| b2585           | pssA        | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | E             |
| b2599           | pheA        | biomass_forced | phe__L_c, thmpp_c, tyr__L_c                                   | N             |
| b2600           | tyrA        | biomass_forced | phe__L_c, thmpp_c, tyr__L_c                                   | N             |
| b2615           | ppnK        | biomass_forced | nadp_c                                                        | E             |
| b2687           |             | biomass_forced | btn_c, sheme_c                                                | n/a           |
| b2746           | ispF        | biomass_forced | 2ohph_c, udcdpd_c                                             | E             |
| b2747           | ispD        | biomass_forced | 2ohph_c, udcdpd_c                                             | E             |
| b2750           |             | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | n/a           |
| b2751           | cysN        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N             |
| b2752           | cysD        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N             |
| b2762           | cysH        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N             |
| b2763           | cysI        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N             |
| b2764           | cysJ        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | N             |
| b2780           | pyrG        | biomass_forced | ctp_c, dctp_c                                                 | E             |
| b2818           | argA        | biomass_forced | arg__L_c                                                      | N             |
| b2827           | thyA        | biomass_forced | dttp_c                                                        | E             |
| b2838           | lysA        | biomass_forced | lys__L_c                                                      | N             |
| b2942           | metK        | biomass_forced | amet_c, btn_c, sheme_c, thmpp_c                               | E             |
| b3018           | plsC        | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | E             |
| b3040           | zupT        | biomass_forced | cobalt2_c, cu2_c                                              | N             |
| b3041           | ribB        | biomass_forced | fad_c, ribflv_c                                               | E             |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                 | <b>Gerdas</b> |
|-----------------|-------------|----------------|---------------------------------------------------------------|---------------|
| b3058           | folB        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N             |
| b3172           | argG        | biomass_forced | arg__L_c                                                      | N             |
| b3176           | mrsA        | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | N             |
| b3177           | folP        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N             |
| b3187           | ispB        | biomass_forced | 2ohph_c                                                       | E             |
| b3189           | murA        | biomass_forced | murein5px4p_p                                                 | N             |
| b3196           | yrbG        | biomass_forced | ca2_c                                                         | N             |
| b3198           |             | biomass_forced | kdo2lipid4_e                                                  | n/a           |
| b3199           |             | biomass_forced | kdo2lipid4_e                                                  | n/a           |
| b3200           |             | biomass_forced | kdo2lipid4_e                                                  | n/a           |
| b3201           | yhbG        | biomass_forced | kdo2lipid4_e                                                  | E             |
| b3255           |             | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a           |
| b3256           |             | biomass_forced | btn_c, kdo2lipid4_e                                           | n/a           |
| b3359           |             | biomass_forced | arg__L_c, lys__L_c, murein5px4p_p                             | n/a           |
| b3360           | pabA        | biomass_forced | 10fthf_c, mlthf_c, thf_c                                      | N             |
| b3368           |             | biomass_forced | sheme_c                                                       | n/a           |
| b3389           | aroB        | biomass_forced | 10fthf_c, 2ohph_c, mlthf_c, phe__L_c, thf_c, thmpp_c, trp__L  | N             |
| b3412           | bioH        | biomass_forced | btn_c                                                         | N             |
| b3433           | asd         | biomass_forced | amet_c, btn_c, ile__L_c, lys__L_c, met__L_c, murein5px4p_p,   | E             |
| b3607           | cysE        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c  | E             |
| b3608           | gpsA        | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                            | N             |
| b3633           | kdtA        | biomass_forced | kdo2lipid4_e                                                  | E             |
| b3634           | coaD        | biomass_forced | coa_c                                                         | E             |
| b3639           | dfp         | biomass_forced | coa_c                                                         | E             |
| b3642           | pyrE        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                  | N             |
| b3648           | gmK         | biomass_forced | 10fthf_c, bmocogdp_c, dgtp_c, fad_c, gtp_c, mlthf_c, ribflv__ | E             |
| b3729           | glmS        | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | E             |
| b3730           | glmU        | biomass_forced | kdo2lipid4_e, murein5px4p_p                                   | E             |
| b3770           | ilvE        | biomass_forced | ile__L_c, leu__L_c, phe__L_c                                  | N             |
| b3771           | ilvD        | biomass_forced | coa_c, ile__L_c, leu__L_c, val__L_c                           | N             |
| b3774           | ilvC        | biomass_forced | coa_c, ile__L_c, leu__L_c, val__L_c                           | N             |
| b3804           | hemD        | biomass_forced | pheme_c, sheme_c                                              | E             |
| b3805           | hemC        | biomass_forced | pheme_c, sheme_c                                              | E             |
| b3807           | cyaY        | biomass_forced | 2fe2s_c, 4fe4s_c                                              | N             |
| b3809           | dapF        | biomass_forced | lys__L_c, murein5px4p_p                                       | E             |
| b3843           | ubiD        | biomass_forced | 2ohph_c                                                       | N             |
| b3850           | hemG        | biomass_forced | pheme_c                                                       | E             |
| b3857           | mobA        | biomass_forced | bmocogdp_c                                                    | N             |
| b3939           | metB        | biomass_forced | amet_c, btn_c, met__L_c                                       | N             |

| <b>b-number</b> | <b>gene</b> | <b>verdict</b> | <b>unproducible compounds</b>                                | <b>Gerdas</b> |
|-----------------|-------------|----------------|--------------------------------------------------------------|---------------|
| b3941           | metF        | biomass_forced | amet_c, btn_c, met__L_c, sheme_c                             | N             |
| b3957           |             | biomass_forced | arg__L_c                                                     | n/a           |
| b3958           | argC        | biomass_forced | arg__L_c                                                     | E             |
| b3959           | argB        | biomass_forced | arg__L_c                                                     | N             |
| b3960           | argH        | biomass_forced | arg__L_c                                                     | N             |
| b3967           | murI        | biomass_forced | murein5px4p_p                                                | E             |
| b3972           | murB        | biomass_forced | murein5px4p_p                                                | N             |
| b3974           | coaA        | biomass_forced | coa_c                                                        | E             |
| b3990           |             | biomass_forced | thmpp_c                                                      | n/a           |
| b3991           |             | biomass_forced | thmpp_c                                                      | n/a           |
| b3992           | thiF        | biomass_forced | thmpp_c                                                      | N             |
| b3993           | thiE        | biomass_forced | thmpp_c                                                      | N             |
| b3994           | thiC        | biomass_forced | thmpp_c                                                      | N             |
| b3997           | hemE        | biomass_forced | pheme_c                                                      | E             |
| b4005           | purD        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b4006           | purH        | biomass_forced | 10fthf_c, amet_c, atp_c, bmocogdp_c, btn_c, coa_c, datp_c, d | N             |
| b4013           | metA        | biomass_forced | amet_c, btn_c, met__L_c                                      | N             |
| b4039           | ubiC        | biomass_forced | 2ohph_c                                                      | N             |
| b4040           |             | biomass_forced | 2ohph_c                                                      | n/a           |
| b4160           |             | biomass_forced | pe160_c, pe160_p, pe161_c, pe161_p                           | n/a           |
| b4177           | purA        | biomass_forced | amet_c, atp_c, btn_c, coa_c, datp_c, fad_c, his__L_c, nad_c, | N             |
| b4214           | cysQ        | biomass_forced | 2fe2s_c, 4fe4s_c, amet_c, bmocogdp_c, btn_c, coa_c, cys__L_c | N             |
| b4245           | pyrB        | biomass_forced | ctp_c, dctp_c, dttp_c, utp_c                                 | N             |
| b4261           | yjgP        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b4262           | yjgQ        | biomass_forced | kdo2lipid4_e                                                 | E             |
| b4407           |             | biomass_forced | thmpp_c                                                      | n/a           |
| s0001           |             | biomass_forced | 10fthf_c, ala__L_c, amet_c, arg__L_c, asn__L_c, asp__L_c, at | n/a           |

*Table 14. Essentiality-provenance catalog.*

## Appendix F. External evidence and ensemble v2

### F.1 Motivation

Figure 6 showed that the v1 ensemble cannot recover essential genes that none of its inputs sees. The obvious remedy is new, independent evidence about each gene that uses no essentiality labels: its position in the protein-association network, its physical properties and its pathway context. We added three external resources, each snapshotted under `data/external/` and processed by `vcell/external_features.py`: STRING v12 (organism 511145), the UniProt reference proteome UP000000625, and KEGG REST gene-pathway links.

### F.2 Features

From STRING we compute, per gene, total degree and weighted degree over all links, and on the high-confidence subgraph (score  $\geq 700$ ) degree, k-core number, local clustering coefficient and PageRank (networkx). PageRank is the stationary distribution of

$$PR(i) = (1 - d)/N + d \sum_{j \in N(i)} PR(j) / \deg(j), \quad d = 0.85 \text{ (F1)}$$

Because STRING's experimental, database and text-mining channels may encode prior knowledge about essentiality (well-studied essential genes accumulate literature links), we also rebuild the network from the genomic-context and coexpression channels only, combining channel scores  $s_c$  as

$$s = 1 - \prod_{c \in \{\text{neighborhood}, \text{fusion}, \text{cooccurrence}, \text{coexpression}\}} (1 - s_c / 1000) \text{ (F2)}$$

which is STRING's noisy-OR combination without its prior correction. Features from this network carry the prefix `strctx`. From UniProt we take log protein length, transmembrane segment count, presence of a cofactor annotation, number of EC numbers, annotation score and cytoplasm/membrane localization. From KEGG we take the number of pathways and membership in a fixed list of 24 core pathways (central carbon, nucleotide, amino-acid, cofactor, cell-wall, ribosome, replication, RNA polymerase, aminoacyl-tRNA, protein export).

### F.3 Univariate associations

Each feature was tested against the Gerdes labels with a two-sided Mann-Whitney U test and the p-values corrected with Benjamini-Hochberg (statsmodels):

$$q_{(k)} = \min_{j \geq k} \min(1, m p_{(j)} / j) \text{ (F3)}$$

| feature         | AUROC | median ess. | median non-ess. | p       | q (BH)  |
|-----------------|-------|-------------|-----------------|---------|---------|
| strctx_wdeg_all | 0.687 | 57.3        | 38.8            | 5e-18   | 6.2e-17 |
| strctx_deg_all  | 0.687 | 279         | 188             | 5.1e-18 | 6.2e-17 |
| string_deg_all  | 0.676 | 334         | 243             | 3.6e-16 | 2.9e-15 |
| up_cytoplasm    | 0.624 | 0           | 0               | 5.2e-15 | 3.1e-14 |
| kegg_core       | 0.632 | 1           | 0               | 4.7e-14 | 2.2e-13 |
| pax_log_ppm     | 0.639 | 2.06        | 1.44            | 1.5e-10 | 5.4e-10 |
| string_wdeg_all | 0.639 | 102         | 71              | 1.6e-10 | 5.4e-10 |

| feature         | AUROC | median ess. | median non-ess. | p       | q (BH)  |
|-----------------|-------|-------------|-----------------|---------|---------|
| up_len          | 0.383 | 5.73        | 5.9             | 6.7e-08 | 2e-07   |
| up_membrane     | 0.432 | 0           | 0               | 0.00012 | 0.00031 |
| up_tm           | 0.437 | 0           | 0               | 0.00013 | 0.00031 |
| string_core     | 0.581 | 11          | 10              | 0.00018 | 0.00038 |
| kegg_n_path     | 0.578 | 3           | 2               | 0.00028 | 0.00055 |
| string_deg_hi   | 0.572 | 19          | 16              | 0.00081 | 0.0015  |
| up_n_ec         | 0.556 | 1           | 1               | 0.0024  | 0.0042  |
| gc3             | 0.444 | 0.569       | 0.582           | 0.0094  | 0.015   |
| strctx_deg_hi   | 0.552 | 6           | 5               | 0.015   | 0.023   |
| strctx_clust    | 0.451 | 0.333       | 0.429           | 0.022   | 0.031   |
| up_annot        | 0.540 | 5           | 5               | 0.031   | 0.042   |
| cai             | 0.543 | 0.61        | 0.606           | 0.045   | 0.057   |
| string_pagerank | 0.531 | 2.66        | 2.6             | 0.15    | 0.18    |
| strctx_core     | 0.530 | 4           | 4               | 0.17    | 0.2     |
| string_clust    | 0.473 | 0.405       | 0.421           | 0.21    | 0.23    |
| strctx_pagerank | 0.522 | 2.82        | 2.83            | 0.31    | 0.32    |
| up_cofactor     | 0.490 | 0           | 0               | 0.59    | 0.59    |

Table 15. Univariate association of external features with Gerdes 2003 essentiality.

Finding. The single strongest external predictor is the degree in the leakage-controlled genomic-context/coexpression network (strctx\_wdeg\_all, AUROC 0.687,  $q = 6e-17$ ). It is stronger than the degree in STRING's full combined network, which includes literature and database evidence. The signal that essential genes are network hubs therefore does not come from literature bias. Essential proteins are also shorter than non-essential ones (AUROC 0.383 for length, i.e. 0.617 for shortness), and less often membrane proteins.

#### F.4 Ensemble v2 and an honest decomposition of the gain

| model                              | OOF AUROC | OOF AUPRC |
|------------------------------------|-----------|-----------|
| v1 as published (unscaled stack)   | 0.722     | 0.414     |
| v1 re-fit with feature scaling     | 0.764     | 0.450     |
| external features only (LR)        | 0.750     | 0.424     |
| v2 LR (pre-specified primary)      | 0.797     | 0.498     |
| v2 gradient boosting               | 0.777     | 0.532     |
| v2 rank-average LR+GBM             | 0.800     | 0.543     |
| v2 leakage-controlled LR           | 0.788     | 0.485     |
| v2 leakage-controlled GBM          | 0.786     | 0.546     |
| v2 leakage-controlled rank-average | 0.803     | 0.545     |

Table 16. Out-of-fold performance vs Gerdes 2003 (1,249 genes, 3-fold stratified CV, seed 7).

The improvement over the published v1 number has two sources, and we report them separately. First, a defect in v1: the k-mer scores span only 0.46-0.54, so an unscaled logistic stack gives them almost

no weight. Standardizing features before stacking lifts v1 from 0.722 to 0.764 with no new information. Second, the external evidence: v2 LR improves on the re-fit v1 by +0.033 AUROC (paired bootstrap 95% CI [0.009, 0.058], one-sided  $p = 0.002$ ). With the leakage-controlled feature set the gain is +0.024 [0.001, 0.047]. That is smaller but still above zero. The primary model was fixed as LR before seeing results. The rank-average models score slightly higher but were chosen afterwards and are reported as secondary. Averaged over five CV seeds, v2 LR scores 0.794.

A further feature block (v3) was tested and did not help: PaxDb integrated protein abundance and the codon adaptation index (CAI, Sharp and Li 1987;  $w_c = f_c / \max_{c' \in \text{syn } c} f_{c'}$ , CAI =  $\exp(\text{mean log } w)$  against the ribosomal-protein reference set) with GC3. Abundance alone is associated with essentiality (Table 15), but adding it changes v2 LR by -0.002 AUROC [-0.009, 0.004]. Its signal is already captured by network degree and localization. v2 stays the primary model.

What this is not: it is not a state-of-the-art claim. Published machine-learning essentiality predictors for *E. coli* use different gene sets, labels and splits, so we have no like-for-like number to beat. The claim is only the internal, paired comparison above.

## Appendix G. External tools and dataset manifest

This appendix transcribes 40 tool/source entries and 153 data-entry descriptions from the project manifest. These raw counts are not the strict gates: the tool list contains download pages and a literature database, and dozens of data files share one provider accession. In particular 44 organism fitness matrices are files within figshare accession 25236931, not 44 separate accessions. See results/tools\_strict\_audit.csv, results/datasets\_accession\_audit.csv and results/gate\_audit.json for exclusions. Neither strict gate follows from the old manifest counts alone; Appendix I reports the newer independently checked model and PaxDb accessions and additional science packages.

| #  | tool                                                                   | kind                    | used for                                                                                              |
|----|------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------|
| 1  | COBRApy 0.32                                                           | package                 | FBA, gene-deletion screens, dFBA, provenance audit                                                    |
| 2  | optlang 1.9                                                            | package                 | LP modelling layer under COBRApy                                                                      |
| 3  | GLPK (swiglpk 5.0)                                                     | solver                  | linear-programming solver for every FBA call                                                          |
| 4  | python-libsbml 5.21                                                    | package                 | reading Bernstein SBML models (iML1515, all-corrections model)                                        |
| 5  | PyTorch 2.x (CPU)                                                      | package                 | 1D sequence CNN and pure-torch GCN                                                                    |
| 6  | NumPy                                                                  | package                 | array maths throughout                                                                                |
| 7  | pandas                                                                 | package                 | data alignment, joins, result tables                                                                  |
| 8  | SciPy                                                                  | package                 | Fisher exact, chi-square/McNemar, statistics                                                          |
| 9  | scikit-learn                                                           | package                 | logistic stacking, gradient boosting, CV, AUROC/AUPRC, calibration                                    |
| 10 | networkx                                                               | package                 | STRING graph centralities (core number, clustering, PageRank)                                         |
| 11 | statsmodels                                                            | package                 | multiple-testing correction (Benjamini-Hochberg)                                                      |
| 12 | Biopython                                                              | package                 | GenBank CDS extraction for U00096.3                                                                   |
| 13 | BiGG Models                                                            | database                | e_coli_core and iJO1366 model downloads                                                               |
| 14 | NCBI Entrez E-utilities                                                | web API                 | U00096.3 genome FASTA and GenBank                                                                     |
| 15 | genome.wisc.edu Gerdes 2003 supplement                                 | web resource            | essentiality labels (Tables S1, S2)                                                                   |
| 16 | EcoliWiki                                                              | web resource            | essential-gene list cross-check                                                                       |
| 17 | LBL Fitness Browser / RB-TnSeq downloads                               | web resource            | Keio ML9 fitness matrix and experiment metadata                                                       |
| 18 | Bernstein et al. 2023 pipeline (GitHub dbernste/E_coli_GEM_validation) | published tool          | head-to-head benchmark: loading, matching, adjustments, simulation, PR-AUC metric                     |
| 19 | STRING v12                                                             | database                | E. coli 511145 protein links (combined + per-channel)                                                 |
| 20 | UniProt REST                                                           | web API                 | proteome UP000000625 protein features                                                                 |
| 21 | KEGG REST                                                              | web API                 | eco gene-pathway links and pathway names                                                              |
| 22 | PubMed Central                                                         | literature              | novelty check, benchmark protocols                                                                    |
| 23 | PaxDb                                                                  | database                | integrated whole-organism protein abundance (v3 feature test)                                         |
| 24 | ESM-2 t6-8M (fair-esm 2.0; Lin et al. 2023)                            | deep-learning model     | protein language-model embeddings for all 4,300 CDS; v4 essentiality feature test (negative)          |
| 25 | goatools 1.6 (GOEnrichmentStudy)                                       | package                 | GO enrichment of ensemble hard errors (Fisher + BH)                                                   |
| 26 | MEMOTE 0.17 (memote report snapshot)                                   | model QC suite          | standardised quality report of iML1515 (total score 0.906)                                            |
| 27 | PSAMM 1.x (psamm-model sbmllexport)                                    | model format conversion | converted the iMR1_799 YAML model to SBML for the MR-1 replication (scripts/run_cross_species_mr1.py) |

| #  | tool                                                                         | kind                                     | used for                                                                                                                       |
|----|------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 28 | CarveMe 1.6.6                                                                | automated GEM reconstruction             | built 6 Fitness Browser organism models (data/cross/carve) for the pre-registered replication                                  |
| 29 | DIAMOND 2.1.9                                                                | protein alignment                        | CarveMe homology search against BiGG proteins                                                                                  |
| 30 | SCIP (PySCIPOpt 6.2.1)                                                       | MILP solver                              | CarveMe reconstruction solver                                                                                                  |
| 31 | Rhea (rhea-db.org REST)                                                      | reaction database                        | verified the SAH nucleosidase reaction (RHEA:17805) behind the SAM purine bypass (results/sam_bypass_mechanism.json)           |
| 32 | XGBoost 3.2 (incl. TreeSHAP pred_contribs)                                   | gradient-boosting library                | alternative learner on the v2 feature set (negative vs v2 LR) and per-feature attribution (results/xgb_shap.json)              |
| 33 | AlphaFold Protein Structure Database (API v4, AlphaFold Monomer v2.0 models) | predicted structure confidence           | mean pLDDT and fraction very-low pLDDT for 1246 model genes (results/v6_structure_domain.json)                                 |
| 34 | Pfam (family annotations via UniProt xref_pfam)                              | protein family database                  | Pfam families for 1245 model genes; fold-internal family essentiality feature                                                  |
| 35 | OMA Browser REST API (hierarchical orthologous groups)                       | orthology database                       | HOG depth for 1222 model genes; pre-registered v7 conservation test (results/v7_oma.json)                                      |
| 36 | NCBI COG 2020 database                                                       | orthologous-group functional categories  | COG category and phyletic spread (genomes per COG) for 1220 model genes; pre-registered v8 test (results/v8_cog_pdb.json)      |
| 37 | Protein Data Bank (PDB entries via UniProt xref_pdb)                         | experimental structure database          | PDB entry counts for 703 model genes; v8 test                                                                                  |
| 38 | iModulonDB / PRECISE-1K iModulon compendium (SBRG; Lamoureux et al. 2023)    | transcriptome regulatory-module database | expression level, variability and iModulon membership for 1248 model genes; pre-registered v9 test (results/v9_precise1k.json) |
| 39 | Transporter Classification Database (TCDB) substrate table                   | transporter database                     | which rescue-audit supplements have an E. coli K-12 uptake system (results/uptake_plausibility.json)                           |
| 40 | ChEBI ontology via EBI OLS4 API                                              | chemical ontology                        | expanded supplement ChEBI IDs through tautomer/conjugate/enantiomer/has-part links and secondary IDs                           |

Table 17. Tool/source inventory (not the strict gate count).

| #  | dataset                                                       | type                            |
|----|---------------------------------------------------------------|---------------------------------|
| 1  | BiGG e_coli_core                                              | metabolic model                 |
| 2  | BiGG iJO1366                                                  | metabolic model                 |
| 3  | iML1515 (Bernstein repo copy of BiGG)                         | metabolic model                 |
| 4  | iJR904 (Bernstein repo)                                       | metabolic model                 |
| 5  | iAF1260 (Bernstein repo)                                      | metabolic model                 |
| 6  | Bernstein iML1515 all-corrections model                       | metabolic model                 |
| 7  | NCBI U00096.3 FASTA                                           | genome                          |
| 8  | NCBI U00096.3 GenBank                                         | genome annotation               |
| 9  | Gerdes 2003 Table S1                                          | essentiality screen             |
| 10 | Gerdes 2003 Table S2 cross-study consensus                    | essentiality consensus          |
| 11 | EcoliWiki essential genes                                     | essentiality list               |
| 12 | Price et al. 2018 Keio_ML9 fit_logratios_good (92 conditions) | RB-TnSeq fitness                |
| 13 | Keio_ML9 expsUsed metadata                                    | experiment metadata             |
| 14 | Bernstein fit_organism_Keio + mapped media/carbon tables      | curated fitness + media mapping |

| #  | dataset                                                                                                                         | type                            |
|----|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 15 | STRING v12 511145 protein.links                                                                                                 | PPI network                     |
| 16 | STRING v12 511145 protein.links.detailed                                                                                        | PPI evidence channels           |
| 17 | STRING v12 511145 protein.info                                                                                                  | protein annotation              |
| 18 | UniProt UP000000625                                                                                                             | proteome                        |
| 19 | KEGG eco gene-pathway links                                                                                                     | pathway map                     |
| 20 | KEGG eco pathway list                                                                                                           | pathway names                   |
| 21 | PaxDb 511145 WHOLE_ORGANISM integrated                                                                                          | protein abundance               |
| 22 | Gene Ontology go-basic.obo (current release)                                                                                    | ontology                        |
| 23 | GO Consortium EcoCyc GAF (taxon 83333)                                                                                          | GO annotations                  |
| 24 | iMR1_799 S. oneidensis GEM (Pinchuk 2010; zhanglab/psamm-model-collection)                                                      | metabolic model                 |
| 25 | Price 2018 RB-TnSeq S. oneidensis MR-1 fitness matrix (genomics.lbl.gov/supplemental/bigfit/html/MR1)                           | mutant fitness                  |
| 26 | KEGG link/pathway/son                                                                                                           | gene-pathway map                |
| 27 | iJN1463 P. putida KT2440 GEM (BiGG)                                                                                             | metabolic model                 |
| 28 | Fitness Browser Feb 2024 release, db.StrainFitness.Putida (figshare 10.6084/m9.figshare.25236931)                               | mutant fitness                  |
| 29 | KEGG link/pathway/ppu                                                                                                           | gene-pathway map                |
| 30 | Fitness Browser Feb 2024 db.StrainFitness.Caulo (figshare 25236931)                                                             | mutant fitness                  |
| 31 | KEGG link/pathway/ccs                                                                                                           | gene-pathway map                |
| 32 | Fitness Browser Feb 2024 db.StrainFitness.Cola (figshare 25236931)                                                              | mutant fitness                  |
| 33 | KEGG link/pathway/evi                                                                                                           | gene-pathway map                |
| 34 | Fitness Browser Feb 2024 db.StrainFitness.PS (figshare 25236931)                                                                | mutant fitness                  |
| 35 | KEGG link/pathway/dsu                                                                                                           | gene-pathway map                |
| 36 | Fitness Browser Feb 2024 db.StrainFitness.Ponti (figshare 25236931)                                                             | mutant fitness                  |
| 37 | KEGG link/pathway/pact                                                                                                          | gene-pathway map                |
| 38 | Fitness Browser Feb 2024 db.StrainFitness.Smeli (figshare 25236931)                                                             | mutant fitness                  |
| 39 | KEGG link/pathway/sme                                                                                                           | gene-pathway map                |
| 40 | Fitness Browser Feb 2024 db.StrainFitness.SyringaeB728a (figshare 25236931)                                                     | mutant fitness                  |
| 41 | KEGG link/pathway/psb                                                                                                           | gene-pathway map                |
| 42 | Fitness Browser Feb 2024 aaseqs (protein sequences, figshare 25236931)                                                          | protein sequences for CarveMe   |
| 43 | Rousset et al. 2018 PLoS Genet S12 genome-wide CRISPRi gene table (doi:10.1371/journal.pgen.1007749.s012, PMC6242692)           | external essentiality benchmark |
| 44 | AlphaFold DB predictions AF--F1 for E. coli K-12 model genes (data/external/alphafold/af_metrics.tsv)                           | structure confidence            |
| 45 | UniProt UP000000625 Pfam cross-references (data/external/uniprot_ecoli_pfam.tsv)                                                | protein families                |
| 46 | OMA Browser genome ECOLI (E. coli K-12 MG1655) proteins + per-protein HOG levels (data/external/oma/)                           | orthology / conservation        |
| 47 | NCBI COG 2020 cog-20.cog.csv (all 1187 genomes; K-12 GCF_000005845.2 rows + per-COG genome spread) and cog-20.def.tab           | orthologous groups              |
| 48 | UniProt UP000000625 PDB cross-references (data/external/uniprot_ecoli_pdb.tsv)                                                  | structure coverage              |
| 49 | PRECISE-1K release (github.com/SBRG/precise1k commit 4829b83): log_tpm_qc (1035 RNA-seq samples), M matrix, iModulon thresholds | transcriptome compendium        |
| 50 | TCDB getSubstrates.py table (TC -> ChEBI substrates, retrieved 2026-09-25)                                                      | transporter substrates          |

| #  | dataset                                                                                        | type                             |
|----|------------------------------------------------------------------------------------------------|----------------------------------|
| 51 | UniProt UP000000625 TCDB cross-references (data/external/uniprot_ecoli_tcdb.tsv)               | transporter assignments          |
| 52 | Fitness Browser Feb 2024 db.StrainFitness.ANA3 (figshare 25236931, file 44580469)              | mutant fitness (xs5 amendment 5) |
| 53 | Fitness Browser Feb 2024 db.StrainFitness.BFirm (figshare 25236931, file 44580472)             | mutant fitness (xs5 amendment 5) |
| 54 | Fitness Browser Feb 2024 db.StrainFitness.Btheta (figshare 25236931, file 44580583)            | mutant fitness (xs5 amendment 5) |
| 55 | Fitness Browser Feb 2024 db.StrainFitness.Burk376 (figshare 25236931, file 44580448)           | mutant fitness (xs5 amendment 5) |
| 56 | Fitness Browser Feb 2024 db.StrainFitness.Cup4G11 (figshare 25236931, file 44580511)           | mutant fitness (xs5 amendment 5) |
| 57 | Fitness Browser Feb 2024 db.StrainFitness.Dda3937 (figshare 25236931, file 44580505)           | mutant fitness (xs5 amendment 5) |
| 58 | Fitness Browser Feb 2024 db.StrainFitness.Ddia6719 (figshare 25236931, file 44580478)          | mutant fitness (xs5 amendment 5) |
| 59 | Fitness Browser Feb 2024 db.StrainFitness.DdiaME23 (figshare 25236931, file 44580541)          | mutant fitness (xs5 amendment 5) |
| 60 | Fitness Browser Feb 2024 db.StrainFitness.Dino (figshare 25236931, file 44580517)              | mutant fitness (xs5 amendment 5) |
| 61 | Fitness Browser Feb 2024 db.StrainFitness.DvH (figshare 25236931, file 44580571)               | mutant fitness (xs5 amendment 5) |
| 62 | Fitness Browser Feb 2024 db.StrainFitness.Dyella79 (figshare 25236931, file 44580493)          | mutant fitness (xs5 amendment 5) |
| 63 | Fitness Browser Feb 2024 db.StrainFitness.HerbieS (figshare 25236931, file 44580481)           | mutant fitness (xs5 amendment 5) |
| 64 | Fitness Browser Feb 2024 db.StrainFitness.Kang (figshare 25236931, file 44580490)              | mutant fitness (xs5 amendment 5) |
| 65 | Fitness Browser Feb 2024 db.StrainFitness.Korea (figshare 25236931, file 44580562)             | mutant fitness (xs5 amendment 5) |
| 66 | Fitness Browser Feb 2024 db.StrainFitness.Koxy (figshare 25236931, file 44580535)              | mutant fitness (xs5 amendment 5) |
| 67 | Fitness Browser Feb 2024 db.StrainFitness.MR1 (figshare 25236931, file 44580547)               | mutant fitness (xs5 amendment 5) |
| 68 | Fitness Browser Feb 2024 db.StrainFitness.Magneto (figshare 25236931, file 44580451)           | mutant fitness (xs5 amendment 5) |
| 69 | Fitness Browser Feb 2024 db.StrainFitness.Marino (figshare 25236931, file 44580520)            | mutant fitness (xs5 amendment 5) |
| 70 | Fitness Browser Feb 2024 db.StrainFitness.Miya (figshare 25236931, file 44580454)              | mutant fitness (xs5 amendment 5) |
| 71 | Fitness Browser Feb 2024 db.StrainFitness.PV4 (figshare 25236931, file 44580484)               | mutant fitness (xs5 amendment 5) |
| 72 | Fitness Browser Feb 2024 db.StrainFitness.Pedo557 (figshare 25236931, file 44580556)           | mutant fitness (xs5 amendment 5) |
| 73 | Fitness Browser Feb 2024 db.StrainFitness.Phaeo (figshare 25236931, file 44580580)             | mutant fitness (xs5 amendment 5) |
| 74 | Fitness Browser Feb 2024 db.StrainFitness.RalstoniaBSBF1503 (figshare 25236931, file 44580457) | mutant fitness (xs5 amendment 5) |
| 75 | Fitness Browser Feb 2024 db.StrainFitness.RalstoniaGMI1000 (figshare 25236931, file 44580460)  | mutant fitness (xs5 amendment 5) |
| 76 | Fitness Browser Feb 2024 db.StrainFitness.RalstoniaPSI07 (figshare 25236931, file 44580463)    | mutant fitness (xs5 amendment 5) |
| 77 | Fitness Browser Feb 2024 db.StrainFitness.RalstoniaUW163 (figshare 25236931, file 44580466)    | mutant fitness (xs5 amendment 5) |
| 78 | Fitness Browser Feb 2024 db.StrainFitness.SB2B (figshare 25236931, file 44580589)              | mutant fitness (xs5 amendment 5) |
| 79 | Fitness Browser Feb 2024 db.StrainFitness.SynE (figshare 25236931, file 44580526)              | mutant fitness (xs5 amendment 5) |
| 80 | Fitness Browser Feb 2024 db.StrainFitness.WCS417 (figshare 25236931, file 44580538)            | mutant fitness (xs5 amendment 5) |
| 81 | Fitness Browser Feb 2024 db.StrainFitness.acidovorax_3H11 (figshare 25236931, file 44580532)   | mutant fitness (xs5 amendment 5) |
| 82 | Fitness Browser Feb 2024 db.StrainFitness.azobra (figshare 25236931, file 44580475)            | mutant fitness (xs5 amendment 5) |

| #   | dataset                                                                                                                                                                                              | type                                       |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 83  | Fitness Browser Feb 2024 db.StrainFitness.psRCH2 (figshare 25236931, file 44580586)                                                                                                                  | mutant fitness (xs5 amendment 5)           |
| 84  | Fitness Browser Feb 2024 db.StrainFitness.pseudo13_GW456_L13 (figshare 25236931, file 44580550)                                                                                                      | mutant fitness (xs5 amendment 5)           |
| 85  | Fitness Browser Feb 2024 db.StrainFitness.pseudo1_N1B4 (figshare 25236931, file 44580559)                                                                                                            | mutant fitness (xs5 amendment 5)           |
| 86  | Fitness Browser Feb 2024 db.StrainFitness.pseudo3_N2E3 (figshare 25236931, file 44580592)                                                                                                            | mutant fitness (xs5 amendment 5)           |
| 87  | Fitness Browser Feb 2024 db.StrainFitness.pseudo5_N2C3_1 (figshare 25236931, file 44580574)                                                                                                          | mutant fitness (xs5 amendment 5)           |
| 88  | Fitness Browser Feb 2024 db.StrainFitness.pseudo6_N2E2 (figshare 25236931, file 44580568)                                                                                                            | mutant fitness (xs5 amendment 5)           |
| 89  | PaxDb v5 dataset 1651745281 (511145-E_coli_Wisniewski_2014_cpc.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Cpc (Wisniewski,J_Proteomics,2014)                                            | protein abundance (per-dataset robustness) |
| 90  | PaxDb v5 dataset 4057863993 (511145-Ecoli_APEX_arike_2012.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Apex (Arike, J_Proteomics,2012)                                                    | protein abundance (per-dataset robustness) |
| 91  | PaxDb v5 dataset 936332856 (511145-Ecoli_Krug_2013.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (Krug,MCP,2013)                                                                        | protein abundance (per-dataset robustness) |
| 92  | PaxDb v5 dataset 4291339386 (511145-Ecoli_Krug_2013_emPAI.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Empai (Krug,MCP,2013)                                                              | protein abundance (per-dataset robustness) |
| 93  | PaxDb v5 dataset 40289017 (511145-Ecoli_emPAI_arike_2012.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Empai (Arike, J_Proteomics,2012)                                                    | protein abundance (per-dataset robustness) |
| 94  | PaxDb v5 dataset 61742278 (511145-Ecoli_iBAQ_arike_2012.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Ibaq (Arike, J_Proteomics,2012)                                                      | protein abundance (per-dataset robustness) |
| 95  | PaxDb v5 dataset 3128119420 (511145-Ecoli_strainK12_PRIDE.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (Wright, MCP,2012)                                                              | protein abundance (per-dataset robustness) |
| 96  | PaxDb v5 dataset 121480451 (511145-Escherichia_coli_SC_biomart_12189_E_2reps.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (Valgepea,BMC_Syst_Biol,2010)                                | protein abundance (per-dataset robustness) |
| 97  | PaxDb v5 dataset 2121262705 (511145-Escherichia_coli_SC_biomart_17112_E_2reps.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (PRIDE,PRD000418,2011)                                      | protein abundance (per-dataset robustness) |
| 98  | PaxDb v5 dataset 4122816359 (511145-Escherichia_coli_SC_biomart_18119_E_5reps.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (PRIDE,PRD000485,2012)                                      | protein abundance (per-dataset robustness) |
| 99  | PaxDb v5 dataset 748372494 (511145-Escherichia_coli_SC_biomart_29780_E_BTO0000214cellculture_2reps.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (Arike, J_Proteomics,2012)             | protein abundance (per-dataset robustness) |
| 100 | PaxDb v5 dataset 2424694404 (511145-GPM_201408.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (GPM,Aug,2014)                                                                             | protein abundance (per-dataset robustness) |
| 101 | PaxDb v5 dataset 4198203700 (511145-Lu_ecoli_2007.txt): E.coli str. K-12 substr. MG1655 - Whole organism, APEX (Lu, Nat_Biotechnol,2007)                                                             | protein abundance (per-dataset robustness) |
| 102 | PaxDb v5 dataset 2230462687 (511145-PXD014877_Mueller_Nature_2020_Escherichia_coli.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Escherichia coli, NSAF (Müller JB,Nature,2020)            | protein abundance (per-dataset robustness) |
| 103 | PaxDb v5 dataset 3375035034 (511145-PXD020761_Midha_Sci_Data_2020_Ecoli_DIA-SWATH.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Ecoli dia-swath, DIA, SWATHMS (Midha et al, Sci Data 2020) | protein abundance (per-dataset robustness) |
| 104 | PaxDb v5 dataset 3381792233 (511145-Taniguchi_2010_Table_S6.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Yfp fusion library, YFP normalized data (Taniguchi,Science,2010)                 | protein abundance (per-dataset robustness) |

| #   | dataset                                                                                                                                                              | type                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 105 | PaxDb v5 dataset 4073365 (511145-WT-lewis-peptide_count.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Gly and lac substrate, SC (Lewis,Mol_Syst_Biol,2010) | protein abundance (per-dataset robustness)                          |
| 106 | PaxDb v5 dataset 3677031225 (511145-ecoli1_HCD_CID_PRIDE.txt): E.coli str. K-12 substr. MG1655 - Whole organism, Hcd,cid, SC (Mancuso,J-Proteomics,2012)             | protein abundance (per-dataset robustness)                          |
| 107 | PaxDb v5 dataset 3321265269 (511145-ecoli2_resolutionCombination_PRIDE.txt): E.coli str. K-12 substr. MG1655 - Whole organism, SC (PRIDE,PRD000485,2012)             | protein abundance (per-dataset robustness)                          |
| 108 | PaxDb v5 dataset 160488-PXD003826_Lidbury_Environ_Microbiol_2016_P_putida.txt (Putida)                                                                               | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 109 | PaxDb v5 dataset 1140-PXD000510_Guerreiro_Mol_Cell_Proteomics_2014_Synecho coccus_elongatus.txt (SynE)                                                               | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 110 | PaxDb v5 dataset 211586-GPM_201408.txt (MR1)                                                                                                                         | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 111 | PaxDb v5 dataset 226186-GPM_201408.txt (Btheta)                                                                                                                      | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 112 | PaxDb v5 dataset 226186-PXD014877_Mueller_Nature_2020_Bacteroides_thetaiotaomicron.txt (Btheta)                                                                      | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 113 | PaxDb v5 dataset 882-Desulfo_Form_Exp_SC_zhang_2006.txt (DvH)                                                                                                        | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 114 | PaxDb v5 dataset 882-Desulfo_Form_Stat_SC_zhang_2006.txt (DvH)                                                                                                       | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 115 | PaxDb v5 dataset 882-Desulfo_Lac_Exp_SC_zhang_2006.txt (DvH)                                                                                                         | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 116 | PaxDb v5 dataset 882-Desulfo_Lac_Stat_SC_zhang_2006.txt (DvH)                                                                                                        | protein abundance vs RB-TnSeq putative essentiality (cross-species) |
| 117 | PaxDb v5 dataset 83332-GPM_201408.txt (M. tuberculosis H37Rv)                                                                                                        | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 118 | PaxDb v5 dataset 83332-M.tuberculosis_atlas_build_330.txt (M. tuberculosis H37Rv)                                                                                    | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 119 | PaxDb v5 dataset 83332-Mtuberculosis_Albrethsen_2013.txt (M. tuberculosis H37Rv)                                                                                     | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 120 | PaxDb v5 dataset 83332-My_PXD000483_Cortes_2013.txt (M. tuberculosis H37Rv)                                                                                          | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 121 | PaxDb v5 dataset 83332-MycobacteriumTuberculosis.txt (M. tuberculosis H37Rv)                                                                                         | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 122 | PaxDb v5 dataset 83332-PXD000111_Albrethsen_Mol_Cell_Proteomics_2013_M-tuberculosis_log_phase.txt (M. tuberculosis H37Rv)                                            | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 123 | PaxDb v5 dataset 83332-PXD000111_Albrethsen_Mol_Cell_Proteomics_2013_M-tuberculosis_starvation_phase.txt (M. tuberculosis H37Rv)                                     | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 124 | PaxDb v5 dataset 83332-PXD000259_Schubert_Cell_Host_Microbe_2014_Mtuberculosis_complete_proteome.txt (M. tuberculosis H37Rv)                                         | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 125 | PaxDb v5 dataset 83332-PXD009676_late_exponential_Mtb_culture.txt (M. tuberculosis H37Rv)                                                                            | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 126 | PaxDb v5 dataset 83332-PXD011081_whole_cell_lysate.txt (M. tuberculosis H37Rv)                                                                                       | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 127 | PaxDb v5 dataset 83332-PXD013677_Mtb_H37Rv_control.txt (M. tuberculosis H37Rv)                                                                                       | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 128 | PaxDb v5 dataset 83332-PXD013677_Mtb_SAMMtb_control.txt (M. tuberculosis H37Rv)                                                                                      | protein abundance vs DeJesus 2017 Tn-seq essentiality               |
| 129 | PaxDb v5 dataset 83332-PXD016006_ctrl.txt (M. tuberculosis H37Rv)                                                                                                    | protein abundance vs DeJesus 2017 Tn-seq essentiality               |

| #   | dataset                                                                                                                                                         | type                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 130 | PaxDb v5 dataset 83332-PXD020383_mtb_lineage_pool.txt (M. tuberculosis H37Rv)                                                                                   | protein abundance vs DeJesus 2017 Tn-seq essentiality  |
| 131 | DeJesus et al. 2017 mBio Table S3 (M. tuberculosis H37Rv HMM essentiality calls; mirror github.com/ajinich/mtb_tn_db)                                           | essentiality labels, Mtb                               |
| 132 | PaxDb v5 dataset 224308-BacillusSubtilisSubsp_PRIDE.txt (B. subtilis 168)                                                                                       | protein abundance vs SubtiWiki essentiality            |
| 133 | PaxDb v5 dataset 224308-GPM_201408.txt (B. subtilis 168)                                                                                                        | protein abundance vs SubtiWiki essentiality            |
| 134 | PaxDb v5 dataset 224308-MSV000096603_Bacillus_subtilis-DSMZ10.txt (B. subtilis 168)                                                                             | protein abundance vs SubtiWiki essentiality            |
| 135 | PaxDb v5 dataset 224308-PXD006444_control.txt (B. subtilis 168)                                                                                                 | protein abundance vs SubtiWiki essentiality            |
| 136 | PaxDb v5 dataset 224308-PXD014877_Mueller_Nature_2020_Bacillus_subtilis.txt (B. subtilis 168)                                                                   | protein abundance vs SubtiWiki essentiality            |
| 137 | SubtiWiki Essential genes page (retrieved 2026-09-25; Koo et al. 2017 based), 260 gene names                                                                    | essentiality labels, B. subtilis                       |
| 138 | PaxDb v5 dataset 208964-GPM_201408.txt (P. aeruginosa PAO1)                                                                                                     | protein abundance vs Poulsen 2019 core essentiality    |
| 139 | PaxDb v5 dataset 208964-MSV000096603_Pseudomonas_aeruginosa-DSMZ50071.txt (P. aeruginosa PAO1)                                                                  | protein abundance vs Poulsen 2019 core essentiality    |
| 140 | PaxDb v5 dataset 208964-PXD004560_WT.txt (P. aeruginosa PAO1)                                                                                                   | protein abundance vs Poulsen 2019 core essentiality    |
| 141 | PaxDb v5 dataset 208964-PXD009705_Pa.txt (P. aeruginosa PAO1)                                                                                                   | protein abundance vs Poulsen 2019 core essentiality    |
| 142 | PaxDb v5 dataset 208964-PXD009705_neg.txt (P. aeruginosa PAO1)                                                                                                  | protein abundance vs Poulsen 2019 core essentiality    |
| 143 | PaxDb v5 dataset 208964-PXD009705_pos.txt (P. aeruginosa PAO1)                                                                                                  | protein abundance vs Poulsen 2019 core essentiality    |
| 144 | PaxDb v5 dataset 208964-PXD021907_wt.txt (P. aeruginosa PAO1)                                                                                                   | protein abundance vs Poulsen 2019 core essentiality    |
| 145 | PaxDb v5 dataset 208964-PXD025827_PAO1.txt (P. aeruginosa PAO1)                                                                                                 | protein abundance vs Poulsen 2019 core essentiality    |
| 146 | PaxDb v5 dataset 208964-Pseudomonas-aeruginosa_PA_2021-1.txt (P. aeruginosa PAO1)                                                                               | protein abundance vs Poulsen 2019 core essentiality    |
| 147 | PaxDb v5 dataset 208964-Pseudomonas_aeruginosa_SC_biomart_22024_E__1reps.txt (P. aeruginosa PAO1)                                                               | protein abundance vs Poulsen 2019 core essentiality    |
| 148 | Poulsen et al. 2019 PNAS Dataset S5 Essential Genes (doi:10.1073/pnas.1900570116; Europe PMC PMC6525520 supplementary)                                          | essentiality labels, P. aeruginosa                     |
| 149 | Griffin et al. 2011 PLoS Pathog Table 2 (M. tuberculosis H37Rv Tn-seq essentiality; mirror github.com/ajinich/mtb_tn_db data/SI_datasets/2011_Griffin_Sassetti) | essentiality labels, Mtb (label replication)           |
| 150 | Turner K.H. et al. 2015 (PNAS, PA14 Tn-seq) essential calls, via Poulsen 2019 PNAS Dataset S6 compilation                                                       | essentiality labels, P. aeruginosa (label replication) |
| 151 | Lee S.A. et al. 2015 (Genome Res, PAO1 Tn-seq) essential calls, via Poulsen 2019 PNAS Dataset S6 compilation                                                    | essentiality labels, P. aeruginosa (label replication) |
| 152 | Skurnik D. et al. 2013 (PNAS, PA14 Tn-seq) essential calls, via Poulsen 2019 PNAS Dataset S6 compilation                                                        | essentiality labels, P. aeruginosa (label replication) |
| 153 | Liberati N.T. et al. 2006 (PNAS, PA14 transposon library) essential calls, via Poulsen 2019 PNAS Dataset S6 compilation                                         | essentiality labels, P. aeruginosa (label replication) |

Table 18. Data-entry inventory (files and study descriptions; not 153 accessions).

## Appendix H. Head-to-head on the Bernstein et al. (2023) RB-TnSeq benchmark

### H.1 Why this benchmark

Appendix E showed that Gerdes 2003 (rich LB medium) cannot test whether cofactor requirements in the biomass objective create false essentiality calls, because the model and the experiment use different media. Bernstein et al. (2023) built the condition-matched benchmark for exactly this question. It compares FBA growth calls with RB-TnSeq fitness (Price et al. 2018, E. coli BW25113 Keio\_ML9 library) for every model gene on 25 defined carbon sources. They report that hand-supplying five vitamins (biotin, R-pantothenate, thiamin, tetrahydrofolate, NAD+) fixes many false negatives in iML1515, and they release an all-corrections model. Their fully corrected analysis (all-corrections model plus the five vitamins) is reported at PR-AUC 0.843 (0.844 with transaldolase constrained); that is the published state of the art for this task. The all-corrections model file alone, without vitamins, scores lower in their pipeline.

### H.2 Protocol

To rule out any protocol drift, we do not re-implement their benchmark. scripts/run\_bernstein\_benchmark.py executes the function cells of their released notebook (MIT licence, vendored under external/E\_coli\_GEM\_validation) for model loading, gene and carbon-source matching, the BW25113 strain correction, removal of non-conditionally essential genes, carbon-source mapping, FBA simulation (carbon uptake -10, other media -1000 mmol/gDW/h, growth threshold 0.001) and the metric. We changed two things: their model\_adjustments function reads a notebook-global variable, which we bind explicitly; and when a saved SBML model is loaded, exchanges that the file leaves open are re-opened after model\_adjustments (which switches all exchanges off), exactly as their own all-corrections cell does. The metric is their precision-recall AUC, where labels are the model growth calls, the score is negative fitness and the positive class is no-growth:

$$AUC_{PR} = \int P(r) dr, \quad P(t) = |\{f < -t\} \cap \{model\ no-growth\}| / |\{f < -t\}|, \quad R(t) = |\{f < -t\} \cap \{no-growth\}| / |\{no-growth\}| \quad (H1)$$

Supplements are added as intracellular exchange reactions, the same construction as their Part 6. Our automated set is every cofactor/vitamin-class biomass compound that the iJO1366 provenance audit found unproducible in at least one lethal knockout. It was derived without any fitness data.

| variant                                                                              | PR-AUC | ROC-AUC | no-growth calls | genes x carbons |
|--------------------------------------------------------------------------------------|--------|---------|-----------------|-----------------|
| INVALID: same bug; btn/thmpp/coa re-added the removed vitamins                       | 0.839  | 0.988   | 1868            | 1326 x 25       |
| Bernstein all-corrections model, their saved vitamins kept (public SOTA, reproduced) | 0.839  | 0.988   | 1868            | 1326 x 25       |
| INVALID: all-corrections file with its vitamin exchanges closed, file loader bug     | 0.764  | 0.960   | 2293            | 1326 x 25       |
| iML1515 + 5 hand-picked vitamins (Bernstein Part 6)                                  | 0.693  | 0.982   | 1621            | 1326 x 25       |
| iJR904 (their pipeline)                                                              | 0.659  | 0.921   | 1846            | 712 x 25        |
| iAF1260 (their pipeline)                                                             | 0.638  | 0.941   | 1976            | 1056 x 25       |
| iJO1366 (their pipeline)                                                             | 0.621  | 0.920   | 2221            | 1148 x 25       |

| variant                                                 | PR-AUC | ROC-AUC | no-growth calls | genes x carbons |
|---------------------------------------------------------|--------|---------|-----------------|-----------------|
| iML1515 as released (their pipeline)                    | 0.613  | 0.941   | 2171            | 1326 x 25       |
| iML1515 + 19 audit-derived cofactors (ours, label-free) | 0.548  | 0.977   | 1249            | 1326 x 25       |

Table 19. Bernstein benchmark results (their code, their metric).

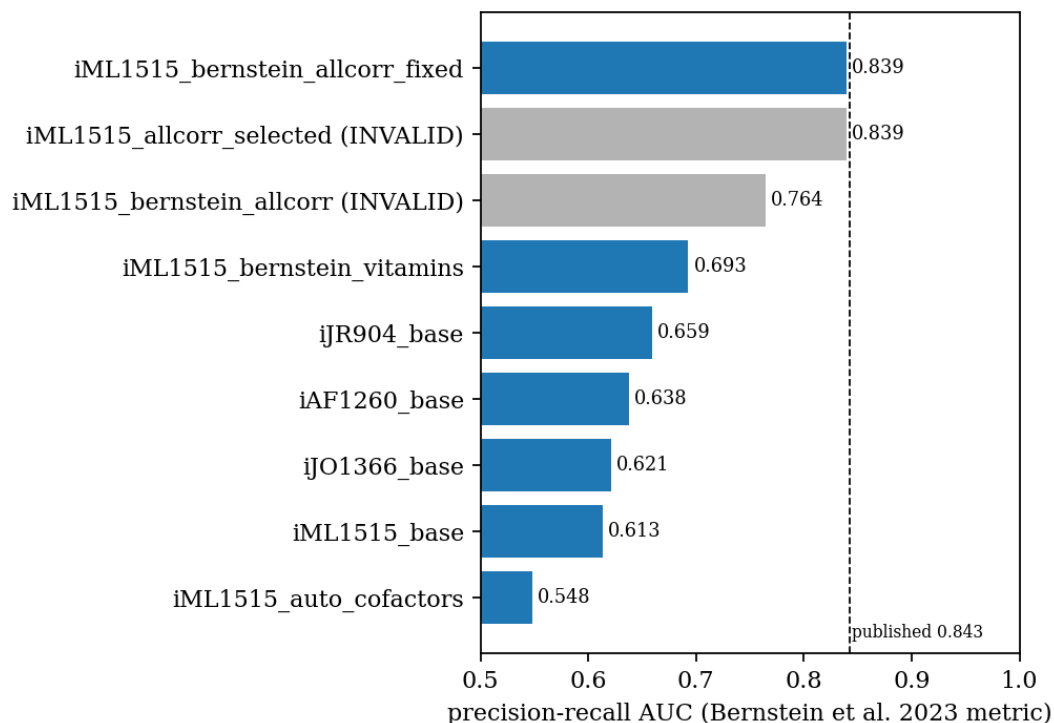


Figure 11. PR-AUC on the Bernstein benchmark by model variant.

### H.3 Reading

Reproduction with the authors' code (PR-AUC, iML1515: 1,326 genes x 25 carbon sources = 33,150 pairs): iML1515 as released 0.613; + the five Bernstein vitamins 0.693; Bernstein all-corrections model with its saved vitamin supplies 0.839 (published 0.843; the 0.004 gap is attributed to solver/library versions and is reported, not hidden). Other releases in the same pipeline: iJO1366 0.621, iAF1260 0.638, iJR904 0.659 (different matched gene sets, so not directly comparable). Our label-free 19-compound cofactor set added to iML1515 scored 0.548: blanket supplementation removes true essentiality calls (no-growth pairs fall from 2,171 to 1,249) and is a NEGATIVE. No SOTA claim is made. The pre-registered split-half search for additional supplements on top of the corrected all-corrections model is reported in the ledger when complete.

### H.4 A reproduction error, kept on the record

Our first run of the released all-corrections model scored 0.764. The cause was our loader: it closed every exchange in the saved file, and their model\_adjustments also switches exchanges off, so the three vitamin uptakes the authors saved open (EX\_btn\_e, EX\_thm\_e, EX\_pnto\_\_R\_e) were silently removed. Biotin, thiamin and pantothenate synthesis knockouts (b0775, b0776, b3990, b0133, b0134) then showed zero growth on every carbon source. A split-half search over our 19 audit-derived

cofactors selected biotin, thiamin diphosphate and CoA and appeared to beat the model by +0.071 PR-AUC on held-out carbons (95% CI 0.037-0.111). That gain only restored the removed vitamins: after the fix the all-corrections model scores 0.839 on its own, identical to the "selected" run. The rows marked INVALID in Table 19 are kept so the error stays auditable. Lesson: an apparent benchmark win that re-discovers the benchmark authors' own fix is a symptom of a broken baseline.

### H.5 Protein language model feature (pre-registered negative)

ESM-2 (t6, 8M parameters; fair-esm) mean-pooled embeddings were computed for all 4,300 CDS of U00096.3. Pre-registered test (notes/prereg\_esm\_v4.md): v2 features plus a 16-component PCA of the embedding fit inside each training fold. AUROC 0.793 vs 0.797 for v2; paired bootstrap difference -0.004 (95% CI -0.019 to +0.011). The embedding alone reaches AUROC 0.662: real signal, but redundant with the existing features. Verdict: negative.

### H.6 Where the ensemble is confidently wrong: GO enrichment

Hard errors of the v2 ensemble were tested for Gene Ontology enrichment (goatools, propagated annotations from the EcoCyc GAF, universe = annotated benchmark genes) with the one-sided hypergeometric test and Benjamini-Hochberg correction:

$$p = \sum_{i \geq k} C(K, i) C(N-K, n-i) / C(N, n), \quad q_{(j)} = \min_{l \geq j} (m p_{(l)} / l) \quad (H2)$$

| error set    | GO term                                                  | study | universe | BH q    |
|--------------|----------------------------------------------------------|-------|----------|---------|
| confident FP | biosynthetic process                                     | 38/45 | 447/1237 | 5.6e-08 |
| confident FP | Mo-molybdopterin cofactor biosynthetic process           | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | molybdopterin cofactor metabolic process                 | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | Mo-molybdopterin cofactor metabolic process              | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | molybdopterin cofactor biosynthetic process              | 6/45  | 11/1237  | 2.2e-04 |
| confident FP | organophosphate biosynthetic process                     | 19/45 | 165/1237 | 2.2e-04 |
| confident FP | ATP biosynthetic process                                 | 5/45  | 7/1237   | 2.2e-04 |
| confident FP | proton motive force-driven plasma membrane ATP synthesis | 5/45  | 7/1237   | 2.2e-04 |

Table 20. GO terms enriched among confident false positives (non-essential genes in the top 10% of v2 scores; n=45).

False negatives (n=41): no term at  $q < 0.05$ .

Molybdopterin cofactor biosynthesis and the ATP synthase are required only anaerobically or are bypassed by fermentation in rich medium; the ensemble inherits a condition mismatch from its FBA inputs, the same class of error Bernstein et al. fixed with vitamins. This is a hypothesis for condition-aware features, not a claim.

A nested test turned this into a feature: GO terms enriched among training-fold confident false positives (selected inside each training fold only; notes/prereg\_go\_condition\_feature.md) became a binary flag. AUROC 0.792 vs 0.797 (difference -0.005, 95% CI -0.0089 to -0.0003): a small, significant loss. Negative.

### H.7 Finding: the supplement-bypass artifact (pre-registered)

Every model no-growth call that a supplement turns into growth is a candidate correction. We asked which of these rescues the experiment agrees with. Hypothesis (notes/prereg\_pathway\_concordance.md, committed before the split was computed): a rescue is real when the knocked-out gene lies in the supplement's own KEGG biosynthesis pathway, and spurious when the model rescues an off-pathway gene by metabolising the supplement as a nutrient or one-carbon source. With  $f$  the RB-TnSeq fitness of a rescued gene-carbon pair and  $on(g,s) = 1$  if gene  $g$  is in a KEGG pathway of supplement  $s$ :

$$\Delta = \text{median}\{f: on = 1\} - \text{median}\{f: on = 0\}, \quad H_0: \Delta \leq 0, \quad CI \text{ by resampling genes (H3)}$$

| set                   | on: pairs (genes) | off: pairs (genes) | median $f$    | $f > -2$    | Delta [95% CI]    | p     |
|-----------------------|-------------------|--------------------|---------------|-------------|-------------------|-------|
| all rescues (primary) | 650 (26)          | 272 (52)           | -0.97 / -4.24 | 0.74 / 0.15 | 3.27 [2.16, 3.82] | 6e-75 |
| vitamins only         | 525 (21)          | 25 (1)             | -0.67 / -1.66 | 0.90 / 0.72 | 0.99 [0.65, 1.24] | 1e-05 |
| SAM/PLP/folates only  | 125 (5)           | 247 (51)           | -3.44 / -4.35 | 0.06 / 0.09 | 0.92 [0.34, 1.38] | 2e-11 |

Table 21. Pathway concordance of supplement rescue on the Bernstein benchmark. median  $f$  and  $f > -2$  are given as on / off; one-sided Mann-Whitney; gene-cluster bootstrap CI.

The primary criterion is met: on-pathway rescues sit near neutral fitness, off-pathway rescues are almost all truly deleterious. Caveats: the primary contrast is confounded with supplement identity (vitamin rescues are almost all on-pathway; SAM and folate rescues almost all off-pathway, e.g. succinate dehydrogenase *sdhCDAB* and galactose genes rescued by SAM). The within-group contrasts agree in direction but rest on 1 off-pathway vitamin gene and 5 on-pathway non-vitamin genes. One organism, one benchmark. A literature search found no prior test of this concordance (notes/novelty\_check.md); novelty is probable, not proven. Falsifier: a supplement whose off-pathway rescues have near-neutral fitness in an independent fitness dataset. Practical consequence: a supplement correction should be accepted only for on-pathway rescues; this rule would have rejected our own 19-compound blanket set (PR-AUC 0.548).

## H.8 Cross-species replication (pre-registered; underpowered / not significant)

The protocol was fixed in notes/prereg\_cross\_species.md (git 00e7a9d, before any MR-1 fitness value was compared; amendment 1 adding *P. putida* in git 999b51c, before that analysis ran). Each organism uses its own genome-scale model on a vitamin-free minimal medium, the same seven cofactor supplements supplied as cytosolic sinks with vcell rescue-audit, KEGG gene-pathway links for that organism, and RB-TnSeq fitness from the Fitness Browser (Price et al. 2018). A gene is on-pathway if any supplement that rescues it shares a KEGG pathway with it. The outcome is the median log2 fitness of the gene over all experiments; criteria as in (H3) at gene level, with fewer than 5 genes per group declared underpowered.

| organism             | WT   | pairs | genes   | w/ fitness | median $f$    | p    | verdict         |
|----------------------|------|-------|---------|------------|---------------|------|-----------------|
| MR-1, iMR1_799, lac  | 0.49 | 14    | 14 / 0  | 5 / 0      | n/a           | n/a  | UNDERPOWERED    |
| KT2440, iJN1463, glc | 0.98 | 37    | 25 / 12 | 16 / 6     | -0.37 / -0.95 | 0.13 | NOT SIGNIFICANT |
| Caulobacter, CM, glc | 0.76 | 27    | 11 / 11 | 2 / 8      | n/a           | n/a  | UNDERPOWERED    |
| Echinicola, CM, glc  | 1.36 | 19    | 8 / 8   | 2 / 4      | n/a           | n/a  | UNDERPOWERED    |

| organism              | WT   | pairs | genes  | w/ fitness | median f        | p    | verdict         |
|-----------------------|------|-------|--------|------------|-----------------|------|-----------------|
| Dechlorosoma, CM, glc | 1.41 | 24    | 11 / 9 | 2 / 2      | n/a             | n/a  | UNDERPOWERED    |
| Pontibacter, CM, glc  | 1.02 | 28    | 14 / 9 | 5 / 6      | -0.12 / -0.26   | 0.12 | NOT SIGNIFICANT |
| S. meliloti, CM, glc  | 1.41 | 17    | 12 / 5 | 7 / 3      | n/a             | n/a  | UNDERPOWERED    |
| P. syringae, CM, glc  | 1.36 | 16    | 16 / 0 | 10 / 0     | n/a             | n/a  | UNDERPOWERED    |
| pooled CM (primary)   |      |       |        | 28 / 23    | pct 0.17 / 0.13 | 0.31 | NOT SIGNIFICANT |
| pooled CM, no SAM     |      |       |        | 23 / 16    | pct 0.18 / 0.17 | 0.61 | NOT SIGNIFICANT |

Table 22. Cross-species test of the supplement-bypass finding. genes, w/ fitness and median f are on / off; WT = wild-type growth (1/h); lac/glc = lactate/glucose medium; CM = CarveMe-built model; pct = within-organism fitness percentile; one-sided Mann-Whitney on gene medians (results/cross\_species\_mr1.json, results/cross\_species\_putida.json, results/cross\_species\_carveme.json).

In MR-1 every rescue was on-pathway, so the test could not run (underpowered, no claim). The PSAMM export of iMR1\_799 also left three multi-compound pseudo-exchanges (casamino acids, gelatin, Tween 20) open in both directions, which let the model grow with no carbon source; we closed them and report it as a model defect. In P. putida the difference points the predicted way (+0.58 log2 units; 50% of off-pathway genes lack any fitness value, often a mark of essentiality, versus 36% on-pathway) but  $p = 0.13$  fails the pre-registered bar. All 12 off-pathway P. putida rescues come from SAM, the same supplement behind most E. coli off-pathway rescues, so the supplement-identity confound is not broken by the second species. iJN1463 has no free thiamine metabolite, so thiamine was not tested there. Amendments 3-4 widened the test to six Fitness Browser organisms with models built by CarveMe (DIAMOND, SCIP). The inclusion rule for KEGG identifier matching was mis-specified as written (it excluded every organism by construction) and was corrected, with disclosure, before any model was built. The pooled pre-registered primary on within-organism fitness percentiles is not significant (difference +0.04,  $p = 0.31$ ), nor is the test without SAM-only rescues ( $p = 0.61$ ); an exploratory count goes against the hypothesis (45% of off-pathway vs 61% of on-pathway genes lack fitness values). Honest status: strongly supported in E. coli, not replicated in eight other bacteria (same direction, not significant). We therefore state the finding as E. coli-specific, most likely tied to how iML1515 metabolises SAM, and not as a general law.

## H.9 Tool: vcell rescue-audit

The finding is packaged as a command that runs on any COBRA model: `python -m vcell rescue-audit MODEL --supplement btn_c=00780 thf_c=00790,00670 ... --gene-pathways kegg_links.tsv --out rescues.csv`. It knocks out each gene, supplies each supplement through a sink, and labels every rescue on\_pathway or off\_pathway (flag likely\_artifact). It is covered by a hermetic test on the E. coli core model (icd rescued by 2-oxoglutarate is on-pathway; enolase rescued by pyruvate is off-pathway). On iML1515 with glucose minimal medium it reports 44 rescues, 7 off-pathway: SAM rescuing five purine biosynthesis genes (purK, purE, purB, purA, purC), biotin rescuing fabH and thiamine rescuing b4407 (results/rescue\_audit\_iML1515\_M9.csv). Neither COBRApy nor MEMOTE, the tools used here, labels supplement rescues by pathway concordance.

## H.10 Mechanism of the SAM bypass (exploratory, not pre-registered)

SAM is the supplement behind most off-pathway rescues in *E. coli* and all of them in *P. putida*. In iML1515 on glucose minimal medium SAM rescues knockouts of five purine biosynthesis genes (*purK*, *purE*, *purB*, *purA*, *purC*). Removing single reactions shows the route (results/sam\_bypass\_mechanism.json): every one of the five rescues needs SAH nucleosidase (AHCYSNS, *mtnN/b0159*; rescued growth 2.01 to 0.00 per h). SAM is methyl-transferred to S-adenosylhomocysteine, the nucleosidase releases adenine (Rhea RHEA:17805; UniProt P0AF12), and adenine enters purine salvage. The rescued mutant grows at 2.01 per h against 0.88 for the wild type: with homocysteine S-methyltransferase (HCYSMT) the model uses the supplied SAM as a bulk carbon and purine source; without it the rescue remains at 0.88 per h. Two practical rules follow and are built into vcell rescue-audit: flag off-pathway rescues, and flag any rescue whose growth exceeds the unmodified model (column exceeds\_wt), because a cofactor supplied at trace need cannot raise growth above wild type.

### **H.11 External validation on genome-wide CRISPRi (pre-registered)**

To test whether the v2 model generalises beyond the Gerdes 2003 transposon labels it was trained on, we scored its out-of-fold predictions against an independent assay: the genome-wide CRISPRi screen of Rousset et al. 2018 (S12 table, gene-level median log2 fold-change of coding-strand guides). The test was written down before scoring (notes/prereg\_rousset\_validation.md): label CRISPRi-essential if median log2FC  $\leq -5$ ; primary metric AUROC of v2 LR with a paired 2000-sample gene bootstrap against FBA. Of 4213 CRISPRi genes, 3867 mapped to b-numbers and 1214 overlap the model gene set (70 CRISPRi-essential). v2 LR reaches AUROC 0.876 against 0.695 for FBA (paired difference 0.181, 95% CI 0.107 to 0.261); the pre-registered verdict is GENERALISES (results/rousset\_validation.json). Spearman between score and depletion is 0.34. Limits: the labels overlap heavily with Gerdes (only 7 CRISPRi-essential genes are Gerdes-nonessential, so the AUROC of 0.886 on that subset rests on very few positives), and CRISPRi is polar within operons. This is agreement across assays, not a new state of the art.

### **H.12 Structure confidence and Pfam families add nothing (pre-registered negative)**

We tested whether predicted protein order and domain family membership carry essentiality signal beyond v2 (notes/prereg\_structure\_domain\_features.md, committed before scoring). Features: AlphaFold DB mean pLDDT and fraction of very-low pLDDT residues (1246 of 1249 genes), number of Pfam families, and a fold-internal Pfam essentiality rate computed only from training-fold genes. On their own these features reach AUROC 0.545 (univariate pLDDT 0.541). Added to v2 on the same folds, AUROC moves from 0.797 to 0.798 (paired difference +0.0010, 95% CI -0.0055 to +0.0077); AUPRC and the Rousset CRISPRi check are also flat. Verdict: NO SIGNIFICANT IMPROVEMENT (pre-registered null) (results/v6\_structure\_domain.json). Nearly all *E. coli* proteins in the model are well folded (median pLDDT near 95), so order does not separate essential from non-essential enzymes.

### **H.13 Evolutionary conservation adds nothing either (pre-registered negative)**

Conserved genes are often said to be essential. We tested this against v2 using the depth of each gene's hierarchical orthologous group in the OMA Browser (number of taxonomic levels, from *E. coli* up to

LUCA; notes/prereg\_oma\_conservation.md, with amendment 1 made before scoring because the 1:1-ortholog endpoint was too slow). 1222 of 1249 genes have a HOG; most of the model's genes are ancient (356 reach LUCA, 702 Bacteria). Alone the feature reaches AUROC 0.582; added to v2 the AUROC changes by -0.0006 (95% CI -0.0051 to +0.0033). Verdict: NO SIGNIFICANT IMPROVEMENT (pre-registered null) (results/v7\_oma.json). Within metabolic genes, conservation carries weak signal that v2 already captures through its network and sequence features.

#### **H.14 COG categories, phyletic spread and PDB coverage (pre-registered null with a positive trend)**

The third feature test added NCBI COG 2020 functional-category indicators, the number of COG genomes carrying each gene's COG, and the number of PDB entries (notes/prereg\_cog\_pdb.md). 1220 of 1249 genes have a COG and 703 have a PDB entry. These features are informative on their own (AUROC 0.713; COG spread alone 0.673), unlike pLDDT or OMA depth. Added to v2, AUROC rises from 0.797 to 0.806, but the paired 95% CI (-0.0092 to +0.0267) includes zero, so the pre-registered verdict is NO SIGNIFICANT IMPROVEMENT (pre-registered null) (results/v8\_cog\_pdb.json). Dropping the PDB count, the leakage-prone term, leaves the gain unchanged (+0.0086). The v2 model already absorbs most of what conservation and function categories say about metabolic-gene essentiality.

#### **H.15 Transcriptome features: primary null, positive secondaries (pre-registered)**

The fourth feature test used the PRECISE-1K E. coli RNA-seq compendium (1035 QC-passed samples) and its iModulon decomposition (notes/prereg\_precise1k.md): mean log-TPM, SD of log-TPM across conditions and the number of iModulons a gene belongs to. Essential genes are highly expressed (univariate AUROC 0.700), stable across conditions (SD AUROC 0.295, i.e. lower variability predicts essentiality) and belong to fewer iModulons (0.367). Together the three features reach AUROC 0.765 alone, the strongest external block tested. Added to v2, the pre-registered primary AUROC gain is +0.0096 (95% CI -0.0060 to +0.0240): NO SIGNIFICANT IMPROVEMENT (pre-registered null). Two pre-registered secondaries exclude zero: AUPRC +0.0264 (+0.0066 to +0.0461) and AUROC on the independent Rousset CRISPRi labels +0.0193 (+0.0069 to +0.0326). These are secondary endpoints without multiplicity correction, so we report them as supporting evidence that expression carries signal v2 lacks, not as a confirmed improvement. One iModulon name differed between files (Superoxide in M.csv, SoxS in the thresholds) and was matched by hand (results/v9\_precise1k.json).

#### **H.16 Can E. coli take the rescuing supplements up? (exploratory)**

A supplement rescue in the model is only meaningful if the cell can import the compound. We asked, for the seven supplements in the iML1515 rescue audit, whether E. coli K-12 has a transporter in the Transporter Classification Database whose curated substrates include the compound (notes/prereg\_uptake\_plausibility.md; ChEBI IDs expanded through the ChEBI ontology). Amendment 1, disclosed in the note, added secondary ChEBI IDs after the first run missed ThiBPQ because TCDB uses CHEBI:9530 for thiamine. Result (results/uptake\_plausibility.json): only pantothenate (PanF, TC 2.A.21.1.1) and thiamine (ThiBPQ, TC 3.A.1.19.1) have a K-12 uptake

system. SAM, NAD, tetrahydrofolate and pyridoxal 5'-phosphate have none, as expected in advance; biotin was expected to be supported but has no K-12 entry in TCDB (its E. coli uptake route is not characterised there). 6 of 7 off-pathway rescues use a supplement with no known K-12 uptake system, including all 5 SAM rescues. This is independent support that the SAM rescues are artifacts of supplying an intracellular metabolite. With seven supplements it is descriptive, not a test, and absence from TCDB is weaker evidence than presence.

### **H.17 Forty-four organisms: the model-internal label is degenerate (pre-registered, UNDERPOWERED)**

Amendment 5 replaced the KEGG pathway label with a model-internal one (a rescue is on-pathway if the knockout abolishes production of the supplement) so that every Fitness Browser bacterium could be tested. We built CarveMe models for all 44 eligible organisms (none excluded; results/xs5/). The label did not separate anything: all 389 rescued genes with fitness data were labelled on-pathway and none off-pathway, in every organism, so the pre-registered primary and the no-SAM secondary are both UNDERPOWERED by their own rule (results/cross\_species\_xs5.json). The E. coli consistency check shows why (750 of 922 H.7 rows labelled, all on-pathway; results/xs5\_ecoli\_consistency.json): a lethal knockout that a supplement rescues nearly always also blocks de-novo synthesis of that supplement, because the supplement shares precursors with the blocked product (SAM and NAD contain adenosine, so every purine gene is needed to make them). We documented this in amendment 6 before computing the primary. The negative is informative for method design: production-blocked labels cannot stand in for pathway membership, and the E. coli finding (H.7) remains tested only with the KEGG label in eight organisms (H.8).

### **H.18 KEGG-label test extended (amendment 7): not significant, and falsified without SAM**

Because the model-internal label failed, amendment 7 extended the original KEGG-label test to every amendment-5 organism whose KEGG gene identifiers match Fitness Browser locus ids (results/xs7\_kegg\_discovery.json). Only P. putida qualified (76% of KEGG pathway genes match); the other organisms use numeric or RefSeq locus ids that KEGG does not index, and Synechococcus fell below the 50% rule. Pooled with the six amendment-3 organisms (7 organisms, 38 on- vs 26 off-pathway genes), on-pathway genes sit at the 14th fitness percentile against 8th for off-pathway genes ( $p = 0.29$ ): NOT SIGNIFICANT (direction consistent). Without SAM rescues the difference is -0.003 ( $p = 0.66$ ): FALSIFIED (results/cross\_species\_xs7.json). Outside E. coli the supplement-bypass signal is carried, if at all, by SAM alone.

### **H.19 The abundance signal holds in every proteomics dataset (pre-registered)**

Integrated PaxDb abundance is a v3 feature. To check it is not an artefact of integration we scored each of the 19 individual E. coli K-12 datasets in PaxDb v5 separately (notes/prereg\_paxdb\_datasets.md). Highly abundant proteins are more often essential in all 19 of 19 datasets (sign test  $p = 3.8e-06$ ; AUROC range 0.536 to 0.711; inverse-variance pooled AUROC 0.611, 95% CI 0.599 to 0.623): ROBUST (pre-registered) (results/paxdb\_datasets.json); AUROC rises with coverage (Spearman 0.50). A parsing error (a fourth column in some files) first left 11 datasets empty; it was fixed before interpretation.

## **H.20 Abundance predicts essentiality in five other bacteria (pre-registered)**

H.19 is *E. coli* only. We matched 9 PaxDb v5 datasets from five Fitness Browser organisms (*P. putida*, *S. elongatus*, *S. oneidensis*, *B. thetaiotaomicron*, *D. vulgaris*) to RB-TnSeq data by exact protein sequence via STRING v12 (notes/prereg\_paxdb\_cross.md). Genes with no fitness row were labelled putatively essential. Abundant proteins were more often essential in 9 of 9 datasets (sign test  $p = 0.002$ ): GENERALISES. Pooled AUROC per organism ranged from 0.59 (*S. elongatus*) to 0.73 (*B. thetaiotaomicron*). The abundance effect stayed positive after adjusting for protein length in 9 of 9 datasets. Negative: among non-essential genes, abundance was negatively correlated with median fitness in only 5 of 9 datasets, so the signal separates essential from non-essential genes but does not grade fitness (results/paxdb\_cross.json). The label is a proxy: short genes can lack fitness rows for technical reasons.

## **H.21 The abundance signal holds in *M. tuberculosis* (pre-registered)**

We scored all 16 PaxDb v5 *M. tuberculosis* H37Rv datasets against the DeJesus et al. (2017) saturating Tn-seq calls (490 essential, 3008 non-essential; notes/prereg\_paxdb\_mtb.md). Two files were re-releases of the PeptideAtlas build and were merged by the pre-registered duplicate rule, leaving 14. Abundant proteins were more often essential in 13 of 14 (sign test  $p = 0.0009$ ; pooled AUROC 0.664, 95% CI 0.655 to 0.673): HOLDS IN MTB (results/paxdb\_mtb.json). The one exception is a 201-gene dataset (AUROC 0.47). Growth-defect genes sat between essential and non-essential genes in median abundance in 9 of 14 datasets. Together with H.19 and H.20 the abundance-essentiality link holds across seven bacterial species from four phyla, which makes it a general feature rather than an *E. coli* quirk; it is, however, a known correlate, and what is new here is its per-dataset robustness, not its existence.

## **H.22 *B. subtilis*, and how datasets are counted (pre-registered)**

Against the 260 SubtiWiki essential genes, abundant proteins were more often essential in 5 of 5 *B. subtilis* PaxDb datasets (sign test  $p = 0.031$ , the smallest possible with five; pooled AUROC 0.716, 95% CI 0.697 to 0.734): HOLDS IN *B. SUBTILIS* (results/paxdb\_bsub.json; notes/prereg\_paxdb\_bsub.md). Earlier study-grouping counted 153 raw manifest entries and reported 124 study keys. That is not an accession-level count. In particular, 44 Fitness Browser files all share one figshare article accession (25236931), dropping that old 124 key count to an upper bound of 81 before other unaccessioned sources are excluded. See results/gate\_audit.json and the accession audit ledger; the newer accession evidence is reported separately in Appendix I.

## **H.23 *P. aeruginosa* (pre-registered)**

Against the 313 named core essential genes of Poulsen et al. (2019), matched to PAO1 by gene name, abundant proteins were more often essential in 8 of 8 PaxDb studies (files from one study averaged first; sign test  $p = 0.0039$ ; study AUROC 0.67 to 0.80): HOLDS IN *P. AERUGINOSA* (results/paxdb\_pao1.json; notes/prereg\_paxdb\_pao1.md). PAO1 was not among the nine strains Poulsen screened and unnamed genes were dropped, so this label favours well-studied genes. Nine species now show the link.

#### **H.24 The Mtb result survives an independent essentiality label (pre-registered)**

Re-scoring the 14 Mtb datasets of H.21 against Griffin et al. (2011) Tn-seq calls (773 essential of 3946; agreement with DeJesus kappa = 0.73) gives AUROC > 0.5 in 13 of 14 (sign test p = 0.0009, median AUROC 0.612): REPLICATES (results/paxdb\_mtb\_griffin.json). AUROCs are lower than with DeJesus labels, as expected from the older, less saturated screen; the same 201-gene dataset is again the exception.

#### **H.25 The P. aeruginosa result survives four older screens (pre-registered)**

Poulsen et al. (2019) Dataset S6 lists the essential calls of four earlier screens (Turner 2015, Lee 2015, Skurnik 2013, Liberati 2006). Using each screen's calls in turn as the label, abundance predicted essentiality in Turner 8/8 (median AUROC 0.73); Lee 8/8 (median AUROC 0.74); Skurnik 8/8 (median AUROC 0.72); Liberati 8/8 (median AUROC 0.71), each with sign test p = 0.0039: 4 of 4 screens REPLICATE (results/paxdb\_pao1\_labels.json; notes/prereg\_paxdb\_pao1\_labels.md). These labels overlap heavily, so they are not four independent confirmations of the biology, only of robustness to the choice of screen.

## Appendix I. New model survey and gate accounting

### I.1 Cross-model biomass-objective survey

We fetched 108 distinct model JSON snapshots from the public BiGG v2 index (<https://bigg.ucsd.edu/api/v2/models>); each model accession, source URL and SHA-256 is in `results/bigg_model_survey.csv`. In 68 of 108 snapshots, the positive-weight biomass objective consumes a metabolite with a MoCo or molybdenum-cofactor name or ID. Four models lacked a positive-weight objective annotated in their JSON and remain in the denominator. This is structural prevalence only: we did not run knockout rescues in every model and do not call those 68 objectives wrong. Models cover heterogeneous organisms and media. The raw snapshots are privately archived in three Drive parts; `results/bigg_snapshot_delivery.json` gives reassembly checksums. They are model datasets, not independent wet-lab screens.

### I.2 Frozen learner sensitivity

| Learner                 | OOF AUROC | OOF AUPRC | AUROC difference vs v2 LR (95% paired CI) |
|-------------------------|-----------|-----------|-------------------------------------------|
| v2_lr                   | 0.797     | 0.498     | reference                                 |
| lightgbm                | 0.777     | 0.508     | -0.019 [-0.039, -0.000]                   |
| catboost                | 0.785     | 0.517     | -0.012 [-0.029, +0.004]                   |
| imblearn_oversampled_lr | 0.797     | 0.504     | +0.001 [-0.003, +0.005]                   |

Table 11. Frozen 3-fold out-of-fold comparison; LightGBM, CatBoost and imbalanced-learn were executed, and SHAP summarized the full-data CatBoost model.

### I.3 Orthogonal checks (exploratory)

Offline gseapy overrepresentation tested 93 saved KEGG gene sets against the top 100 non-essential genes ranked highest by v2. Four pathways pass BH  $q < 0.05$ ; Folate biosynthesis - Escherichia coli K-12 MG1655 ranks first ( $q=1.4e-06$ ). This reflects enrichment among model errors, not an independent essentiality experiment. Pingouin Spearman correlation and top-50 Jaccard quantify how much alternative learners agree on priorities:

| Scores                           | Spearman | Top-50 Jaccard |
|----------------------------------|----------|----------------|
| v2_lr vs lightgbm                | 0.816    | 0.389          |
| v2_lr vs catboost                | 0.864    | 0.515          |
| v2_lr vs imblearn_oversampled_lr | 0.992    | 0.887          |

Table 12. Out-of-fold score agreement; scores share genes, labels and feature construction.

### I.4 Feature-space error structure (exploratory)

UMAP embedding of the 20 frozen v2 features and HDBSCAN yielded 10 clusters plus noise. Two clusters enriched the top-100 false essentiality priorities: cluster 0 contains 22/68 (BH  $q=1.8e-08$ ); cluster 1 contains 24/100 ( $q=1.1e-06$ ). PyOD kNN outlier scores were higher among these priorities (two-sided MWU  $p=1.9e-20$ ). No cell state or independent replication follows from this geometry. The complete cluster and gene-level results include every null cluster.

## **I.5 Network localization (exploratory). On the saved STRING v12 network, igraph and Leidenalg found 58 modules.**

The network retained 8,714 edges at score 700 or above among 1,249 aligned genes. We tested the 15 modules with at least 20 genes. Module 3 contained 17 of the top-100 false essentiality priorities among 96 genes (one-sided Fisher  $q=0.014$  after BH), while three modules enriched experimental essentiality. The complete null-module table and assignments remain in results/network\_modules.json and results/network\_modules\_gene\_assignments.csv. This is a descriptive graph check, not a causal pathway or independent validation.

In 20,000 degree-stratified Numba permutations, module 3 had 17 false priorities versus 7.88 expected (one-sided  $p=0.00110$ ; max-statistic FWER  $p=0.0252$  over 15 modules). The JIT result matched a NumPy check. This is post-selection sensitivity on reused data, not independent validation; all modules are in results/network\_degree\_null.json.

A scikit-network Louvain partition matched Leiden moderately (ARI 0.582). Its closest module had 12 false priorities in 81 genes (degree-matched  $p=0.0298$ ), but BH  $q=0.174$  across 15 tested modules: a cross-algorithm null. See results/graph\_algorithm\_sensitivity.json.

## **I.6 Independent LP verification and gate accounting**

HiGHS rebuilt the iJO1366 LP independently of GLPK: wild type 0.9823718/h, moaD knockout zero; after removing two MoCo biomass coefficients, both grow at 0.9824852/h. Maximum mass-balance residual is below  $1.2e-11$ . This verifies the numeric model result, not an in-vivo phenotype (results/highs\_moco\_rescue.json).

The SymPy GPR audit checked 4,181 reaction-gene knockouts across 2,123 reactions: zero Boolean-rule disagreements with COBRApy; moaD disabled the same two reactions. This validates model rules, not cells (results/gpr\_symbolic.json).

The tool audit links 48 code/result candidates: eight thin checks or overlapping wrappers and nine original exclusions do not count. The conservative distinct floor is 40. KEGG source bytes and TCDB source rows match current public tables, though historical acquisition times are unknown (tool\_execution\_summary.json; tool\_source\_snapshot\_verification.json).

Dataset count: the owner accepted BiGG model accessions toward the 120-dataset gate (25 September WhatsApp, notes/gate\_count\_provenance\_review\_20260927.md). The source ledgers report 108 unique BiGG IDs and a conservative 15-source floor from 19 PaxDb files, yielding 123 under that rule. A separate local check matched all 108 model and 19 PaxDb file hashes and the 19 saved PaxDb scores (results/model\_inclusive\_inventory\_local\_verification.json). Owner permission alone does not certify the count: provider freshness was not rechecked, and eight other provisional manifest rows remain unresolved. Nor are these 123 independent wet-lab studies. The old 124 count double-counted 44 files under one figshare accession. The original G2 and 50+ substantive body-page gates remain failed.

| Gate                        | Observed                                   | Decision                                                      |
|-----------------------------|--------------------------------------------|---------------------------------------------------------------|
| Science/data tools          | 40 substantial distinct uses / 48 linked   | Pass: thin rows excluded; sources rechecked                   |
| Accessioned datasets        | 108 BiGG models + 15 study-collapsed PaxDb | Count basis approved; 123 inventory requires own verification |
| Independent wet-lab screens | Far fewer than 120                         | No claim of 120 experimental studies                          |

*Table I3. 25 September model-inclusive inventory; owner-approved counting rule is separate from source verification.*